MIL-R-55219B(EL)

31 AUGUST 1965
Superseding
MIL-R-55219A(EL)
14 September 1964

MILITARY SPECIFICATION

RADIO SET AN/GRC-50(V) (UNITS OF)

1. SCOPE

1.1 This specification covers the following units of one type of tactical radio relay equipment for use in the frequency ranges of 601.5 to 999.5 and 1,350.5 to 1,849.5 megacycles (mc). This equipment, designated Radio Set AN/GRC-50(V) ( ), is Variable, in the sense that different amounts of the following units (and of unnomenclatured plug-in modules) are used to compose each specific set. (See 6.1 and 6.6).

Transmitter, Radio T-893(P)()/GRC
Amplifier-Converter AM-1955 ( )/GRC
Amplifier-Converter AM-1955A ( )/GRC
Amplifier-Converter AM-1956 ( )/GRC
Amplifier-Converter AM-1956A ( )/GRC
Amplifier-Oscillator AM-1957( )/GRC
Amplifier-Oscillator AM-1958( )/GRC
Amplifier-Oscillator AM-1958A ( )/GRC
Power Supply PP-2054( )/GRC
Receiver, Radio R-1148(P) ( )/GRC
Receiver, Radio R-1331 (P) ( )/GRC
Case, Standardized Components,
Electrical CY-2582( )/GRC
Case, Standardized Components,
Electrical CY-2583 ( )/GRC
Dummy Load, Electrical DA-189( )/GRC

FSC 5820

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids, or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

QQ-S-571	Solder: Lead Alloy, Tin Lead Alloy, and Tin Alloys: Flux Cored Ribbon and Wire, and Solid Form.
QQ-S-781	Strapping, flat; Steel
TT-A-580	Anti-Seize Compound, White Lead-Base,General-Purpose (For Threaded Fittings)
T-T-871	Twine, Cotton, Wrapping
UU-T-111	Tape; paper, gummed (Sealing and Securing)
PPP-T-97	Tape; pressure-sensitive adhesive, filament reinforced
PPP-P-291	Paperboard, wrapping' cushioning
PPP-B-585	Boxes, Wood, Wirebound
PPP-B-591	Boxes, Fiberboard, Wood-Cleated
PPP-B-601	Boxes, Wood, Cleated Plywood
PPP-B-621	Boxes, Wood, Nailed and Lock-Corner.
PPP-B-636	Box, Fiberboard
PPP-B-640	Boxes, Fiberboard, Corrugated, Triple Wall
PPP-S-760	Strapping, Nonmetallics, (And Connectors)

MILITARY

MIL-E-1	Electron Tubes and Crystal Rectifiers.
MIL-C-17	Cables, Radio Frequency, Coaxial, Dual Coaxial, Twin Conductor, and Twin Lead.
MIL-E-75	Electron Tubes, packaging, packing and container-marking of,general specifications for.
MIL-C-104	Crates, Wood; Lumber and Plywood Sheathed, Nailed and Bolted.
MIL-P-116	Preservation, Methods of.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

SPECIFICATIONS (Contd)

MIL-C-132	Crates, Open Wood, Maximum Capacity 2,500 Pounds.
MIL-V-173	Varnish, Moisture and Fungus Resistant, for the Treatment of Communications, Electronic and Associated Electrical Equipment.
MIL-I-631	Insulation, Electrical, Synthetic-Resin Composition, Nonrigid.
MIL-I-3190	Insulation Sleeving, Electrical, Flexible.
MIL-C-3702	Cable, Power, Electrical, Ignition, High Tension.
MIL-C-3774	Crates, Open, Wood (2,500 to 10,000 lbs).
MIL-C-3885	Cable Assemblies and Cord Assemblies, Electrical (For Use in Electronic, Communication, and Associated Electrical Equipment).
MIL-W-6858	Welding, Resistance; Aluminum, Magnesium, Nonhardening Steels or Alloys, Nickel Alloys, Head-Resisting Alloys, and Titanium Alloys, Spot and Seam.
MIL-I-7444	Insulation Sleeving, Electrical, Flexible.
MIL-B-7883	Brazing of Steels, Copper, Copper Alloys, and Nickel Alloys.
MIL-T-7928	Terminals, Lug and Splice, Crimp Style, Copper (ASG).
MIL-L-10547	Liners, Case, Waterproof
MIL-M-10578	Metal Conditioner and Rust Remover (Phosphoric Acid Base).
MIL-P-11268	Parts, Materials and Processes
MIL-M-13231	Marking of Electronic Items.
MIL-F-14072	Finishes for Ground Signal Equipment.
MIL-F-14256	Flux, Soldering, Liquid (Rosin Base).
MIL-E-17555	Electronic and Electrical Equipment and Associated Repair Parts, Preparation For Delivery of
MIL-I-18057	Insulation Sleeving, Electrical, Flexible, Glass Fiber, Silicone Rubber Treated.
MIL-S-19500	Semiconductor Devices, General Specifications for.
MIL-B-43291	Box, Fiberboard, corrugated, double wall, Weather Resistant
MIL-E-55301	Electromagnetic Compability

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

STANDARDS

MILITARY (Contd)

MIL-STD-22	Welded-Joint Designs.
MIL-STD-105	Sampling Procedures and Tables for Inspection by Attributes.
MIL-STD-109	Inspection Terms and Definitions
MIL-STD-129	Marking for Shipment and Storage.
MIL-STD-169	Extreme-Temperature Cycle.
MIL-STD-170	Moisture Resistance Test Cycle for Ground Signal Equipment.
MIL-STD-200	Electron Tubes.
MIL-STD-210	Climatic Extremes For Equipment.
MIL-STD-252	Wired Equipment, Classification of Visual and Mechanical Defects.
MIL-STD-280	Definitions of Terms For Equipment Divisions
MIL-STD-701	Preferred and Guidance List of Transistors.
MS75027	Cable-Cord, Standard Lengths and Standard Tolerances for Communication, Power and Radio Frequency Cable and Cord Assemblies.

DRAWINGS

ELECTRONICS COMMAND

SC-C-33073	Form to be used for Pilot Pack Process Sheet.
SC-A-46439	List of Accessories for Package Tester.
SC-B-61578	Grounding to Chassis.
SC-A-362100	Requirements For Preparation of Packaging and Packing Illustrations And Bills of Material
SC-DL-422467	Drawing and Data List for Transmitter, Radio T-893(P) ( )/GRC.
SC-DL-422492	Drawing and Data List for Amplifier-Converter AM-1955( )/GRC or Amplifier-Converter AM-1955A ( )/GRC.
SC-DL-422494	Drawing and Data List for Amplifier-Converter AM-1956( )/GRC or Amplifier-Converter AM-1956A( )/GRC.
SC-DL-422474	Drawing and Data List for Amplifier-Oscillator AM-1957( )/GRC.
SC-DL-422476	Drawing and Data List for Amplifier-Oscillator AM-1958( )/GRC or Amplifier-Oscillator AM-1958A ( )/GRC.
SC-DL-422478	Drawing and Data List for Power Supply PP-2054( )/GRC.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

DRAWINGS ELECT COMMAND (Contd)

SC-DL-422486	Drawing and Data List for Receiver, Radio R-1148(P) ( )/GRC.
SC-DL-498320	Drawing and Data List for Receiver, Radio R-1331(P) ( )/GRC.
SC-DL-424180	Drawing and Data List for Case, Standardized Components, Electrical Cy-2582( )/GRC
SC-DL-423910	Drawing and Data List for Case, Standardized Components, Electrical CY-2583( )/GRC.
SC-DL-423519	Drawing and Data List for Dummy Load, Electrical DA-189( )/GRC.

EXHIBITS

ELECTRONICS COMMAND

PMD-38 Preparation For Delivery of Signal Corps Equipment, Inspection Of

(Copies of specification, standards, drawings, and publications required by contractors in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer. Both the title and number or symbol should be stipulated when requesting copies.)

2.2 Other publications.- The following documents form a part of this specification to the extent specified herein. Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

CONSOLIDATED CLASSIFICATION COMMITTEE

Consolidated Freight Classification.
Uniform Freight Classification.

(Application for copies should be addressed to the Consolidated Classification Committee, 202 Union Station, Chicago 6, Illinois)

NATIONAL BUREAU OF STANDARDS

Handbook H28 Screw-Thread Standards for Federal Services.

POST OFFICE DEPARTMENT

Parcel Post Regulations.

(Application for copies should be addressed to the Superintendent of Documents, Government Printing Office, Washington 25, D.C.)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

2.2 (Contd)

AMERICAN TRUCKING ASSOCIATION, INC.

National Motor Freight Classification

(Application for copies should be addressed to the National Classification Board, 1424 16th Street, NW, Washington, 6, D.C.)

3. REQUIREMENTS

3.1 Description.-

3.1.1 Functional.- The equipment covered by this specification are units of a complete, compact, rugged, radio relay equipment whose major operating components may be used either as a terminal or modulation frequency patch repeater for intermediate or forward area systems. The radio set can be used to provide simultaneous two-way (duplex) transmission over eight jumps in series. The system provides 399 radio-frequency (rf) channel assignments in the 601.5 to 999.5 mc band and 500 rf channel assignments in the 1350.5 to 1849.5 mc band. The channels have one megacycle spacing. When used in combination, either as a repeater or as a terminal, it is possible to tune and operate the receivers and transmitters on independent frequencies over the entire frequency range. One set covers both frequency ranges, with the two bands provided by means of plug-in units in the transmitter and receiver. The basic radio set contains a maximum of common equipment with the plug-in units containing only those circuits which are peculiar to each frequency range.

3.1.2 Radio Set AN/GRC-50(V), physical.- The (V) 1 version provides for multichannel, line-of-sight, two-way communication in the ultra high-frequency range. It provides 399 operating channels in the 601.5 to 999.5 mc Low Band range and 500 channels in the 1350.5 to 1849.5 mc High Band range. This version is intended primarily for use as a radio link in a communication network, which may include carrier telephone and teletypewriter sets. The radio set is used in pairs; one set is used at each end of the line-of-sight path. The radio set may be used with 4-, 12-, or 24 channel frequency-division multiplex (fdm), or 12-or 24-channel pulse-code modulation (pcm) equipment.

3.1.2.1 Radio Transmitter T-893(P) ( )/GRC.-The transmitter for the AN/GRC-50(V) ( ), in a self-contained carrying case (CY-2429( )/GRC, consists of a main frame and either of two amplifier-oscillator rf heads, the AM-1957( )/GRC, AM-1958( )/GRC or AM-1958A( )/GRC. The AM-1957( )/GRC is the Low Band head; the AM-1958( )/GRC or AM-1958A( )/GRC is the High Band head. The main frame contains four plug-in subchassis; the current regulator, the baseboard, the automatic frequency control (afc), and the modulator.

3.1.2.2 Receiver, Radio -R-1148(P) ( )/GRC.- The receiver for the AN/GRC-50(V) ( ), in a self-contained carrying case (CY-2429( )/GRC, consists of a main frame and one of four amplifier-converter rf heads, the AM-1955( )/GRC, AM-1955A( )/GRC, AM-1956( )/GRC or AM-1956A( )/GRC. AM-1955 ( )/GRC and AM-1955A( )/GRC are low band heads; AM-1956( )/GRC and AM-1956A( )/GRC are high band heads. The main frame contains four plug-in subchassis; the baseband, the automatic frequency control, the signalling unit, and the second intermediate-frequency (IF) unit. The receiver power is provided by a self-contained power supply located on the lower deck of the main frame.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.1.2.3 Radio Receiver R-1331(P) ( )/GRC - The receiver for the AN/GRC-50(V) ( ), in a self-contained carrying case (CY-2429( )/GRC, consists of a main frame and either of two amplifier-converter rf heads, the AM-1955A()/GRC or the AM-1956A( )/GRC.

The AM-1955A( )/GRC is the Low Band head and the AM-1956A( )/GRC is the high band head. The main frame contains three plug-in subchassis; the baseband, the signalling unit, and the second intermediate frequency (IF) unit. The receiver power is provided by a self-contained power supply located on the lower deck of the main frame.

3.1.2.4 Power Supply PP-2054/GRC - This transmitter power supply consists of a front panel with a conventional single-frame chassis; it fits into a carrying case (CY-2428 ( )/G) designed for stack-mounting on top of the other radio set components. It operates from a single phase, nominal 115-volt, 50- to 60-cycle per second (cps) power source.

3.2 Description, other units.-

3.2.1 Cases.-

3.2.1.1 Transmitter and receiver.- A Case, Standardized Components, Electrical CY-2429( )/GRC is part of each transmitter and of each receiver.

3.2.1.2 Power supply. - A Case, Power Supply CY-2428( )/GRC is part of Power Supply PP-2054( )/GRC.

3.2.1.3 Spare parts.- Spare parts for the radio set are shipped in Case, Standardized Components, Electrical CY-2583( )/GRC.

3.2.1.4 Alternate Amplifiers.- Alternate tuning unit Amplifier-Converter AM-1955( )/GRC or AM-1955A( )/GRC, Amplifier-Oscillator AM-1957( )/GRC and AM-1958( )/GRC or AM-1958A( )/GRC are housed in Case, Standardized Component, Electrical CY-2582( )/GRC.

3.2.2 Cable Interconnections.-

3.2.2.1 Cable Assembly, Special Purpose, Electrical CX-4557( )/GRC. This cable assembly is used to interconnect the transmitter and receiver.

3.2.2.2 Cable Assembly, Power, Electrical CX-4558( )/U.- This cable assembly is used to interconnect the transmitter and the transmitter power supply.

3.2.2.3 Cable Assembly, Power, Electrical CX-4559( )/U. - This cable assembly is used to connect either the transmitter power supply or receiver to Regulator, Voltage CN-514( )/U.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

3.2.2.4 Maintenance cables.- Cables for maintenance of the equipment are supplied to provide cable extensions between the transmitter and receiver and their respective subassemblies as part of Test Facilities Kit MK-715( )/GRC-50.

3.2.2.5 Cable Assembly, Radio Frequency CG-718( )/U.- A 3 foot cable is used to connect AM-1955( )/GRC or AM-1955A( )/GRC to an AM-1957( )/GRC, an AM-1956( )/GRC or AM-1956A( )/GRC to an AM-1958( )/GRC; or AM-1958A( )/GRC.

3.2.3 Connector Adapter U-211( )/U.- This connector adapter is used to connect the CX-4559( )/U to a utility outlet.

3.3. Preproduction samples.- The contractor shall furnish two preproduction samples of the equipment for approval, provided the invitation for bids and contract require samples. (See 4.3)

3.4 Construction.-

The following equipment shall be constructed in accordance with the drawings specified below:

UNIT	DRAWINGS ON:
Transmitter, Radio T-893(P)( )/GRC	SC-DL-422467
Amplifier-Converter AM-1955( )/GRC	SC-DL-422492
Amplifier-or Converter AM-1955A( )/GRC	SC-DL-422492
Amplifier-Converter AM-1956( )/GRC	SC-DL-422494
Amplifier-Converter or AM-1956A( )/GRC	SC-DL-422494
Amplifier-Oscillator AM-1957 ( )/GRC	SC-DL-422474
Amplifier-Oscillator AM-1958( )/GRC	SC-DL-422476
Amplifier-Oscillator AM-1958A( )/GRC	SC-DL-422476
Power Supply PP-2054( )/GRC	SC-DL-422478
Receiver, Radio R-1148(P ( )/GRC	SC-DL-422486
Receiver, Radio R-1331(P) ( )/GRC	SC-DL-498320
Case, Standardized Components, Electrical CY-2582( )/GRC	SC-DL-424180
Case, Standardized Components, Electrical CY-2583()/GRC	SC-DL-423910
Dummy Load, Electrical DA-189( )/GRC	SC-DL-423519

3.5 Parts, materials, and processes.-

3.5.1 Cable assemblies (see 4.4).- The length of electrical cable assemblies shall conform to Military Standard MS75027 unless otherwise specified in the bid request and contract or drawings. Power and audio-frequency (af) cable and cord assemblies used for voltages up to and including 600 volts alternating-current (ac) or corresponding direct-current(dc) shall conform to all requirements of Specification MIL-C-3885. Cable assemblies used for over 600 volts ac or equivalent dc shall be subject to approval by the contracting officer. Where applicable, high-voltage cables shall conform to Specification MIL-C-3702. Radio frequency coaxial cables shall be in accordance with Specification MIL-C-17.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.5.2 Finish, Protective (see 4.4) - The items on order shall be given protective finish in accordance with Specification MIL-F-14072 and the equipment drawings. The final paint film on Type 1 surfaces of the equipment shall be finish P513, per MIL-F-14072, final film E, semigloss, light gray enamel per Table IA. The equipment cases shall be green color (olive drab), semigloss enamel matching a color chip provided by the procuring agency (see 6.3).

3.5.3 Plastic materials and parts.-Where not machined, plastic materials and parts shall have the original smooth or polished surfaces. Surfaces that have been sawed, cut, punched, or otherwise machined, should be as smooth as practicable, in accordance with good manufacturing practice for the intended application.

3.5.4 Cleaning.-

3.5.4.1 Parts. - After fabrication, parts shall be cleaned in accordance with good commercial practice, or as specified in an applicable document. Cleaning processes shall have no deleterious effect. Corrosive material shall be removed completely before the parts are mounted on chassis, panels, etc.

3.5.4.2 Units.- After assembly, units shall be cleaned thoroughly and shall be free from particles of solder, flux, and other foreign material. In addition, when necessary, such cleaning shall also be performed before final assembly of the units.

3.5.5 Marking.-

3.5.5.1 Conformance.-Marking shall conform to Specification MIL-M-13231. (See 4.4) Front panel marking shall be group 1 as described in Specification MIL-M-13231.

3.5.5.2 Visibility of markings.- Wherever practicable, parts shall be so mounted that their identification markings will be readily visible with minimum disassembly of the equipment.

3.5.5.3 Serial numbers.- The following items shall have serial numbers unless otherwise specified by the contracting officer:

Transmitter, Radio T-893(P) ( )/GRC
Amplifier-Converter AM-1955( )/GRC
or
Amplifier-Converter AM-1955A( )/GRC
Amplifier-Converter AM-1956( )/GRC
or
Amplifier-Converter AM-1956A( )/GRC

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.5.5.3 Serial Numbers (Contd)

Amplifier-Oscillator AM-1957( )/GRC
Amplifier-Oscillator AM-1958( )/GRC
Amplifier-Oscillator AM-1958A( )/GRC
Power Supply PP-2054( )/GRC
Receiver, Radio R-1148(P) ( )/GRC
Receiver, Radio R-1331(P) ( )/GRC
Case, Standardized Components, Electrical CY-2582( )/GRC
Case, Standardized Components, Electrical CY-2583( )/GRC
Dummy Load, Electrical DA-189( )/GRC

In addition, other units shall be serial numbered if required by the contracting officer or individual specifications.

3.5.6 Riveting.- Thickness of countersunk metal which accepts the heads of flush rivets shall be not less than the height of the rivet heads. After riveting, the joints shall be tight, the joined parts shall be undamaged, and the rivet heads shall be properly seated and tight against their bearing surfaces.

3.5.7 Securing of parts.-

3.5.7.1 Mounting components.- Brackets, lugs, flanges, inserts, bolts, and other mounting arrangements shall be such as to retain items securely when subjected to specified service conditions. Parts shall be so mounted that loosening, wear of mounting arrangements, or permanent separation of parts will not occur when the items in which they are used are subjected to specified service conditions. Any permanent deformation resulting from the tests shall not cause degradation of performance below full specification requirements. When practicable, friction between mating surfaces shall not be employed as the sole means of preventing fixed parts from rotating. Parts shall be secured in such manner that failure of a single rivet or screw will not free the part completely.

3.5.7.2 Securing of parts by threaded fasteners.- Securing of parts by threaded fasteners to hard material, or to soft material (aluminum, magnesium, zinc, plastic, etc) except wood, shall conform to Specification MIL-P-11268. Except as specified therein, a flatwasher shall be used between any screwhead and soft material unless the shape of the head is such that the flatwasher is unnecessary. Nylok or other screws with plastic devices, and Loctite or similar sealants, shall not be subjected to temperatures in excess of 250° Fahrenheit (as during baking of paint). Loctite or similar sealants shall not be used where the connection must be electrically conductive; where used, such materials shall be applied in accordance with the manufacturer's instructions.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.5.7.3 Brittle material.- Brittle castings, or parts made of ceramic or other brittle material, shall not be over-constrained by the means used to secure them. Not more than three points of contact shall be used between the brittle part and the mounting surface, unless the mating faces are accurately machined. Mounting washers of suitable plastic, rubber, or soft copper having slight compressibility shall be provided to prevent breakage or cracking of the parts. Lead washers shall not be used. Threaded holes in ceramic material shall not be employed for assembly or mounting of parts.

3.5.8 Brazing.- Brazing of steels, copper, copper alloys and nickel alloys shall conform to Specification MIL-B-7883. (See 4.4)

3.5.9 Soldering.-

3.5.9.1 Solder.- Solder shall conform to Specification QQ-S-571. For soldering electrical connections, composition Sn60, type AR or S, shall be used for general purposes; and composition Sn62 or Sn63, type AR or S, for special purposes. When type S solder is used for soldering electrical connections, the flux shall conform to Specification MIL-F-14256.

3.5.9.2 Acid or acid salts.-No acid or acid salts shall be used in preparation for or duing soldering; however, exception is permitted for preliminary tinning of electrical connections and for tinning or soldering of mechanical joints not used to complete electrical circuits, but in no case shall acid or acid salts be used where they can come in contact with insulation material. Where acid or acid salts are used, as permitted above, they shall be completely neutralized and removed immediately after use.

3.5.9.3 Process.-There shall be no sharp points or rough surfaces resulting from insufficient heating. The solder shall feather out to a thin edge, indicating proper flowing and wetting actions, and shall not be crystallized, overheated or underheated. The minimum necessary amount of flux and solder shall be used for electrical connections. Any means employed to remove an unavoidable excess of flux shall not incur the risk of loose particles of flux, brush bristles, or other foreign material remaining in the equipment; flux being spread over a larger area; or damage to the equipment. Insulation material that has been subjected to heating during the soldering operation shall be undamaged and parts fastened thereto shall not have become loosened.

3.5.10 Welding (see 4.4)- When spot or other intermittent welds are used to hold a part, the number of welds shall be at least two. Welded joints shall conform with procurement requirements. (See 3.4) Applicable welded joints shall conform to Standard MIL-STD-22 and shall be such that grinding on the finished weld will be unnecessary. Other spot, stitch, and seam welds shall conform to Specification MIL-W-6858.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

3.5.10.1 Cleaning.- Surfaces to be welded shall be cleaned in accordance with good commercial practice and shall be free from rust, scale, paint, grease, and other foreign material. Welded assemblies shall be cleaned to remove scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Acid used for cleaning shall be completely neutralized and removed.

3.5.10.2 Process.- Preheating shall be employed where distortion is likely to result from welding. Welds shall have thorough penetration and good fusion and shall be free from scabs, blisters, abnormal pock marks, cracks, voids, slat inclusions, and other harmful defects. Where undesirable internal stresses are likely to result from welding, welded items shall be stress-relieved. Inert-gas-shielded arc welding shall be used, when practicable, for welding of aluminum, magnesium, or stainless steel.

3.5.10.3 Cleaning after welding.-Welded assemblies shall be cleaned to remove rust, scale, oxidation products, and excess flux by sandblasting, wire brushing, or other suitable means. Prior to painting, steel parts that have been arc welded or acetylene welded shall, in addition, be subjected to vat passivation or a phosphoric acid etch in accordance with Specification MIL-M-10578. Acid used for cleaning shall be completely neutralized and removed.

3.5.11 Screws, other threaded devices, and related parts.-

3.5.11.1 Form and fit.- Unless otherwise specified, screw threads shall conform to Handbook H28; the Unified Thread Form and Series shall be Class 2A or 2B fit, and the American National Thread Series shall be Class 2 fit.

3.5.11.2 Self-tapping screws.- Chips formed in through holes by thread-cutting screws, where used, shall be removed.

3.5.11.3 Thread engagement and extension.- The length of thread engagement for tapped holes in steel or hard brass shall at least equal the nominal thickness of corresponding machine-screw nuts in Sizes No. 12 and smaller and semifinished hexagon regular nuts in sizes 1/4 inch and larger, as shown in Handbook H28. Thread engagement for tapped holes in aluminum, magnesium, or other soft material shall be at least 50 percent greater than the thread engagement specified for steel or hard brass. Where permitted by design of the equipment, screws and bolts shall extend at least 1-1/2 threads beyond the nut, but no more than 1-1/2 threads plus 1/8 inch for screws up to 1 inch in length and 1-1/2 threads plus 1/4 inch for screws over 1 inch in length.

3.5.11.4 Torque.- Threaded parts of steel shall be tightened to the torque values shown in table 1. Threaded parts of other size or material shall be tightened to a suitable design value established by the contractor for the particular application. For critically stressed applications, the torque for tightening threaded parts shall be controlled by use of torque measuring or controlling devices.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

Table 1.- Torque for tightening steel threaded parts

Thread	Torque (lb-in.)
6-32	7 to 9
8-32	14 to 18
10-24	19 to 23
10-32	25 to 30
12-24	32 to 30

3.5.11.5 Anti-seize compound.- Unless otherwise specified, anti-seize compound conforming to Federal Specification TT-A-580 shall be used to prevent seizing of threads where the threaded surface is aluminum or magnesium.

3.5.12 Terminals, fixed.-Fixed terminals (including binding posts) shall properly support the wires connected thereto and shall not turn or become loose when the items in which the terminals are used are subjected to specified service conditions.

3.5.13. Wiring and cabling.-Wiring and cabling shall be neat and sturdy.

3.5.13.1 Protection.-Wires and cables shall be so placed and protected as to avoid contact, under specified service conditions, with rough or irregular surfaces or sharp edges. Wires shall not be bent sharply where they enter insulated material. Where wires run through holes in metal partitions, shields, or similar items less than 1/8 inch in thickness, the wires shall be protected by suitable grommets or bushings. Where wires run through holes in metal 1/8 inch in thickness or greater, the wires shall be protected by grommets or bushings or shall have the edges of the holes rounded to a minimum radius of 1/16 inch.

3.5.13.2 Support.-Wire and cable shall be properly supported and secured, to prevent undue stress on the conductors and terminals and undue change in position of the wire or cable (i) during and after subjection of the equipment to specified service conditions or (ii) after service or repair of the equipment in a normal manner. Note: When shielding on wire or cable is unprotected by an outer insulation, adequate support is necessary to prevent the shielding from coming in contact with exposed terminals or conductors.) Twine or tape shall not be used for securing wire and cable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.5.13.3 Connections.- Before being soldered to terminal lugs or fixed terminals, wires shall be mechanically secured so that the connections are not dependent for strength on solder alone. Where practicable, wires to be soldered to fixed terminals shall be looped at least one-half turn but not more than three-quarters turn around the terminal before soldering. Bared ends of wire leads to be terminated in solder-type terminal lugs shall be tinned, silver plated, or lead-alloy coated. Electrical connections shall not be made by clamping between a metallic and nonmetallic material. Fraying of textile ends of wires shall be prevented mechanically or by application of varnish conforming to Specification MIL-V-173. No varnish, lacquer, inspection paint, or other coating shall be applied to completed electrical connections.

3.5.13.4 Grounding.- Ground connection to shields and to other mechanical parts, except the chassis or frame, shall not be made to complete electrical circuits but only to eliminate high-potential ac points. (See Drawing SC-B-61578).

3.5.13.5 Wire and cable shielding.- Wire or cable shielding shall be grounded to the chassis or frame of the equipment unless otherwise specified in the drawing. The shielding shall end at a sufficient distance from exposed conductors to insure against shorting or arcing between the conductor and the shielding.

3.5.13.6 Crimping of solderless terminal lugs.-Crimping of solderless terminal lugs shall be so accomplished that the connections will meet the resistance (voltage drop) and tensile strength requirements and tests of Specification MIL-T-7928. (See 4.4.)

3.5.13.7 Clearance.- Clearance between wires or cables, and units such as electron tubes, resistors, and power diodes, shall be provided to avoid deterioration of the wires or cables because of the heat dissipated by such units, and vibration and motion of movable units, when the equipment is subjected to specified service conditions. Clearance between solder connections or bare conductors, on terminal boards or other parts, shall be such that no accidental contact can occur between adjacent connections when subjected to specified service conditions.

3.5.13.8 Splicing.- Wires in a continuous run between two terminals shall not be spliced during the wiring operation.

3.5.13.9 Insulating sleeving.- Insulating sleeving shall not be used unless specified for a particular application or specifically approved by the contracting officer. Approval will not be granted when other means of insulation are practicable. If insulating sleeving is used, it shall conform to Specificatio MIL-I-631, MIL-I-3190, MIL-I-7444, or MIL-I-18057, as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.5.13.10 Inductive and capacitive effects.- Wires and cables shall be so located that inductive and capacitive effects, unless used as a design feature, will be the minimum practicable.

3.5.13.11 Slack.- Wires and cables shall be as short as practicable except that sufficient slack shall be provided to prevent undue stress on cable forms, wires, and connections; to enable parts to be removed or replaced during servicing without disconnecting other parts; and to facilitate field repair of broken or cut wires.

3.5.14 Controls and rotating components.- Controls shall conform to the following requirements:

3.5.14.1 Effect of vibration, bounce and shock.- Controls shall maintain their setting when the items in which they are used are subjected to specified operating service conditions.

3.5.14.2 Mechanical operation.- Play and backlash shall not cause poor contact or inaccurate setting. Controls shall operate freely and smoothly, and shall be lubricated when lubrication does not interfere with operation.

3.5.14.3 Locking devices.- Locking devices shall be capable of retaining controls in any given setting within the range of the control. The locking and unlocking actions shall be easily accomplished and shall not affect the setting of the control. When in the unlocked position, the locking device shall not interfere with the normal operation of the control. Where vernier controls are used, the locking device shall operate on both main and vernier controls if necessary to prevent damage.

3.5.14.4 Nonturn device.- Controls shall be equipped with a positive device to prevent their turning in respect to the panel on which they are mounted.

3.5.14.5 Tuning dial stability. - Friction in the tuning dial mechanism shall be sufficient to maintain the tuning knob in any position over the operating range of the mechanism without excessive Torque. The Torque shall not exceed 35 inch ounce.

3.5.14.6 Gears and cams.- Gears shall be properly alined and marked, and shall operate without interference, tight spots, or other irregularities.

3.5.14.7 Chains.- Where used, excessive play shall not be allowed.

3.5.14.8 Blower motors.- The blower motors shall rotate in the direction that will force air out of the front grill.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.5.15 Adjustment and repair.- The equipment shall be so constructed that parts, terminals, wiring, etc, are accessible for circuit checking, adjustment, maintenance, repair, and replacement with minimum disturbance to other parts and wiring. After normal maintenance and repair, the equipment shall meet specified performance requirements.

3.5.16 Tropicalization of material.- Material shall be treated as follows:

3.5.16.1 Treating materials.- Treating materials containing a mercury-bearing fungicide shall not be used. The contractor shall determine that the treating material is compatible with the material or surface to be treated. Selection of treating materials shall be such that any increase in flammability of treated material will be held to the practical minimum.

3.5.16.2 Toxicity.- Treating materials shall cause no skin irritation or other injury to personnel handling the treated material during fabrication, transportation, operation, or maintenence of the equipment, or during use of the finished items when used for purpose intended.

3.5.16.3 Flexibility.- Treatment shall not affect the flexibility of treated materials, to the extent that the equipment may fail to meet specified requirements when subjected to specified service conditions.

3.5.16.4 Statement of treatment.- The contractor shall submit for approval, to the contracting officer, a statement describing in detail the materials to be treated and the treating materials and processes that he proposes to use. (See 6.2(h)).

3.5.17 Fabric and thread.- The color of any required fabric shall be Olive-Drab No. 7. The color of thread shall match shade S-1 of the United States Army Standard Color Card for Official Standardization Shades of Sewing Thread. (See 6.7) Fabric and thread shall be preshrunk, or allowance shall be made for shrinkage, in order to provide for satisfactory fit of finished items both before and after they are immersed in water and then dried.

3.5.18 Interchangeability.- Like units, assemblies, subassemblies, and replaceable parts shall be physically and functionally interchangeable, without modification of such items or of the equipment. (See 4.17.) Individual items shall not be hand-picked for fit or performance; however; matched pairs or sets, when permitted, may be interchangeable as such. Reliance shall not be placed on any unspecified dimension, rating, characteristic, etc.

3.5.19 Tubes, electron; and semiconductors, transistor and diode.- Except as otherwise provided by Standard MIL-STD-200 and MIL-STD-701, these electron devices (tubes and semiconductors) shall conform to Specifications MIL-E-1 and MIL-S-19500.

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:14Z
Check the source to verify that this is the current version before use.

3.5.19.1 Choice, use and application.- Except as otherwise specified in a and b below, the choice, use, and application of these electron devices shall conform to Standards MIL-STD-200 and MIL-STD-701:

a. When substitution of a preferred or guidance type of electron devide for a type required by the equipment specification would degrade equipment performance or ruggedness below specified requirements, pertinent data shall be furnished to the contracting officer, to justify retaining the type required by the equipment specification.

b. When the foregoing standards require modification of equipment circuitry, the modification shall be subjected to prior approval by the contracting officer.

3.5.19.2 Circuit location.- The electron devices furnished in equipment on contract shall be the individual devices used when the equipment was inspected for acceptance and, unless separate packaging of the devices is specified, shall remain in the respective locations occupied during the inspection.

3.5.20 Power - The equipment shall operate from a 115 Volts, rms, 47 cycles to 63 cycles, single phase, with a voltage variation of plus or minus (±5%) percent.

3.6 Service conditions.- The equipment shall meet the following service conditions; where a test is referenced, meeting the test shall be considered as compliance with the requirement. Minor units having no tests specified shall pass full specification requirements after subjection to environmental tests.

3.6.1 Operation.- Continuous use for a period of 6 months, under the world-wide environmental conditions specified in 3.6.2 through 3.6.7, with a duty-cycle of 24 hours on and with no maintenance or replacement of parts, other than tubes after each successive 500 hours of operation. Where the contractor is required to make a selection of parts, materials, processes, construction methods, etc., he shall be guided by this requirement. Approval of the preproduction sample (see 3.3) shall be considered as compliance with this requirement.

3.6.2 Temperature (see 4.6).-

a. Operating: Ambient temperature in the range of +132° Fahrenheit (F) to -20°F.

b. Nonoperating: Continuous exposure in the range of +160°F to -80°F.

3.6.3 Relative humidity.- Up to 97 percent relative humidity. (See 6.10). Meeting the tests of 4.7 shall be considered compliance with this requirement.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.6.4 Elevation (see 4.8).-

a. Operating: Up to 12,000 feet above sea level.

b. Nonoperating: Up to 50,000 feet above sea level.

3.6.5 Orientation (Except DA-189( )/GRC).-

a. Operating (see 4.9).- Any orientation up to 20 degrees from normal operating position (that is: upward, downward, left, or right).

b. Nonoperating: Storage in any position for a period of two years. Approval of the preproduction sample (see 3.3) shall be considered as compliance with this requirement.

3.6.6 Rain (see 4.14).- Two inches of rainfall per hour, with wind at nominal 20 miles per hour. No leakage is permissible.

3.7 Electromagnetic Compatibility.- Radio Set AN/GRC-50 (V) (Units of) procured under this specification for Army usage, shall conform to the Tactical requirements of MIL-E-55301 (EL).

3.8 Vibration, bounce, and shock.-

3.8.1 Vibration, internal.- With the equipment shock-mounts (if any) blocked the amplitude of vibration of any part, subassembly, or structural member of the equipment shall not exceed twice the amplitude of the vibration applied to the equipment at any frequency between 10 and 55 cycles per second (See 4.10).

3.8.2 Bounce and shock.- The equipment shall be capable of meeting the requirements of table II.

TABLE II - Bounce and Shock

Paragraph heading	Para	Performance after test
Bounce	4.11	Specified performance. (Note A) No physical damage; superficial surface abrasion is permissible.
Shock, bench-handling	4.12	Specified performance. (Note A). No physical damage.
Shock; drop	4.13	Specified performance. (Note A) Any physical damage shall be minor only.

NOTE A: The equipment shall meet specified performance for the following:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.8.2 Bounce and shock (Contd)

TEST NR.	TEST	Req para.
1	PCM Signal to Noise	3.10.3.4
2	FDM Signal to Noise (Wideband)	3.10.3.3(b)
3	Orderwire Signal to Noise	3.10.5.2
4	Level Adj. & Metering	3.10.6.2
5	Transmitter Freq. Range and Power Output	3.10.1.1
6	Transmitter Freq. Settability	3.10.1.5
7	Receiver Frequency Settability	3.10.2.11
8 *	Receiver Automatic Frequency Control	3.10.2.5
9 *	Receiver Secondary Automatic Freq. Control	3.10.2.7
10	Remote Alarm	3.10.5.1
11	Tone Generator	3.10.6.6
12	Dummy Load VSWR	3.10.7.1
13	Dummy Load Power Calibration accuracy	3.10.7.2
*	As applicable	

3.9 Bounce preconditioning.- The equipment shall be capable of meeting the requirements specified herein, without subsequent processing, after subjection to the bounce preconditioning of 4.15. (Also see 4.5).

3.10 Electrical requirements (see 4.16).-

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.1 Transmitter (see 4.16.3).-

3.10.1.1 (a) Frequency range and power output - When tested as specified in 4.16.3.1(a)the transmitter power output measured at the duplexer output shall be at least 12 watts from channels 1-100 (601.5 through 700.5 mc), at least 15 watts from channels 101-399 (701.5 through 999.5 mc), and at least 8 watts from channels 400-899(1,350.5 through 1,849.5 mc) under standard operating conditions (see 4.16.1).

3.10.1.1 (b) The power output in either band shall not decrease more than 2 db from the power obtained per paragraph 3.10.1.1 (a) under standard conditions at any given frequency under any combination of operating service conditions, or with an input supply of 109.3 volts at 47.5 cps applied to the input of Power Supply PP-2054( )/GR. (see 4.16.3.1 (b).

3.10.1.2 Overload (see 4.16.3.2).- With the mixer and amplifier stages detuned, the cathode current shall not exceed the level originally established by the respective potentiometer controls.

3.10.1.3 Frequency stability (see 4.16.3.3).- The transmitter frequency, when set to any value between 601.5 mc to 999.5 mc or 1,350.5 mc to 1,849.5 mc after 30 minutes warmup and at any ambient temperature between -20 degrees F and +132 degrees F, shall not vary beyond plus 0.005% plus 90 kilocycles (kd) to minus 0.005% minus 90 kc of this frequency for the eight hours, with the ambient temperature during any 8 hour period not removed more than 54 degrees F from the starting temperature but still within the above temperature limits.

3.10.1.4 Short time afc change (see 4.16.3.4).- The frequency of the transmitter shall not be abruptly or suddenly changed by switching of the automatic frequency control.

3.10.1.5 Transmitter frequency settability (see 4.16.3.5).- It shall be possible to align the radio transmitter to within ± 0.005% ± 90 KC of the desired frequency by the use of the automatic frequency control system.

3.10.1.6 Transmitter Spurious output (see 4.16.3.6).- All spurious outputs shall be at least 50 db below the desired output (reference point).

3.10.1.7 Transmitter Alarm. - When tested as specified in 4.16.3.7, the alarm shall be activated when the low band channel output drops below 10 watts, the high band below 5 watts, or when cooling air stops circulating.

3.10.1.8 Insertion loss.- When tested as specified in 4.16.3.8, the insertion loss shall not exceed 2 db over the pass band of one channel.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.1.9 Transmitter FDM Deviation Sensitivity (See 4.16.3.9). With the input attenuator set to mid-position a 72 KC signal applied to the 135 ohm FDM input terminals at a level of +13 dbm shall produce rated deviation of 375 Kc ± 2db (300 Kc + 470 Kc). Correspondingly a level of +11 dbm shall produce a deviation of 300 Kc +2db. (240 Kc to 375 Kc). NOTE: At any other modulating frequency the deviation sensitivity shall be modified by the transmitter pre-emphasis characteristic.

3.10.1.10 Transmitter PCM Deviation Sensitivity (See 4.16.3.10). - The peak deviation of a video train of peak-to-peak amplitude of 1.5 volts, and of such characteristics as produced by 3 marks followed by 3 spaces applied to the PCM input shall be 375 KC ± 1 db followed by 3 spaces applied to the PCM input shall be 375 KC ± 1 db with the PCM input attenuator set to mid-position. A 10 KC sinusoidal signal of 0.53 volts rms (1.5 volts peak to peak) applied to the PCM nput terminal with the attenuator in mid-position shall be used to simulate the signal and shall be 375 Kc ±2 db (300 Kc to 470 Kc).

3.10.1.11 AC Power Interruption (Transmitter). - When tested as specified in 4.16.3.11, the transmitter shall have the capability such that the output frequency returns to within ± 0.005 percent ± 90 KC of the assigned channel frequency following A-C power interruptions of durations from two (2) seconds to ninety (90) seconds.

3.10.2 Receiver (see 4.16.4).- The following requirements shall be met without the rf duplexer (see 3.10.1.8):

3.10.2.1 Frequency range and Noise figure.- When measured as specified in 4.16.4.1, for frequencies from 601.5 mc to 999.5 mc, inclusive, the noise figure shall not exceed 13 db; for frequencies from 1,350.5 mc and 1,849.5 mc, inclusive, the noise figure shall not exceed 12 db. (See 6.12.)

3.10.2.2 Selectivity.- The selectivity of the receiver shall be such that the requirements of 3.10.2.4 and 3.10.6.5 are met.

3.10.2.3 Receiver Deviation Sensitivity. - (see 4.16.4.2)

3.10.2.3.1 Receiver FDM Deviation Sensitivity (See 4.16.4.2).

The output signal shall be at a level of +13 dbm ±2db with the output attenuator at mid position when measured at the 135 ohm FDM output terminals for an rf output, signal at rated deviation of 375 Kc at the crossover modulation frequency of 72 KC. Correspondingly the output shall be at a level of 11 dbm ±2 db for a deviation of 300 KC. (Note: To simulate 24 and 12 channel loading respectively).

3.10.2.4 Spurious response (see 4.16.4.3).- The ratio of desired signal response to any spurious response, including the image response, shall be at least 50 db.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.2.5 Automatic frequency control.- When tested as specified in 4.16.4.4, with Receiver, Radio R-1148(P) ( )/GRC and Amplifier Converter AM-1955( )/GRC or AM-1956 ( )/GRC, after setting the receiver to a given frequency channel anywhere within the frequency range 600 to 1000 mc. or 1350 to 1850 mc, with a remote signal whose power shall be at any level between minus 80 dbm and minus 40 dbm at the assigned channel frequency, the primary AFC shall keep the receiver local oscillator frequency (and the unmodulated frequency in the IF amplifier) within ± 0.005 percent ± 90 Kc of the assigned channel frequency at any time within an 8 hour period thereafter. This requirement shall be met with the ambient temperature removed any amount up to 54°F from the ambient temperature at the time of initial setting of the receiver. In no case shall the temperature be above plus 132°F or below minus 20°F. After originally locking in on the remote signal, the unmodulated frequency in the if amplifier shall be maintained within the limits specified above for the next week. Operating the afc switch to ODD (or EVEN) shall not cause any momentary frequency change.

3.10.2.6 Frequency Stability - When tested as specified in 4.16.4.5, with Receiver, Radio R-1331 (P) ( )/GRC and Amplifier, Converter AM-1955A( )/GRC or AM-1956A( )/GRC, after setting the receiver to a given frequency channel anywhere within the frequency range of 600 to 1000 mc. or 1350 to 1850 mc., with a remote signal whose power shall be at any level between minus 80 dbm and minus 40 dbm at the assigned channel frequency, the local oscillator shall not exceed ± 750 kc. from its initailly measured value at any time within an 8 hour period. This requirement shall be met with the ambient temperature removed any amount up to 54 degrees F from the ambient temperature at the time of initial setting of the receiver. In no case shall the temperature be above plus 132 degrees F or below minus 20 degrees F. The above limits shall be maintained for a continuous operational period of up to one week.

3.10.2.7 Secondary Automatic Frequency Control.- When tested as specified in 4.16.4.6, with a receiver and Amplifier Converter AM-1955( )/GRC or AM-1956A( )/GRC, after setting the receiver to a given frequency channel anywhere within the frequency range of 600-1000 mc. or 1350 to 1850 mc, with a remote signal w hose power shall be at any level between minus 80 dbm and minus 40 dbm at the assigned channel frequency, the secondary AFC control shall keep the second intermediate frequency (17 mc) variation within ± 60 KC of the discriminator zero crossover frequency from its initially measured value at any time within an 8 hour period. This requirement shall be met with the ambient temperature removed any amount up to 54 degrees F from the ambient temperature at the time of initial zero discriminator frequencies measurement. In no case shall the temperature be above plus 132 degrees F or below minus 20 degrees F. The above limits shall be maintained for a continuous operational period of up to one week.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.2.8 IF bandwidth (see 4.16.4.7).- The IF selectivity shall have a 3 db bandwidth of 1.1 mc, + 0.4 mc., -0.2 mc with approximate gaussian characteristic and with the response down at least 60 db at ± 3.0 mc or more from resonance.

3.10.2.9 Squelch and Alarm.- When tested as specified in 4.16.4.8, the squelch control shall be capable of adjusting the squelch to both the muting and non-muting conditions for any input level from -90 to -70 dbm. When operating at a level of -70 dbm, the dead space shall be less than 3 db. With the squelch operating (muting) the output noise levels at the FDM, PCM, and ROW terminals shall be less than -45 dbm when measured with a flat weighted meter over a detected range of at least 0.300 to 108 KC for FDM and PCM, 300 to 1800 cps for ROW, and the level of 1600 cps at the ORDER WIRE TERMINALS shall be within -15.5 dbm to -25 dbm (0.043 to 0.13 volts).

3.10.2.10 Receiver Local Oscillator Radiation (see 4.16.4.9).- The available power at the receiver input connector from the local oscillator shall be less than 0.125 microwatts (2500 microvolts into 50 ohms) without duplexer filter.

3.10.2.11 Receiver frequency settability. -When tested as specified in 4.16.4.10 with Receiver, Radio R-1148(P)( )/GRC and Amplifier Converter AM-1955( )/GRC or AM-1956( )/GRC, the receiver local oscillator shall be within ± 0.005% ± 90 KC of the desired channel frequency.

3.10.2.11.1 Receiver frequency settability.- When tested as specified in 4.16.4.10.1 and operating with Amplifier Converter AM-1955A( )/GRC or AM-1956A()/GRC, the receiver local oscillator shall be within ± 400 KC of the desired frequency.

3.10.2.12 Receiver Signal Capture Capability.- When testing as specified in 4.16.4.11 with a receiver and amplifier Converter AM-1955A( )/GRC or AM-1956A( )/GRC, the receiver shall be capable of capturing and locking to an input signal of -87 dbm.

3.10.2.13 AC Power Interruption (Receiver) (4.16.4.12).

3.10.2.13.1 AC Power Interruption for Receiver R-1148P( )/GRC with Amplifier Converter AM-1955( )/GRC or AM-1956( )/GRC. - When tested in accordance with 4.16.4.12.1 the receiver shall be capable of restoration to within ± 0.005 percent ± 90 KC of the assigned channel frequency in the event of an AC power interruption of duration from two seconds to ninety seconds.

3.10.2.13.2 AC Power Interruption for receivers with Amplifier Converter AM-1955A( )/GRC or AM-1956A( )/GRC.- When tested in accordance with 4.16.4.12.2, the receiver shall be capable of restoration to the assigned frequency channel such that the second IF is within 60 kc of the zero crossover frequency in the event of an AC power interruption of durations from two seconds to ninety seconds.

3.10.2.14 Receiver PCM Deviation Sensitivity (see 4.16.4.13).- The output signal shall be at a level of 0.265V ± 2db at the PCM output for an RF input signal with a peak deviation of 375 KC at a 10 KC modulating frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.2.15 Receiver FDM S+N/N Ratio (see 4.16.4.14).- With an r. f. signal input of -80 dbm the signal-plus-noise to noise ratio (S+N/N) shall be at least 45 db for 12 channels and 42 db for 24 channels at the 135 ohm FDM output terminals when measured with a flat weighted meter. 12 and 24 channels shall be defined as a noise bandwidth from 4 to 68 KC and 4 to 108 KC respectively. With an r. f. input of -60 dbm, the ratio shall be at least 61 db and 58 db for 12- and 24- channel, respectively.

3.10.3 Broadband modulation (see 4.16.5).-

3.10.3.1 Input impedance.- The impedance of the broadband input terminals of the transmitter and the broadband output terminals of the receiver shall be as follows:

(a) PCM: When tested as specified in 4.16.5.1, the nominal impedance at the PCM broadband terminals shall be 50 ohms resistive ± 10% over the frequency range 0.300 KC to 680 KC.

(b) FDM and balance: When measured as specified in 4.16.5.2, the impedance at the FDM broadband terminals shall be 135 ohms ± 15%, resistive, over the band of 300 to 108,000 cps, and 600 ohms ± 15%, resistive, over the band of 300 to 20,000 cps. The degree of balance to ground at any modulation frequency between 300 and 108,000 cps for the 135-ohm circuit and between 300 and 20,000 cps for the 600 ohm circuit shall be at least 27 db. These requirements shall be met for all settings of the input attenuator of the transmitter or the output attenuator of the receiver.

3.10.3.2 Input and Output Attenuators (see 4.16.5.3)

(a) FDM input attenuator: When tested as specified in 4.16.5.3(a), attenuation shall decrease in 1 db steps with clockwise rotation of the control. The accuracy of attenuation shall be ± 0.5 db for all settings of the control at all frequencies in the band 300 to 108,000 cps. The difference in attenuation between two successive steps shall be 1.0 db plus or minus 0.2 db.

(b) FDM output attenuator:- When tested as specified in 4.16.5.3(b), the attenuation shall decrease in 1 db steps with clockwise rotation of the control. The accuracy of attenuation shall be ± 0.5 db for all settings of the control at all frequencies in the band 0.3 to 108 KC. The difference of attenuation between two successive steps shall be 1.0 db ± 0.2 db.

(c) PCM input attenuator: - When tested as specified in 4.16.5.3(c), the nominal zero-position of the control will be such that the PCM modulation requirements (see 3.10.3.4) shall be met with a reserve gain of at least 6 db. Each successive detented position of the network shall present a gain change of 1.0 ± 0.2 db. The accuracy of attenuation shall be ± .5 db for all settings.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.3.3 FDM signal-plus-noise to noise.- When tested as specified in 4.16.5.4:

(a) The average noise power at the receiver 135-ohm broadband output terminals, measured in any 4 kc channel from 4 kc to 108 kc shall not exceed the values specified in the following table, using a flat-weighted meter:

4-kc channel between	Noise power (-dbm)
4 and 8 kc	56
8 and 12 kc	48
12 and 20 kc	56
20 and 38 kc	52
38 and 80 kc	50
80 and 108 kc	48

(b) The average noise power of the receiver 135-Ohm broadband terminals shall not exceed -45 dbm, using a flat-weighted meter over a detected range of at least 4.0 kc to 108 kc.

3.10.3.4 PCM Signal - plus - Noise to Noise.- When tested as specified in 4.16.5.5:

(a) The ratio of average detected sinusoidal power to the average noise power at the 50-ohm PCM output terminals of the receiver when measured with a flat weighted meter over a detected noise bandwidth of .250 KC to 1 mc shall be at least 22 db with a PCM input of 0.53 volts @ 10 Kc for an RF level of -80 dbm.

(b) The ratio of average detected sinusoidal power to the average noise power at the 50-ohm PCM receiver output terminals, measured with a flat weighted meter over a detected noise bandwidth of 0.25 Kc to 1 megacycle, shall be at least 29 db with a PCM input of 0.53 volts @ 10 Kc for rf level of -73 dbm.

3.10.4 Frequency response (see 4.16.6).-

3.10.4.1 FDM frequency response.- When tested as specified in 4.16.6.1:

(a) The following overall frequency response characteristics shall be met with any combination of transmitter and receiver connected in a full duplex system and with the remote orderwire terminals, (1) open-circuited and (2) terminated with a 600-ohm resistive load:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

3.10.4.1 FDM freq response (contd)

Frequency (kc)	Response relative to 1 kc 600-ohm terminal	135-ohm terminal
0.30 to 3.999	± 1 db	± 1 db
4.0 to 19.999	± 0.5 db	± 0.5 db
20 to 67.999		± 0.8 db
68 to 108		± 1 db

(b) The response shall not vary more than 0.25 db between adjacent 4 kc channels.

3.10.4.2 FDM transmitter frequency response.-When tested as specified in 4.16.6.2, the transmitter fdm frequency response characteristics shall be as follows:

Frequency (kc)	Response relative to 1 kc (in db) 600-ohm terminal	135-ohm terminal
0.30 to 10	0.0 ± .5	0 ± .5
15	0.15 ± .5	0.15 ± .5
20	0.3 ± .5	0.3 ± .5
40		1.2 ± .5
70		3.0 ± .5
100		4.8 ± .5
108		5.1 ± .5

3.10.4.3 PCM Frequency response. - When tested as specified in 4.16.6.3, the overall frequency response characteristics below shall be met with any combination of transmitter and receiver:

Frequency Kc	Response relative to 10 KC (50-ohm terminals) in db Channels 400-899	Channels 1-100	Channels 100-399
0.18	3.0	3.0	3.0
0.5 to 50	0 ± .06	0± .06	0± .06
70	0.5 ± 0.6	0.5 ± 0.6	0.5 ± .06
138	1.0 ± 0.7	1.0 ± 0.7	1.0 ± 0.7

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.4.3 PCM Freq response (Contd)

Frequency (kc)	Channels 400 to 899	Response relative to 10 KC (50 ohm terminals) in db Channels 1-100	Channels 100-399
196	2.0 ± 0.9	2.2± 1.0	2 ± 0.9
240	3.0 ± 1.0	3.3 ± 1.3	3.0 ± 1.0
340	6.0 ± 1.5	6.6 ± 2.0	6.0 ± 1.5
480	12.0 ± 3.0	13.0 ± 4.0	12 ± 3.0
680	24. ± 4.0	27.5 ± 7.0	26.0 ± 6.0

3.10.4.4 PCM Harmonic distortion.- When measured as specified in 4.16.6.4, the total harmonic distortion shall not exceed 2 percent.

3.10.4.5 PCM Phase response. -When tested as specified in 4.16.6.5, the phase angle measured at receiver output terminals shall be 40 degrees ± 10 degrees at 300 cps, referred to the phase angle obtained at 1 kc.

3.10.5 Orderwire (see 4.16.7 and 6.11).-

3.10.5.1 Remote alarm.- When tested as specified in 4.16.7.1, the buzzer and lamp shall be energized by operation of the squelch control or when a 1,600 cps signal is received in the orderwire channel.

3.10.5.2 Orderwire signal-plus-noise to noise.- When measured as specified in 4.16.7.2, signal-plus-noise to noise ratio measured at the remote orderwire terminals, the local orderwire terminals (receive phone terminals), and the receive broadband terminals shall be at least 33 db at any operating frequency, with FIA weighting.

3.10.5.3 Remote Orderwire Impedance . - When measured as specified in 4.16.7.3, the terminal impedance of the remote orderwire 2-wire line shall be 600 ohms + 5% -25% over the band 300 to 1800 cps, (press-to-talk) switch not depressed), with a reflection coefficient not exceeding 20 percent.

3.10.5.4 Peak limiting.- When measured as specified in 4.16.7.4, the limiting action to be provided in the transmitter for the orderwire shall be such as to compress the 1 kc test tone power inserted at the zero transmission level point as follows:

Input power	Compression
-10 dbm	None
0 dbm	0.4 db or less
+15 dbm	10 db ± 3 db

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.5.5 Orderwire frequency response.-

(a) Remote: When tested as specified in 4.16.7.5, the overall frequency attenuation characteristic between the transmitter remote orderwire input and receiver remote orderwire output terminals, shall be within +1 and -3db of the 1 KC ref (0 dbm) from 300 to 1800 cps, when the equipment is employed in a full duplex fdm or pcm system. These requirements shall apply with the remote OW terminals terminated with Telephone Set TA-312( )/PT or equal and with both on-hook and off-hook conditions.

(b) Local:- The characteristics of the local orderwire channel shall be such that it meets the requirements for remote orderwire operation as set forth in (a) above, with requirements referred to the microphone input terminals and phone output terminals instead of the remote orderwire terminals.

3.10.5.6 Orderwire Distortion.- When tested as specified in 4.16.7.6, the distortion shall not exceed 5 percent at any of the three frequencies.

3.10.5.7 PCM Combiner Input Impedance:- When measured as specified in 4.16.7.7, the PCM combiner input impedance shall be 600 ohms, resistive, ± 10% over the frequency range of 300 to 1,800 cps.

3.10.6 Orderwire System Levels (see 4.16.8).

3.10.6.1 Sidetone and Net Loss Tests.- When tested as specified in 4.16.8.1. the system net losses shall not exceed the following:

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.6.1 Orderwire System Levels

When tested at 1 KC as specified in 4.16.8.1, the system levels shall meet the following:

Input Terminals	Output Terminals Rx#1 FDM (135 ohms)	Rx#1 ROW (600 ohms)	Rx#1 PHONE (150 ohms)	Rx#2 FDM 135 ohms	Rx#2 ROW 600 ohms	Rx#2 PHONE 150 ohms
FDM - Step 8 -37v 0 dbm 135 ohms		-20dbm ± 3db 0.055 to 0.111v	-18dbm ± 3db 0.034 to 0.068v	0dbm ± 1db 0.33 to 0.42v	-20dbm ± 3db 0.055 to 0.111v	-18dbm ± 3dbm 0.034 to 0.068v
ROW - Step 11 .775v 0dbm 600 ohms	-2dbm ± 1db 0.26 to 0.33v		-20dbm ± 4.5db 0.023 to 0.065v	-2dbm ± 1db 0.26 to 0.33v	-22dbm ± 4.5db 0.037 to 0.102v	-20dbm ± 4.5db 0.023 to 0.065v
MIKE - Step 14 .245v 0dbm 56 ohms	0dbm ± 1db 0.33 -0.42v	-15.5 to -25dbm 0.044 to 0.13v	-24dbm ± 4.5 db 0.014 to 0.041v	0dbm ± 1db 0.33-0.42v	-15.5 to -25 dbm 0.044 to 0.13v	-18dbm ± 4.5db 0.029 to 0.081v
PCM Combiner .49v -600 ohms - 4dbm Step 17				0 dbm ± 1db 0.33 to 0.42v	-20dbm ± 3db 0.055 to 0.111v	-18dbm ± 3db 0.034 to 0.068v

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.6.2 Level Adjustment and Metering: When tested as specified in 4.16.8.2:

(a) The deviation shall be 59 kc @ 1 kc and 84 kc @ 68 kc.

(b) The system level adjustments shall be accurate within ± 2.0 db.

(c) The color-coded region about the center of the afc meter shall correctly indicate the permissible drift in both transmitter and receiver.

(d) The transmitter general purpose meter shall meet the following:

Measurement	Indication Requirement
(1) Oscillator amplitude	At least 10% of full scale
(2) Main Tune	At least 10% of full scale
(3) Mixer amplitude	At least 10% of full scale
(4) Amplifier Amplitude	At least 10% of full scale
(5) Output frequency	At least 10% of full scale
(6) Power output	At least 40% of full scale
(7) Reflected power	At least 5% full scale for VSWR of 2:1
(8) AFC level	At least 20% of full scale
(9) 1kc input level	± 2 db
(10) 68 kc input level	± 2 db
(11) 1 kc modulation level	± 2 db
(12) 68 kc modulation level	± 2 db
(13) PCM input level	± 2 db
(14) Test	Within ± 10% and ± 30% of midscale for the corresponding maintenance test points

(e) The receiver general purpose meter shall meet the following:

Measurement	Indication Requirement
(1) Oscillator frequency	At least 10% of full scale
* (2) AFC level	At least 20% of full scale
(3) Receiver Signal	At least 10% of full scale for -80 dbm input with increasing indications for levels to -40dbm.
(4) Test Tone Calibrate	±2db
(5) 1 kc output adjust	±2db
(6) 68 kc output adjust	±2db
(7) PCM output	±2db
(8) Test	Within ± 10% and ± 30% of midscale for the corresponding maintenance test points.

* Not applicable for operation with Amplifier Converter AM-1955A ( )/GRC or AM-1956A( )/GRC.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.6.3 Duplexer operation.- When tested as specified in 4.16.8.3, the duplexer shall be such that for any transmitter operating frequency in either the low or high band (see 3.1.2), it shall be possible to normally operate the receiver at any other frequency in the same band, except for 15-mc or lower separation between the transmitter and receiver.

3.10.6.4 Level stability.- When tested as specified in 4.16.8.4, the system gain shall not vary more than ± 1 db in any eight hour period.

3.10.6.5 Crosstalk:- When tested as specified in 4.16.8.5

(a) 12 Channel Ratio. - The ratio of the noise in the 34 KC slot to the 14 KC and 60 KC slots shall be at least 45 db.

(b) Orderwire signal to noise with noise loading.- Crosstalk plus noise shall not exceed -44 dbm with F1A weighting at the order wire output terminals.

(c) 24 Channel Ratio.- The ratio of the noise in the 34 KC slot to the 6 KC and 106 KC slots shall be at least 43 db for RF channels 21 through 899 and at least 40 db for RF channels 1 through 20.

3.10.6.6 Tone generator.- When tested as specified in 4.16.8.6, the buzzer and call lights shall be activated.

3.10.6.7 Dial accuracy.- When tested as specified in 4.16.8.8, the desired signal of -80 dbm shall be detectable (2 ua change) in the RECEIVE SIGNAL position without retuning.

3.10.7 Dummy Load Electrical DA-189( )/GRC. - (see 4.16.9)

3.10.7.1 VSWR test. - The VSWR of the dummy antenna shall be 1.5 or less, over the radio frequency band of 600 to 1,000 mc and from 1350 mc to 1850 mc. (see 4.16.9.1).

3.10.7.2 Power Calibration Accuracy Test.- The power calibration accuracy shall be within ± 1 db over the frequency band specified in 3.10.7.1 above, for power indications from 10 to 40 watts. (see 4.16.9.2)

3.10.8 Power Supply PP-2054( )/GRC (see 4.16.10)

3.10.8.1 Relay Operation. - When tested as specified in paragraph 4.16.10.1, the filament pilot lamps shall light and the blower shall operate when the Standby-Operate switch is in the "operate" position and the circuit breaker switch (CB-1) is in the "on" position. Within 50-80 seconds after turning the breaker switch (CB-1) to the "on" position, the time delay relay (K2) shall operate lighting the HV and LV pilot lamps.

3.10.8.2 H.V. Supply & Ripple test.- When tested as specified in paragraph 4.16.10.2, the HV output at terminals R,W & M of J2 connector shall measure 800 ±40 VDC, 650 ± 33 VDC & 645 ±33 VDC, respectively. In addition, the ripple voltage measured at terminals R, W & M shall not be greater than 21 VAC, 1.6 VAC & 0.21VAC respectively.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

3.10.8.3 LV Supply & Ripple Test. - When tested as specified in paragraph 4.16.10.3, the LV output and ripple voltage at terminals K, V, T & A of J2 connector shall measure:

	INPUT VOLTAGE (AC)	LOAD SETTING	VDC	OUTPUTS VAC
(a)	115	K(360V)	360 ± 15	20 MAX
(b)	115	V(250V)	250 ± 0.5	0.02 MAX
(c)	109	V(250V)	(± 1V of Step b)	0.02 MAX
(d)	121	V(250V)	(±1V of Step b)	0.02 MAX
(e)	115	T(150V)	$150 \pm \frac{5}{10}$	0.02 MAX
(f)	109	T(150V)	(± 2 V of Step f)	0.02 MAX
(g)	121	T(150V)	(±2V of Step g)	0.02 MAX
(h)	115	A(108V)	(108 ±3	0.005 MAX.
(i)	109	A(108V)	(±1V of Step e)	0.005 MAX
(j)	121	A(108V)	(±1V of Step e	0.005 MAX

3.10.8.4 Current.- When tested as specified in test paragraph 4.16.10.4, the current measured at terminals J3, J4, J5, J6,J7,J8 and J9 shall be 25.0 ± 2.5 microamperes (ua).

3.10.8.5 Interlock - When tested as specified in test paragraph 4.16.10.5:

(a) With the condition of Switch (S2) being opened for 10 seconds and then closed, the HV & LV pilot lamps should go out as soon as switch (S2) of dummy load is opened and should go on again immediately as soon as S2 is closed.

(b) With the condition of Switch (S1) being opened for 5 seconds and then closed, the HV & LV pilot lamps should go out as soon as switch (S1) of dummy load is opened and should go on again immediately as soon as (S1) is closed.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

3.10.8.5 (Contd)

(c) With the condition of Switch (S1) being opened for 10 seconds and then closed the HV & LV pilot lamps should go out as soon as switch (S1) is opened and should not go on again until 45 to 90 seconds after S1 is closed.

3.11 Operational check.- When tested in accordance with 4.16.8.7, normal two way communication shall be possible.

3.12 Technical literature and running spare parts.- Technical literature and running spare parts shall be furnished if specified in the contract. Running spare parts shall be furnished if specified in the contract. Running spare parts shall be identical to corresponding parts in the equipment furnished on the order. (See 6.2 (f) and (g).)

3.13 Workmanship.- The equipment shall be manufactured and assembled in accordance with the applicable portions of the following paragraphs.

3.4	Construction
3.5.3	Plastic materials and parts
3.5.4	Cleaning
3.5.5	Marking
3.5.6	Riveting
3.5.7	Securing of parts.
3.5.8	Brazing
3.5.9	Soldering
3.5.10	Welding
3.5.11	Screws, other threaded devices, and related parts.
3.5.12	Terminals, fixed.
3.5.13	Wiring and cabling
3.5.14	Controls and rotating components
3.5.15	Adjustment and repair
3.5.16	Tropicalization of material
3.5.17	Fabric and thread
3.5.19	Tubes, electron; and semi-conductors, transistor and diode

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for Inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. Inspection records of the examination and tests shall be kept complete and available to the Government as specified in the contract. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure that supplies and services conform to prescribed requirements.

4.2 Classification of inspection.- Inspection shall be classified as follows (see 6.4.2).

(a) Preproduction inspection (does not include preparation for delivery). (see 4.3).

(b) Inspection covered by subsidiary documents. (see 4.4).

(c) Quality conformance inspection.

(1) Quality conformance inspection of equipment before preparation for delivery. (see 4.5).

(2) Quality conformance inspection of preparation for delivery. (see 4.19).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.3 Preproduction inspection.-This inspection will be performed by the Government unless otherwise specified in the contract. It shall consist of the preproduction inspection specified in table III, the inspection specified in the subsidiary documents covering the items listed in 4.4, and the inspection specified for group A, group B and group C (see tables IV, V and VI respectively). The preproduction inspection will normally be performed in this order (1) vibration, (2) bounce, (3) shock, bendhandling, (4) shock, drop, and (5) rain; other preproduction inspection may precede, follow or be interspersed between the foregoing.

Table III - Preproduction inspection

Inspection (For additional preproduction inspection see 4.3)	Req para	Insp para
Temperature	3.6.2	4.6
Relative humidity	3.6.3	4.7
Elevation	3.6.4	4.8
Orientation	3.6.5	4.9
Rain	3.6.6	4.14
Electromagnetic Compatibility	3.7	4.20
Vibration, internal	3.8.1	4.10
Bounce	3.8.2	4.11
Shock, bench-handling	3.8.2	4.12
Shock, drop	3.8.2	4.13

4.4 Quality Conformance inspection covered by subsidiary documents.- The following shall be inspected under the applicable subsidiary documents as part of the Quality Conformance before preparation for delivery:

ITEM	Where required
Amplifier-Converter AM-1955()/GRC	3.4
Amplifier-Converter AM-1955A()/GRC	3.4
Amplifier-Converter AM-1956( )/GRC	3.4
Amplifier-Converter AM-1956A ( )/GRC	3.4
Amplifier-Oscillator AM-1957 ( )/GRC	3.4
Amplifier-Oscillator AM-1958( )/GRC	3.4
Amplifier-Oscillator AM-1958A( )/GRC	3.4
Power Supply PP-2054( )/GRC	3.4

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.4 (Contd)

ITEM	Where required
Cable Assemblies	3.5.1
Finish	3.5.2
Marking	3.5.5
Brazing	3.5.8
Welding	3.5.10
Crimping of solderless terminal lugs	3.5.13.6

4.5 Quality conformance inspection of equipment before preparation for delivery. -The contractor shall perform the inspection specified in 4.4 and 4.5.1 through 4.5.4. This does not relieve the contractor of his responsibility for performing any additional inspection which is necessary to control the quality of the product and to assure compliance with all specification requirements. The Government will review and evaluate the contractor's inspection procedures and examine the contractor's inspection records. In addition, the Government may perform all or any part of the specified inspection, to verify the contractor's compliance with specified requirements. (see 6.8) Test equipment for Government verification inspection shall be made available by the contractor. Each unit which will be subjected to group A, group B, or group C inspection shall be bounce preconditioned after final assembly. (see 3.9).

4.5.1 Group A inspection.- This inspection, including sampling, shall conform to table IV and the ordinary inspection procedures of Standard MIL-STD-105. Group A inspection shall be performed in any order which is satisfactory to the Government.

TABLE IV - Group A inspection

Inspection	Req para	Insp para	AQL * Major	AQL * Minor
Transmitter, T-893(P) ( )/GRC				
Visual-Mechanical:	3.13	4.18		
Transmitter main frame			2.5	10.0
Baseband unit			2.5	10.0
Auto freq control unit			2.5	10.0
Modulator unit			2.5	10.0

Source: https://asstst.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.5.1 TABLE IV (Contd)

Inspection	Req para	Insp para	AQL* Major	AQL* Minor
Current regulator unit			1.5	6.5
Case, CY-2429( )/GRC			1.0	4.0
Amp-Osc, AM-1957( )/GRC			1.5	6.5
Amp-Osc, AM-1958( )/GRC			1.5	6.5
Amp-Osc, AM-1958A( )/GRC			1.5	6.5
RECEIVER R-1148(P)( )/GRC				
Visual-Mechanical	3.13	4.18		
Receiver main frame			2.5	10.0
Baseband unit			2.5	10.0
Auto freq control unit			2.5	10.0
Signalling Unit			2.5	10.0
Second IF Unit			2.5	10.0
Case, CY-2429( )/GRC			1.0%	4.0%
Amp-Conv, AM-1955( )/GRC			1.5	6.5
Amp-Conv, AM-1955A( )/GRC			1.5	6.5
Amp-Conv, AM-1956( )/GRC			1.5	6.5
Amp-Conv, AM-1956A( )/GRC			1.5	6.5
RECEIVER R-1331(P)( )/GRC				
Visual-Mechanical	3.13	4.18		
Receiver main frame			2.5	10.0
Baseband Unit			2.5	10.0
Secondary Auto freq control unit			2.5	10.0
Signalling Unit			2.5	10.0
Second IF Unit			2.5	10.0
Case, CY-2429 ( )/GRC			1.0%	4.0%
Amp-Conv, AM-1955A( )/GRC			1.5	6.5
Amp-Conv, AM-1956A( )/GRC			1.5	6.5

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the current version before use.

4.5.1 TABLE IV (Contd)

Inspection	Req para	Insp para	AQL * Major	Minor
Power Supply PP-2054( )/GRC			1.0%	4.0%
Dummy Load, DA-189( )/GRC			1.5%	6.5%
Wavemeter unit			1.5%	6.5%
Case, CY-2428( )/GRC			1.0%	4.0%
Case, CY-2582( )/GRC			1.0%	4.0%
Case, CY-2583( )/GRC			1.0%	4.0%
ELECTRICAL				
Power Supply PP-2054( )/GRC	3.10.8	4.16.10	1.5	**
Amp-Conv, AM-1955( )/GRC	3.4	4.4	1.5	
Amp-Conv, AM-1955A( )/GRC	3.4	4.4	1.5	
Amp-Conv, AM-1956( )/GRC	3.4	4.4	1.5	
Amp-Conv, AM-1956A( )/GRC	3.4	4.4	1.5	
Amp-Osc, AM-1957 ( )/GRC	3.4	4.4	1.5	
Amp-Osc, AM-1958 ( )/GRC	3.4	4.4	1.5	
Amp-Osc. AM-1958A( )/GRC	3.4	4.4	1.5	
Dummy Load, DA-189 ( )/GRC	3.10.7	4.16.9	1.5%	
TRANSMITTER, T-893(P) ( )/GRC Electrical:				
Freq. Range and power output	3.10.1.1 (a)	4.16.3.1 (a)	2.5 for group combined	**
Overload	3.10.1.2	4.16.3.2		
Short time afc	3.10.1.4	4.16.3.4		
Transmitter Freq settability	3.10.1.5	4.16.3.5		
Alarm	3.10.1.7	4.16.3.7		
Transmitter FDM Deviation Sensitivity	3.10.1.9	4.16.3.9		
Transmitter PCM Deviation Sensitivity	3.10.1.10	4.16.3.10		
AC Power Interruption (Trans)	3.10.1.11	4.16.3.11		

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.5.1 TABLE IV (Contd)

Inspection	Req para	Insp para	AQL * Major	DPHU
RECEIVER, R-1148(P) ( )/GRC Electrical:			2.5 for group combined	* *
Receiver PCM Deviation Sensitivity	3.10.2.14	4.16.4.13		
Freq range and noise figure	3.10.2.1	4.16.4.1		
Receiver FDM Deviation Sensitivity	3.10.2.3	4.16.4.2		
Squelch and alarm	3.10.2.9	4.16.4.8		
Receiver Freq. Settability	3.10.2.11	4.16.4.10 (with AM-1955/GRC or AM-1956/GRC)		
Receiver Freq, settability	3.10.2.11.1	4.16.4.10.1 (with AM-1955A/GRC or AM-1956A/GRC)		
Receiver Signal Capture capability test	3.10.2.12	4.16.4.11		
AC Power Interruption (Receiver)	3.10.2.13.1	4.16.4.12.1		
RECEIVER, R-1331 (P) ( )/GRC Electrical:			2.5 for group combined	* *
Freq. range and noise figure	3.10.2.1	4.16.4.1		
Receiver FDM Deviation Sensitivity	3.10.2.3	4.16.4.2		
Squelch and Alarm	3.10.2.9	4.16.4.8		
Receiver Freq. settability	3.10.2.11.1	4.16.4.10.1 (with AM-1955A/GRC or AM-1956A/GRC)		
Receiver Signal Capture Capability Test	3.10.2.12	4.16.4.11		
AC Power Interruption(Receiver)	3.10.2.13.2	4.16.4.12.22		
Receiver PCM Deviation Sensitivity	3.10.2.4	4.16.4.13		

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

Group A Inspection

Inspection	Req par	Insp para	AQL * Major	DPHU
FDM (S + N)/N ratio	3.10.3.3	4.16.5.4		
PCM (S + N)/N ratio	3.10.3.4	4.16.5.5		
FDM freq response	3.10.4.1	4.16.6.1		
PCM freq response	3.10.4.3	4.16.6.3		
PCM Harmonic distortion	3.10.4.4	4.16.6.4	4.0 for the group combined	
Remote alarm	3.10.5.1	4.16.7.1		
Orderwire (S + N)/N	3.10.5.2	4.16.7.2		
Orderwire freq response	3.10.5.5	4.16.7.5		
Orderwire Distortion	3.10.5.6	4.16.7.6		
Sidetone & Net loss	3.10.6.1	4.16.8.1		
Level adj and metering	3.10.6.2	4.16.8.2		
Crosstalk	3.10.6.5	4.16.8.5		
Tone generator	3.10.6.6	4.16.8.6		
Operational check	3.11	4.16.8.7	1.0	

* DPHU unless otherwise indicated.

** No minor classification on electrical inspection.

4.5.2 Group B inspection and sampling.- This inspection, including sampling, shall conform to table V and to the special procedures for small-sample inspection of Standard MIL-STD-105. The reduced inspection procedures shall be R-1. Group B inspection shall normally be performed on inspection lots that have passed group A inspection and on samples selected from units that have been subjected to and met the group A inspection. The acceptable quality level (AQL) shall be 6.5% defective, as specified in table V, and the inspection level shall be L-7 for normal and L-5 for reduced inspection.

TABLE V - Group B Inspection

Inspection	Req para	Insp para	AQL (see 4.5.2)
TRANSMITTER, T-893(P) ( )/GRC			
ELECTRICAL:			
Insertion loss	3.10.1.8	4.16.3.8	6.5% for group combined
FDM xmtr freq response	3.10.4.2	4.16.6.2	

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

TABLE V (Group B inspection) Contd

Inspection	Req para	Insp para	AQL (see 4.5.2)	
RECEIVER, R-1148(P)( )/GRC				
ELECTRICAL:				
IF bandwidth	3.10.2.8	4.16.4.7	6.5%	
RECEIVER, R-1331(P)()/GRC				
ELECTRICAL:				
IF bandwidth	3.10.2.8	4.16.4.7	6.5%	for group combined
Receiver FDM S+N/N Test	3.10.2.15	4.16.4.14		
Systems, trans rcvr				
ELECTRICAL:				
PCM input impedance	3.10.3.1(a)	4.16.5.1		
FDM input and balance	3.10.3.1(b)	4.16.5.2	6.5%	for group combined
Remote orderwire impedance	3.10.5.3	4.16.7.3		
PCM combiner input impedance	3.10.5.7	4.16.7.7		
Interchangeability	3.5.18	4.17	6.5%	for group combined

4.5.3 Group C inspection.- This inspection shall be as listed in Table VI, and shall normally be performed on sample units that have been subjected to and met group A and group B inspection.

TABLE VI. - GROUP C inspection

Inspection	Req para	Insp para
Transmitter, T-893(P)()/GRC		
ELECTRICAL:		
Freq stability	3.10.1.3	4.16.3.3
Spurious outputs	3.10.1.6	4.16.3.6
Frequency Range and Power Output	3.10.1.1(b)	4.16.3.1(b)
Receiver, R-1148(P)()/GRC		
ELECTRICAL:		

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.5.3 TABLE VI (Contd)

Inspection	Req para	Insp para
Spurious response	3.10.2.4	4.16.4.3
Auto freq control	3.10.2.5	4.16.4.4 (For AM-1955( )/GRC & AM-1956( )/GRC)
Secondary Auto freq control	3.10.2.7	4.16.4.6 (For AM-1955A( )/GRC or AM-1956A()/GRC)
Receiver local osc radiation	3.10.2.10	4.16.4.9
Receiver, R-1331(P)()/GRC ELECTRICAL:		
Spurious response - secondary	3.10.2.4	4.16.4.3
Auto freq control	3.10.2.7	4.16.4.6
Receiver, local osc radiation	3.10.2.10	4.16.4.9
Receiver freq settability	3.10.2.11.1	4.16.4.10.1
Receiver freq stability	3.10.2.6	4.16.4.5
System, xmtr & rcvr ELECTRICAL:		
Attenuators	3.10.3.2	4.16.5.3
Peaking limiting	3.10.5.4	4.16.7.4
Duplexer operation	3.10.6.3	4.16.8.3
Level stability	3.10.6.4	4.16.8.4
Dial accuracy	3.10.6.7	4.16.8.8
Lev Adj & Metering	3.10.6.2(d)(6)	4.16.8.2
PCM Phase Response	3.10.4.5	4.16.6.5

4.5.3.1 Sampling for inspection of equipment.- Two of each 40 units or two per month, whichever occurs first, shall be selected for group C inspection without regard to their quality, except that the units inspected at the start of the contract shall be selected from the first units produced.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.5.3.2 Noncompliance.- If a sample unit fails group C inspection, the contractor shall immediately investigate the cause of failure and shall report to the Government Inspector the results thereof and details of the corrective action taken on the process and all units of product which were manufactured with the same conditions, materials, processes, etc. If the Government Inspector does not consider that the corrective action will enable the product to meet specified requirements, or if the contractor cannot determine the cause of failure, the matter shall be referred to the contracting officer. (See 6.5).

4.5.4 Reinspection of conforming group B and group C sample units.- Unless otherwise specified, sample units which have been subjected to and passed group B or group C inspection, or both, may be accepted on contract, provided that they are resubjected to and pass group A reinspection after repair of all known and visible damage.

4.6 Temperature test (see 3.6.2).-The equipment shall be subjected to the temperature cycle shown on Standard MIL-STD-169. Performance shall be as follows:

STEP	TEST NR.		Test	Req para	COMMENTS
1	a		PCM Signal to Noise	3.10.3.4	Initial measurements.
	b		FDM Sig to Noise(Wideband)	3.10.3.3(b)	Complete as rapidly
	c		Orderwire Sig to Noise	3.10.5.2	as possible. Do not
	d		Crosstalk(24 channel)	3.10.6.5(c)	leave power on after
	e		Level Adj & Metering	3.10.6.2	measurements have
	f		Transmitter Freq Range and Power output	3.10.1.1	been completed.
	g		Freq stability	3.10.1.3	
	h		Trans freq settability	3.10.1.5	
	i		Rec freq settability	3.10.2.11	
	j	*	Receiver automatic frequency control	3.10.2.5	
	k	*	Receiver Frequency Stability	3.10.2.6	
	l		Receiver Secondary Automatic Freq Control	3.10.2.7	
	m		Tone generator	3.10.6.6	
	n		Remote alarm	3.10.5.1	
	o		Dummy Load VSWR	3.10.7.1	
	p		Dummy Load Power Calibration accuracy	3.10.7.2	
2A					No measurements

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.6 Temperature Test (Contd)

STEP	TEST NR.	Test	Req para	COMMENTS
3				Omit this step
4		Repeat all tests of step 1		Change to + 132°F at this step. Start measurements 1/2 hour after power is applied to the equipment. Leave power on after measurements are completed.
5		Repeat all tests of Step 1		Do not leave power on after measurements have been completed.
6A				No measurements.
7				Omit this step
8		Repeat all tests of step 1		Change to -20°F at this step. Start measurements 1/2 hour after power is applied to equipment. Leave power on after measurements are completed.
9		Transmitter Freq.Stability	3.10.1.3	Complete as
	*	Receiver Automatic Frequency Control	3.10.2.5	rapidly as possibl
		(Receiver Frequency Stability)	3.10.2.6	
	*	(Receiver Secondary)	3.10.2.7	
		(Automatic Freq Control)		
10		Repeat all tests of step 1		Final measurements.

* As applicable.

4.7 Moisture-resistance test (see 3.6.3).- The units shall be tested as follows:

4.7.1 Test conditions.-

(a) Do not move unit from the humidity chamber for measurements.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.7.1 Contd

(b) Start measurements not more than 5 minutes after power is applied to the unit. Complete measurements as rapidly as possible. Do not leave power on after measurements have been completed.

(c) Test sealed equipment with the seal broken or with the chassis removed from its enclosure.

4.7.2 Test method.-

(a) Dry at 130° ±5°F for 24 hours.

(b) Condition at 77° ±5°F and 40 to 50 percent relative humidity for 24 hours.

(c) Take measurements as specified in 4.7.3, and readjust or realine as necessary to meet full specification requirements.

(d) Subject to continuous cycling for five 48-hour cycles. Temperature, relative humidity, and period of time for each portion of the cycle shall conform to Standard MIL-STD-170. The measurements specified in 4.7.3 shall be made during the periods shown on the standard and shall comply with 4.7.3.

(e) After cycling has been completed, condition the unit for 24 hours at 77° ±5°F and 40 to 60 percent relative humidity. Then adjust for optimum performance, using only those means provided ty the unit. No repair or replacement of parts shall be made. After adjustment, the unit shall meet full specified performance for those measurements specified in 4.7.3.

4.7.3 Performance.- Measurements made during test and specified performance furing cycling shall be as follows:

TEST NR.	TEST	Req para
1	PCM Signal to Noise	3.10.3.4
2	FDM Sig to Noise(Wideband)	3.10.3.3(b)
3	Orderwire Sig to Noise	3.10.5.2
4	Crosstalk(24 channel)	3.10.6.5(c)
5	Level Adj & Metering	3.10.6.2
6	Transmitter Freq Range and Power Output	3.10.1.1
7	Transmitter Freq. Settability	3.10.1.5
8	Receiver Frequency Settability	3.10.2.11
9 *	Receiver Automatic Frequency Control	3.10.2.5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.7.2 Test method (Contd)

TEST NR.	TEST	Req para
10 *	Receiver Secondary Automatic Freq Control	3.10.2.7
11	Remote Alarm	3.10.5.1
12	Tone Generator	3.10.6.6
13	Dummy Load VSWR	3.10.7.1
14	Calibration Accuracy	3.10.7.2

* As applicable.

4.7.4 Failure.- If any equipment fails to meet the performance specified in 4.7.3 during cycling, it shall be realined or readjusted once. If the equipment then fails to meet 4.7.3 or fails subsequently during cycling, it does not pass the test. In addition, if the equipment fails to meet full specification requirements after conditioning and adjustment, as specified in 4.7.2(e), it does not pass the test.

4.8 Elevation test (see 3.6.4) . - Each unit placed in its normal operating position in an altitude chamber, shall be tested as follows: (Note: To check temperature stabilization, place a thermocouple on the largest internal mass centrally located within the equipment.)

(a) With the equipment placed in the chamber, operating and stabilized at room temperature, take the following measurements:

TEST NR.	TEST	Req Para
1	PCM Signal to Noise	3.10.3.4
2	FDM Signal to Noise(Wideband)	3.10.3.3(b)
3	Orderwire Signal to Noise	3.10.5.2
4	Crosstalk(24 channel)	3.10.6.5(c)
5	Level Adj & Metering	3.10.6.2
6	Transmitter Freq Range & Power Output	3.10.1.1
7	Transmitter Freq. Settability	3.10.1.5
8	Receiver Frequency Settability	3.10.2.11

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

4.8 Elevation test (Contd)

TEST NR.	TEST	req para
9 *	Receiver Automatic Freq Control	3.10.2.5
10 *	Receiver Secondary Automatic Freq Control	3.10.2.7
11	Remote Alarm	3.10.5.1
12	Tone Generator	3.10.6.6
13	Dummy Load VSWR	3.10.7.1
14	Dummy Load Power Calibration accuracy	3.10.7.2

* As applicable

(b) Reduce the barometric pressure 19.0 ± 0.1 inches of mercury (simulated 12,000 feet above sea level) and, after temperature and pressure stabilization, repeat the measurements specified in 4.8 (a) 2, 6, 7 and 8. The equipment shall meet specified performance for those measurements.

(c) Remove the power from the equipment and further reduce the barometric pressure to 3.4 + 0.1 inches of mercury (simulated 50,000 feet above sea level). Maintain this condition for 1 hour.

(d) Return the chamber to ambient pressure. Apply power to the equipment and, after stabilization, repeat the measurements specified in 4.8(a). The equipment shall meet specified performance for these measurements.

4.9 Orientation test (except DA-189( )/GRC.- The equipment shall be inclined for a minimum of 5 minutes in each plane (upward, downward, left, and right) to an angle of 20 + 3 degrees . During inclination in each plane, the equipment shall meet specified performance (see 3.6.5) for the following measurements:

TEST NR.	TEST	req para
1	FDM Signal to Noise	3.10.10.3 (b)
2	Transmitter Frequency Range and Power Out.	3.10.10.1
3	Operational	3.11
4	Tone Generator	3.10.6.6
5	Remote Alarm	3.10.5.1
6	Dummy Load Power Calibration Accuracy	3.10.7.2
7	Level Adj. and Metering	3.10.6.2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.10 Test for internal vibration.- Internal vibration of the equipment shall be measured as follows, to determine conformance to 3.8.1:

a. Secure the equipment (shock mounts, if any, removed or blocked) directly to a vibration table that can be controlled within 10 percent of the specified amplitude. Mounting method shall be such that vibration within the equipment can be observed and measured. To facilitate this observation and measurement, sub-assemblies may be tested separately provided they are secured to the table in a manner similar to that used to mount them in the equipment.

b. Vibrate the equipment successively in three mutually perpendicular directions over a frequency range of 10 to 55 cycles per second. The total excursion of the applied vibration shall be not less than 0.016 inch.

c. In each of the three directions, change the frequency in steps of one cycle per second and maintain each frequency for at least 10 seconds.

d. Measure vibration amplitudes by optical means, or by other means provided that vibration of the part is not affected by the measurement.

4.11 Bounce test (see 3.8.3). - Each unit shall be tested on the package tester, type 1000-SC, as made by the L.A.B. Corporation, Skaneateles, New York, or equal. Accessories shall be selected from those listed on Drawing SC-A-46439. The test shall be as follows:

a. Cover the tested bed with a panel of 1/2-inch plywood, with the grain parallel to the drive chain. Space sixpenny nails, with the heads below the surface, at 6-inch intervals around all four edges and at 3-inch intervals in a 6-inch square in the center.

b. Place the equipment in its transit case, if any, prepared for field transport, on the bed of the package tested. Limit the lateral motion, by wooden fences, to not more than 3 inches and not less than 1 inch. Additional barriers may be used to prevent tumbling, provided that the fore-and-aft motion of the equipment against the backstop is not restrained.

c. Operate the package tester, shafts in phase, for a total of 3 hours at 284 ± 2 revolutions per minute. Turn the equipment at the end of each 30 minutes so it will rest on a new face.

4.12 Shock test; bench handling (see 3.8.2).- The chassis-and-front panel assembly shall be removed from its enclosure, as for servicing, and place in a suitable position for servicing on a solid, 2-inch fir bench top. The test shall be performed as follows in a manner simulating shocks liable to occur during servicing:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.12 Shock test; bench handling (see 3.8.2).-(Contd)

a. Tilt up the assembly through an angle of 30 degrees, using one edge of the assembly as an axis, and permit the assembly to drop back freely to the horizontal. Repeat, using other practicable edges of the same horizontal face as axes, for a total of four drops.

b. Repeat (a), with the assembly testing on other faces, until it has been dropped for a total of four times on each face on which the assembly might practically be placed during servicing.

4.13 Shock test, drop (3.8.2).-

4.13.1 Dummy Antenna DA-189( )/GRC.- Each unit, in its transit condition, shall be dropped a distance of 2 feet on each corner, edge, and face (total of 26 drops). Drops shall be made from a quick-release hook, on Acme Drop Tester, or equal; A Conbur Tester (10 degree inclined plane), or equal; or a basket-type pendulum tester. The floor or barrier receiving the impact shall be 2-inch fir backed by concrete or a rigid steel frame.

4.13.2 Other units.- Each unit, in its transit frame assembly shall be tested as in 4.13.1, except that 8 drops, one on each corner shall be used.

4.14 Rain test (see 3.6.6).- The radio equipments and spare parts case shall be tested, step by step, as follows:

4.14 Rain test (see 3.6.6). - Contd)

a. Dry at 150° F±5° for 48 hours.

b. Condition for 4 hours at 77° ± 5°F and 40 to 60 percent relative humidity.

c. Take at least the following measurements to insure that the component meets specified performance:

TEST NR.	TEST	Req Para
1	PCM Signal to Noise	3.10.3.4
2	FDM Signal to Noise(Wideband)	3.10.3.3(b)
3	Orderwire Signal to Noise	3.10.5.2
4	Level Adj & Metering	3.10.6.2
5	Transmitter Freq Range and Power Output	3.10.1.1

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.14 Rain test Contd)

TEST NR.	TEST	Req Para
6	Transmitter Freq Settability	3.10.1.5
7	Receiver Frequency Settability	3.10.2.11
8 *	Receiver Automatic Frequency Control	3.10.2.5
9 *	Receiver Secondary Automatic Freq Control	3.10.2.7
10	Rmote Alarm	3.10.5.1
11	Tone Generator	3.10.6.6
12	Dummy Load VSWR	3.10.7.1
13	Dummy Load Power Calibration accuracy	3.10.7.2

* As applicable.

d. Place the component, non-operating, but in its normal operating position, in a test chamber maintained at 77°F ± 5°F. Allow simulated rainfall to fall on exposed surfaces of the component. The water shall be at a temperature of 65° F ± 5°F and shall fall freely at a rate of 2 inches per hour.

e. With the rainfall continuing at the 2 inch rate, introduce a simulated wind blowing horizontally at 20 ± 5 miles per hour against one side of the component. Maintain this condition for 1 hour.

f. Rotate the component 90 degrees horizontally, to expose a second side of the component to the wind and rain. Maintain this condition for 1 hour.

g. Continue as in (f) above until all four sides of the component have been exposed to the wind and rain, each side for 1 hour.

h. Immediately after completion of the foregoing, repeat the measurement specified in (c) above without removal from the chamber but with the rain and wind discontinued. Complete the measurements within 1 hour. The measurement shall meet specified performance.

4.15 Bounce preconditioning (see 3.9.- Each unit, in its case, shall be placed in its normal operating position on the table of the Model 1000 SC Package Tester as made by the L.A.B. Corporation, Skaneateles, New York, or equal.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.15 Bounce preconditioning (Contd)

The package tester, shafts in phase shall have a speed such that it is just possible to insert a 1/32-inch-thick strip of material under one corner or edge of the unit to a distance of 3 inches as the unit bounces. The unit shall be subjected to this preconditioning for 1 minute. After bounce preconditioning, the unit shall not be repaired, alined, cleaned, or otherwise changed prior to subjection to acceptance inspection.

4.16 Electrical tests (see 3.10.- Equivalent tests may be substituted, if approved by the contracting officer.

4.16.1 Standard conditions.- Unless otherwise specified, all tests shall be performed at 115-volts, ±5 percent from a single-phase power source operating between 47.5 and 63.5 cycles per second. Tests shall be conducted at room ambient temperature, pressure, and humidity unless otherwise specified. The warmup period shall not exceed 30 minutes. All tubes shall be in place unless otherwise specified.

4.16.2 Standard tuning procedures.-

4.16.2.1 Transmitter tuning procedure.-

4.16.2.1.1 High Band.-

Step 1. Preset the following controls as indicated.

a. AC Power switch (Power Supply) to "OFF."

b. STAND BY- OPERATE switch (Power Supply) to "OPERATE."

c. "AFC CORRECTION" to mid-range scale.

d. "INPUT LEVELS-PCM" to mid-range setting.

e. "INPUT LEVELS-FDM" to mid-range setting.

f. "AFC" switch to "TUNE."

g. "AFC TUNE" to the desired channel (as indicated on AFC Channel indicator).

NOTE! Unlock all controls on RF head

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.2.1.1 High Band (Contd)

h. "Power Out" to the desired channel (as indicated on TRANSMIT CHANNEL incicator).

i. "MAIN TUNING" to the desired Channel.

j. "COUPLER" to the desired Channel.

k. "REC SIG 2" to the desired Channel (as indicated on RECEIVER CHANNEL indicator). The Receiver Channel should be detuned at least 50 channels from the Transmit Channel.

l. "WAVEMETER" set to the reading listed under the Main Tune column on the Wavemeter charts corresponding to the desired channel.

m. "TRAFFIC CHAN" to the "12/24 CHAN."

n. Set the AC Power switch "ON" and allow at least 15 minutes warm up. (FIL, LV and HV indicators should light after at least 75 seconds. If the Low Power indicator lights and the buzzer sounds, depress the Buz OFF button to silence the buzzer during tune-up. Depress the "BUZ OFF" button each time the buzzer operates on each power output level change.)

Step 2. Set the multimeter selector switch to "OSC."

Step 3. Adjust "OSC" Control for peak indication on meter."OSC" indication should be at 5 μa Hold MAIN TUNE Knob.

Step 4. Set the multimeter selector switch to "MAIN TUNE."

Step 5. Adjust the "MAIN TUNING" Control for peak indication on meter. (Hold "OSC" Control.)

NOTE: During tune-up, if the meter deflection is off -scale, depress the "METER SHUNT" button to accomplish on -scale adjustment.

Step 6. Repeat steps 2-5 above.

Step 7. Lock the "MAIN TUNING" Control.

Step 8. Set the multimeter selector switch to "MIXER."

Step 9. Adjust the "MIXER" Control for peak meter reading.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the currentversion before use.

4.16.2.1.1 High Band (Contd)

Step 10. Set the "WAVEMETER" to the reading listed under "OUT FREQ" column of the wavemeter chart corresponding to the desired channel.

Step 11. Set the multimeter selector switch to "AMP."

Step 12. Adjust the "AMP" control for peak meter reading.

Step 13. Adjust the "AFC Correction" control for peak meter reading.

Step 14. Set the multimeter selector switch to "POWER OUT."

Step 15. Adjust "POWER OUT" Control for peak meter reading.

Step 16. Adjust "COUPLING: and "AMP" Controls alternately to obtain maximum meter deflection.

Step 17. Adjust "POWR OUT" and "COUPLING: controls alternately for maximum meter deflection.

Step 18. Adjust "AMP" and "POWER OUT" controls alternately for maximum meter deflection.

Step 19. The "LOW POWER" indicator should be extinguished after Step 18 .

NOTE: Initially set the OSC., mixer and Amp. Cathode currents to read 25 microamperes by adjusting potentiometers R-14, R-15 and R-16 using test points J16, J17 and J18. Use the front panel meter in position "TEST" and its test point for the OSC., Mixer and Amp. are tuned up. (There must be sufficient drive on the Cathode.)

Step 20. Retune the transmitter starting with Step 2 and proceed to Step 21.

Step 21. Set the multimeter selector switch to "AFC LEVEL." Adjust the "AFC LEVEL" Control for peak meter reading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.2.1.1 High Band (Contd)

Step 22. Adjust the "AFC TUNE" control for peak meter reading.

Step 23. Repeat steps 22 and 23. The multimeter must indicate at least 10 microamperes.

Step 24. Set the AFC selector switch to "ODD" or "EVEN" depending on the selected channel. Set to "ODD" if the selected channel is odd numbered and to "EVEN" if the selected channel is even numbered.

Step 25. Check AFC circuit for proper lock-on by rotating the "AFC CORRECTION" Control about 20° in either direction and observe that the AFC meter deflects each side of center. The meter indicator should move back towards center and stop. The "AFC CORRECTION" control should return to its original setting and should be within ± 10 degrees of center.

Step 26. Set multimeter selector switch to "OUT FREQ."

Step 27. Tune the "WAVEMETER" control for peak meter reading. The Wavemeter dial should be within 2 1/2 division of the wavemeter chart "OUT FREQUENCY" column for Channels 400 to 499, and 2 division for Channels 500 to 899. (If the difference is greater, measure the output frequency by using a Transfer Oscillator and counter to determine if the Wavemeter is in error or if the AFC is locked to the wrong frequency.)

Step 28. Set the multimeter selector switch to "MIXER". Adjust for peak meter reading. Mixer indication at least 5 ua.

Step 29. Set the multimeter selector switch to "PWR OUT." Adjust the "AMP", "POWER OUT" and "COUPLING" controls for peak meter readings. PWR OUT indication at least 20 ua. Lock controls.

4.16.2.1.2 Low Band.-

Step 1. Preset the controls as indicated in Step 1 of paragraph 4.16.2.1.1 (High Band).

Step 2. Set the multimeter selector switch to "MAIN TUNE."

Step 3. Adjust the "MAIN TUNING" control for peak meter reading. Lock the "MAIN TUNE" control.

NOTE: During tuning, if the meter deflection is off-scale, depress the "METER SHUNT" button for on-scale adjustment.

Step 4. Set the multimeter selector switch to "MIXER." Adjust the "MIXER" control for peak meter reading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.2.1.2 Low Band (Contd)

Step 5. Repeat steps 2-4 above.

Step 6. Set the "WAVEMETER" to the reading listed under "OUT FREQ" column of the wavemeter chart corresponding to the desired channel.

Step 7. Set the multimeter selector switch to "AMP." Adjust the "AMP" control for peak meter reading. If no AMP meter indication is obtained, place the "COUPLING" control to a channel for peak. Before proceeding with Step 8, reset "COUPLER" Control to the desired channel and repeak the "AMP" control.

Step 8. Adjust the "AFC CORRECTION" control for peak meter reading.

Step 9. Repeat steps 4 and 7.

Step 10. Set the multimeter selector switch to "POWER OUT." Adjust the "POWER OUT" control for peak meter reading.

Step 11. Adjust "COUPLING" Control for peak meter reading.

Step 12. Adjust "AMP" control for peak meter reading.

Step 13. Adjust "POWER OUT" for peak meter reading.

Step 14. The "Low POWER" indicator should be extinguished after Step 13.

NOTE: Initially set the OSC. and mixer cathode currents to read 25 microamperes by adjusting potentiometers R-14 and R-15, using test points J16 and J17 respectively. Set the Amp. cathode current to 27 microamperes by adjusting potentiometer R16, using test point J18. Use the front panel meter in position "TEST" and its test point for the OSC., Mixer and Amp. Adjust these potentiometers only when the OSC., mixer and amp are tuned up. (There must be sufficient drive on the cathode).

Step 15. Retune the transmitter starting with Step 4 and proceed to Step 16.

Step 16. Set the multimeter selector switch to "AFC LEVEL." Adjust the "AFC LEVEL" control for peak meter reading.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.2.1.2 Low Band (Contd)

Step 17. Adjust the "AFC TUNE" control for peak meter reading.

Step 18. Repeat steps 16 and 17.

Step 19. Set the AFC selector switch to "ODD" or "EVEN" depending on the selected channel. Set to "ODD" if the selected channel is odd numbered and to "EVEN" if the selected channel is even numbered.

Step 20. Check AFC circuit for proper lock-on by rotating the "AFC CORRECTION" control slightly in either direction until the AFC meter indicates ± 40. The meter indicator should then move back towards center and stop. The"AFC CORRECT" control should return to its original setting.

Step 21. Set multimeter selector switch to "OUT FREQ."

Step 22. Tune the "WAVEMETER" control for peak meter reading. The Wavemeter dial should be within 2 1/2 divisions of the wavemeter chart "OUT FREQUENCY," column for channels 1 to 299, and 2 divisions for channels 300 to 399. (If the difference is greater, measure the output frequency by using a transfer Oscillator and Counter to determine if the Wavemeter is in error or if the AFC is locked to the wrong frequency.)

Step 23. Set the multimeter selector switch to "MIXER". Adjust for peak meter reading.

Step 24. Set the multimeter selector switch to "PWR OUT." Adjust the "AMP," "POWER OUT" and "COUPLING: controls for peak meter readings.

Step 25. Set the multimeter selector switch to "REF PWR." Fine tune the "POWER OUT" control and observe the meter indication. Center the "POWER OUT" control between the skirts of increasing reflected power. If the curve has two dips, set the control for the peak between the two dips. If the curve has a single dip, set the control for the dip. In all cases, the criteria is to center the duplexer over the signal.

Step 26. Set the multimeter selector switch to "PWR OUT." Readjust the "AMP" and "COUPLING" controls for peak meter readings. This peak meter reading is normally greater than the peak obtained in step 25. If less, check the antenna and dummy load connections. If the connections are adequate and the power out indication is greater than 16 microamps lock the "AMP" and "COUPLING" controls.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.2.2 Receiver Tuning Procedure: The receiver shall be tuned by any one of two available methods as determined by the type of Amplifier Converter employed. These are as follows:

4.16.2.2.1 Receiver Tuning procedure for use with Receiver with Amplifier Converter AM-1955( )/GRC or AM 1956 ( )/GRC.-

Step 1. Preset the following controls as indicated:

a. "WAVEMETER" to the desired setting as per wavemeter chart.

b. "OSCILLATOR" to the desired channel.

c. "REC. SIG 1" to the desired channel.

d. "AFC LEVEL" to mid-range setting.

e. "AFC TUNE" to the desired channel.

f. "FDM OUTPUT" TO mid-range setting.

g. "SQUELCH" to maximum CW ("OFF" position).

h. "TEST TONE" to "OFF".

i. "AFC" function switch to "TUNE".

j. "TRAFFIC CHAN SWITCH" to 12/24/

k. "AFC CORRECTION" to mid-position.

Step 2. Turn the AC POWER switch to ON.

Step 3. The AC POWR and RING lamps shall light and the buzzer shall operate. After a few seconds, the buzzer shall stop operating and the RING lamp shall go out.

Step 4. Allow the unit to warm up to a maximum of 30 minutes before proceeding with any further testing.

Step 5. Set the meter selector switch to OSC.

Step 6. Slowly rotate the OSCILLATOR control for peak reading on the front panel meter.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.2.2.1 (Contd)

Step 7. Set the meter selector switch to *AFC LEVEL*.

Step 8. Rotate the AFC LEVEL control for a maximum panel meter reading. This maximum must exceed 10 uamps.

Step 9. Slowly rotate the AFC TUNE control for a peak panel meter reading.

Step 10. Set the AFC function switch to ODD or EVEN, depending on the channel number selected in step 1a.

Step 11. The AFC motor should operate until the AFC panel meter becomes zero. To check that the receiver is properly locked on channel, rotate the OSCILLATOR control. This should cause the panel meter to move off zero. The AFC motor should then operate until the AFC meter returns to zero.

Step 12. Repeat step 11, but rotating the Oscillator control in the opposite direction back to the original setting.

Step 13. Set the selector switch to OSC.

Step 14. Rotate the WAVEMETER control for a maximum panel meter deflection. The wavemeter dial should be within three divisions of the reading indicated on the calibrated chart for channels 1 to 49 and 400 to 499; within 2 1/2 division for channels 50 to 349 and 450 to 679; within 2 division for channels 350 to 399 and 680 to 899. If the difference is greater, measure the output frequency by using a transfer Oscillator and frequency counter to determine if wavementer is in error or if AFC is locked to the wrong frequency.

4.16.2.2.2 Receiver Tuning Procedure. for use with Receivers with Amplifier Converter AM-1955A( )/GRC or AM-1956A( )/GRC.

Step 1. Preset the following controls as indicated:

a. "WAVEMETER" to the desired setting as per wavemeter chart.

b. "OSCILLATOR" to the desired channel.

c. "REC SIG 1" to the desired channel.

d. "REC SIG 2" to the desired channel.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.2.2.2 (Contd)

e. "SQUELCH" to maximum clockwise "OFF" position.

f. "TEST TONE" to "OFF".

NOTE: All AFC control settings and indications may be disregarded if Receiver, Radio R-1148(P) ( )/GRC is being employed.

Step 2. Turn the AC Power switch to on.

Step 3. The AC POWER and RING lamps shall light and the buzzer shall operated. After a few seconds the buzzer shall stop operating and the RING lamp shall go out.

Step 4. Allow the unit to warm up a minimum of 30 minutes.

Step 5. Set the meter selector switch to OSC.

Step 6. Slowly rotate the oscillator control for a peak reading on the front panel meter.

Step 7. Set the meter selector switch to REC-SIGNAL.

Step 8. Connect the rf input signal source to the receiver input terminals and tune REC SIG-1 for maximum meter deflection.

Step 9a Depress AFC DISABLE switch of Receiver Radio R-1331(P)()/GRC and tune OSC for Maximum meter deflection. For Receiver, Radio R-1148(P) ( )/GRC follow step 9b.

Step 9b Loosen receiver front panel screws and slide receiver out of case until J5 and J7 of 2d IF assembly are accessible. Insert a test lead jumper between J5 and J7. Tune OSC for maximum meter reading.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

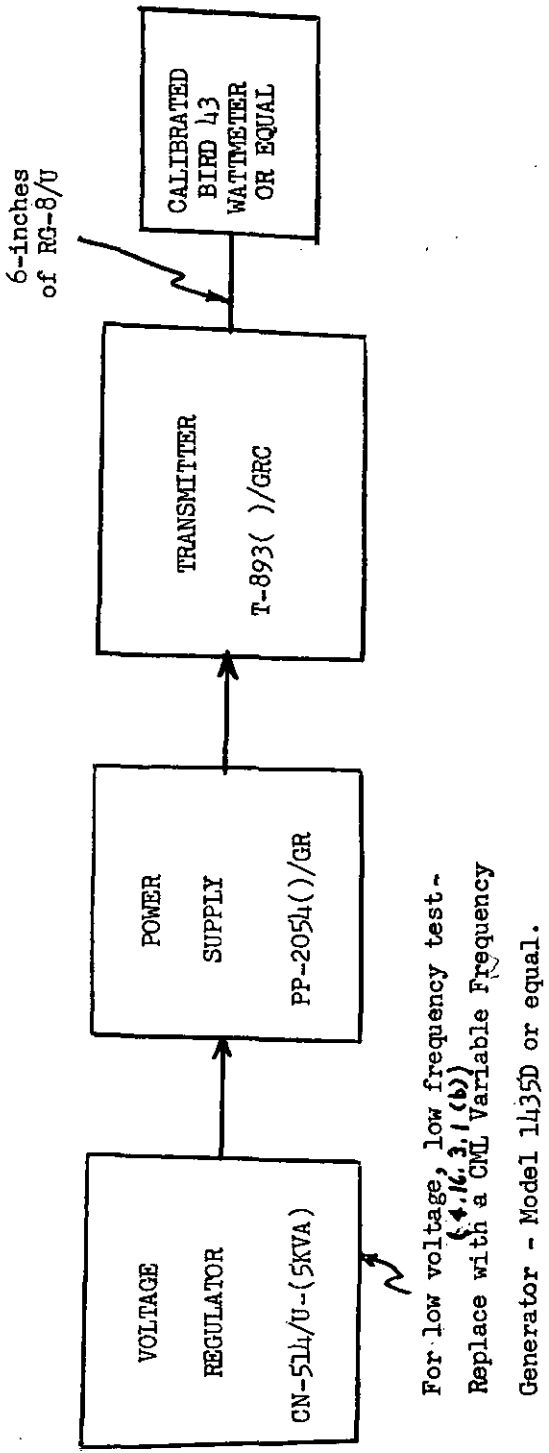


FIGURE 1.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.6.2.2.2 (Contd)

Step 10. Release AFC DISABLE switch in Receiver Radio R-1331(P)( )/GRC or remove test lead jumper in Receiver, Radio R-1148(P)()/GRC and slide receiver into case as the situation required.

4.16.3 Transmitter tests (see 3.10.1)

4.16.3.1. (a) Frequency range and power output test.- The equipment shall be set up as shown in figure 1. Standard test conditions shall apply. (see 4.16.1).

Step 1. The transmitter shall be tuned in accordance with 4.16.2.1

Step 2. Peak for maximum output.

Step 3. Record reading indicated on wattmeter.

Step 4. Repeat steps 1 through 3 above for channels 1, 250, 399, 400, 650 and 899 as applicable. The requirements of 3.10.1.1a shall be met.

4.16.3.1(b) Run the following additional steps for preproduction samples only.

Step 1. Repeat steps 1 through 3 of 4.16.3.1 (a) for every 50 channels. (Set-up and conditions as above).

Step 2. The channel having the lowest relative output shall be retested at 109.3 volts, 47.5 cps. The requirements of 3.10.1.1b shall be met.

4.16.3.2 Overload Test.- The equipment shall be set up as shown in figure 1. Standard test conditions shall apply. (See 4.16.1). The transmitter shall be tuned to Channel 399 or 899, as applicable, for maximum output in accordance with 4.16.2.1

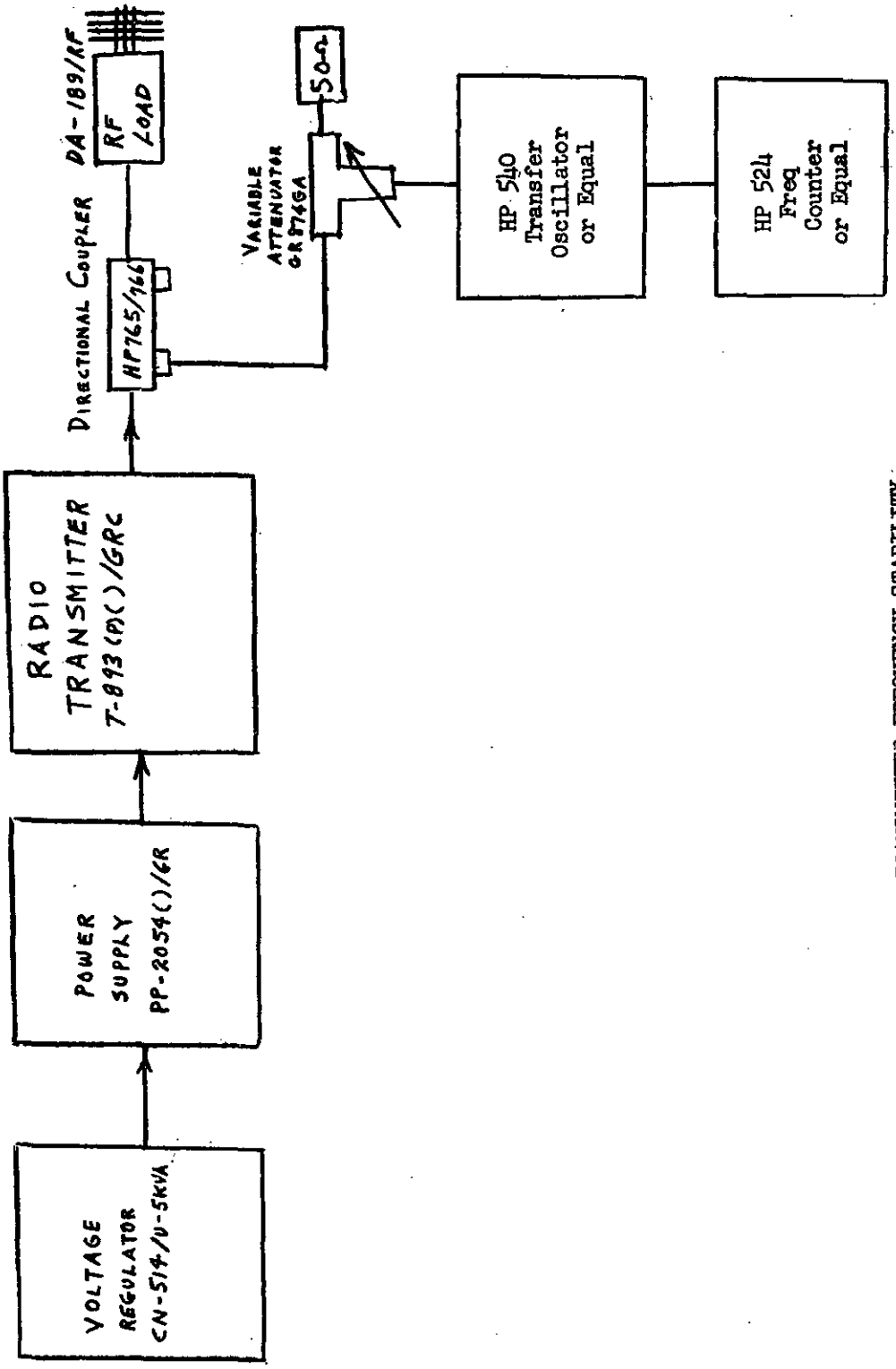
Step 1. Set Multimeter switch to test. Connect ground jumper from J7 to J15. Connect J5 to J16 and note reading; connect J5 to J17 and note reading; connect J5 to J18 and note reading. Meter readings should be within the green area.

Step 2. Detune transmitter controls Main Tune, OSC., MIX and AMP by at least 50 channels.

Step 3. Record cathode current Meter readings from J5 to J16., J5 to J17 and J5 to J18.

Step 4. The requirements of 3.10.1.2 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



TRANSMITTER FREQUENCY STABILITY

FIGURE 2

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.3.3 Transmitter Frequency Stability Test.-

Step 1. The transmitter will be connected as illustrated in Figure 2.

Step 2. Temperature and Humidity will be at the prevailing room ambient condition, not outside the temperature extremes of -20°F to +132°F. Input voltage will be 115v 60 cycles.

Step 3. The transmitter will be tuned to any channel in accordance with 4.16.2.1.

Step 4. The output frequency of the channel selected shall be counted and recorded at the beginning of the eight hour test. The transmitter will be left operating on this channel for the duration of the eight hours.

Step 5. At the end of eight hours of operation, the transmitter output frequency will again be counted and recorded.

Step 6. The requirements of 3.10.1.3 shall be met.

* Step 7. Repeat steps 1-4 above.

Step 8. The ambient temperature shall be changed 54 degrees F from the starting temperature during this period, but still within the above temperature limits.

Step 9. At the completion of the eight hours, the unit output frequency shall again be measured and recorded. The requirements of 3.10.1.3 shall be met. (see 4.6).

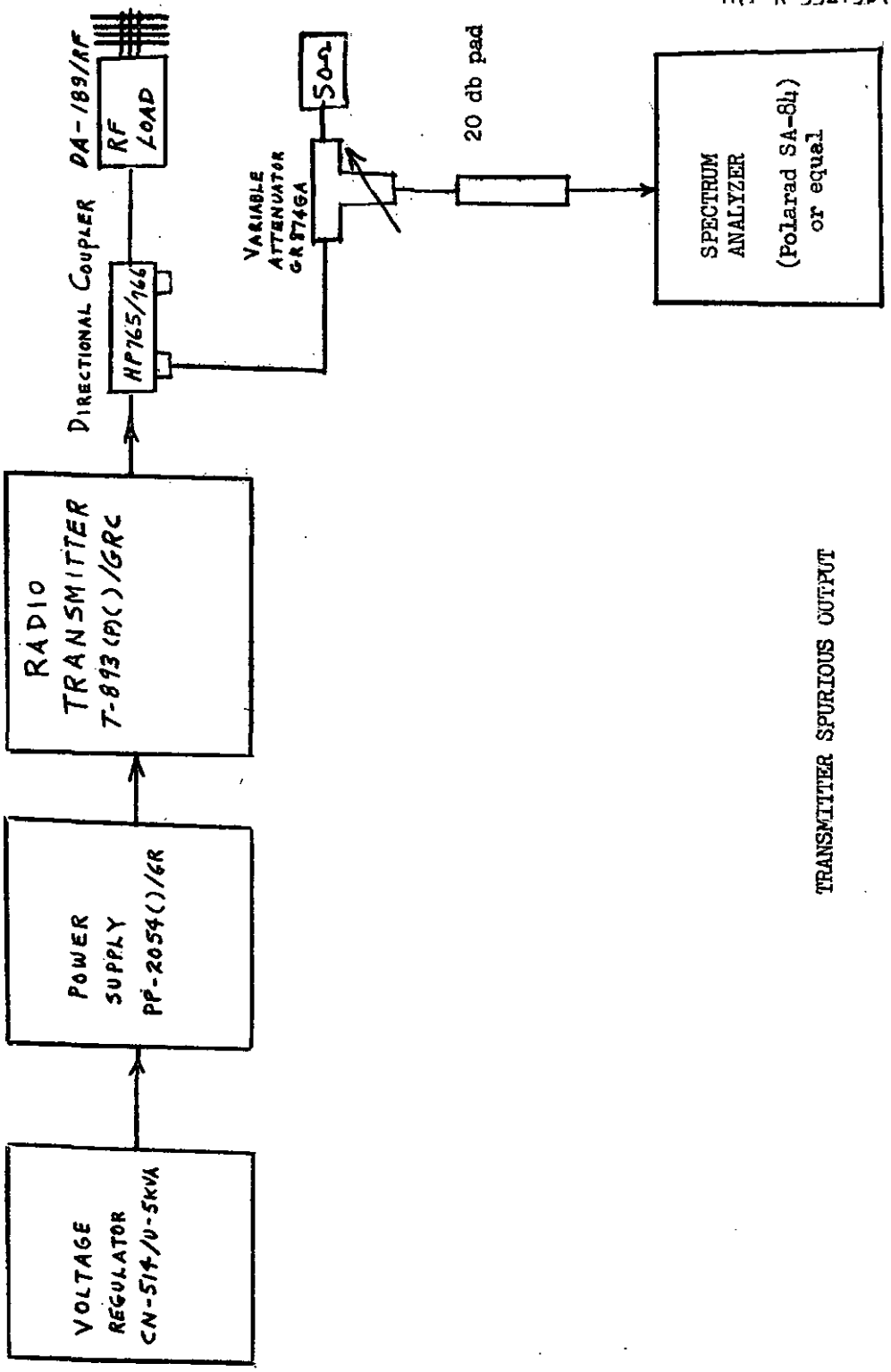
* Preproduction test only (steps 7-9)

4.16.3.4 Short Time AFC Change Test.-With the transmitter set up as for maximum output (4.16.3.1(a), turn the AFC control switch to TUNE and then back to ODD or EVEN as applicable. The requirements of 3.10.1.4 shall be met.

4.16.3.5 Transmitter frequency settability test.- The equipment shall be set up as in figure 2. Standard test conditions shall apply. (See 4.16.1)

Step 1. The transmitter shall be tuned to channel 1/400 in accordance with 4.16.2.1 as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



TRANSMITTER SPURIOUS OUTPUT

FIGURE 3

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.3.5 (Contd)

Step 2. Record the reading indicated on the frequency counter.

Step 3. Repeat for Channels 200, 399, 400, 650 and 899 as applicable.

Step 4. Tune the wavemeter to extreme CW position and retune to Channel 399/899, as applicable, and record reading indicated on frequency counter. Repeat for channels 200, 1, 650, 400 as applicable. The requirements of 3.10.1.5 shall be met.

4.16.3.6 Transmitter Spurious Output Test.- The equipment shall be set up as illustrated in figure 3. Standard test conditions shall apply. (See 4.16.)

Step 1. Tune the transmitter to the desired channel in accordance with 4.16.2.1.

Step 2. Insert at least 50 db of attenuation in the variable attenuator between the directional coupler output and the Spectrum Analyzer input.

Step 3. Tune the Spectrum Analyzer to the Transmitter output frequency and adjust Spectrum Analyzer display for a four centimeter amplitude reference level.

Step 4. With 50 db of attenuation removed between the Directional Coupler output and the Spectrum Analyzer input, tune the Spectrum Analyzer over the range from 600 to 1,000 mc for the low band transmitter and the 1350 to 1850 mc range for the high band transmitter. Responses should be below the established reference level.

Step 5. The requirements of 3.10.16 shall be met.

NOTE: Responses that are to be measured are those that move from right to left on the Spectrum Analyzer display when the tuning dial on the Analyzer is moved counter clockwise, from a higher to a lower frequency. The generation of Spurious responses in the Spectrum Analyzer caused by the high RF Input from the Transmitter can be identified by the following means and can be discounted if they exist. With the questioned response centered on the display, tune the Rec. Sig 2 dial on the Transmitter to the Transmitter operating channel. If the displayed response decreases in amplitude the response is generated in the Analyzer itself. Further proof that this is not a transmitter spurious output can be made by tuning the Rec. Sig. 2 Dial to the apparent frequency of the spurious response. If tuning the Rec. Sig. 2 Dial does not cause the amplitude to decrease it is not a Transmitter Spurious Output.

The NF-105 Noise and Field Intensity Meter - Empire Device can be used in place of the spectrum analyzer as the reference device.

4.16.3.7 Transmitter Alarm Test.- The equipment shall be set up as in figure 1. Standard test conditions shall apply. (See 4.16.1).

Step 1. Select any one channel for each band.

Step 2. The transmitter shall be tuned in accordance with 4.16.2.1.

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.3.7 (Contd)

Step 3. Connect a Voltmeter to REPEATER ALARM CONNECTION on POWER SUPPLY and set to 5 VDC range.

Step 4. Set low power "ALARM ADJ" full CCW.

Step 5. Detune the transmitter by adjusting the COUPLING Control for (a) 10 watts low band and (b) 5 watts-high band.

Step 6. Rotate "ALARM ADJ" CW for low Power lamp indication, and buzzer operation at this level.

Step 7. Swing COUPLING control above and below the 5 watt level and note that buzzer operates, voltage of 2 VDC minimum appears at REPEATER ALARM and LOW POWER lamp lights.

Step 8. Push BUZ OFF. Note buzzer silences and repeater voltage nulls.

Step 9. Repeat COUPLING control for 8 watts or more. Note LOW POWER l amp goes out and buzzer operates.

Step 10. Push BUZ OFF to silence buzzer.

Step 11. Turn the power supply switch to "Standby" and remove transmitter head from main frame.

Step 12. Remote transmitter head using cable with a normally closed switch in series with pin 30 of the transmitter head.

Step 13. Using a standard head-frame bolt, depress head-frame interlock switch (S-6).

Step 14. Restore power to the transmitter head by placing the power supply switch in operate position (a 75 second delay will occur).

Step 15. Repeak transmitter head for maximum power-out at the channel to which transmitter was tuned. (Be sure Low Power indications are deactivated when transmitter is peaked-up).

Step 16. By removing the head frame bolt (step 3) observe that high and low voltage to the transmitter head is turned off)appropriate power supply lights turned off) and that the low power alarms are activated.

Step 17. Restore power to the transmitter head by replacing the head frame interlock bolt(a delay that will occur if the interruption was greater than 6 seconds). Low Power indications should be deactivated when power is restored.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

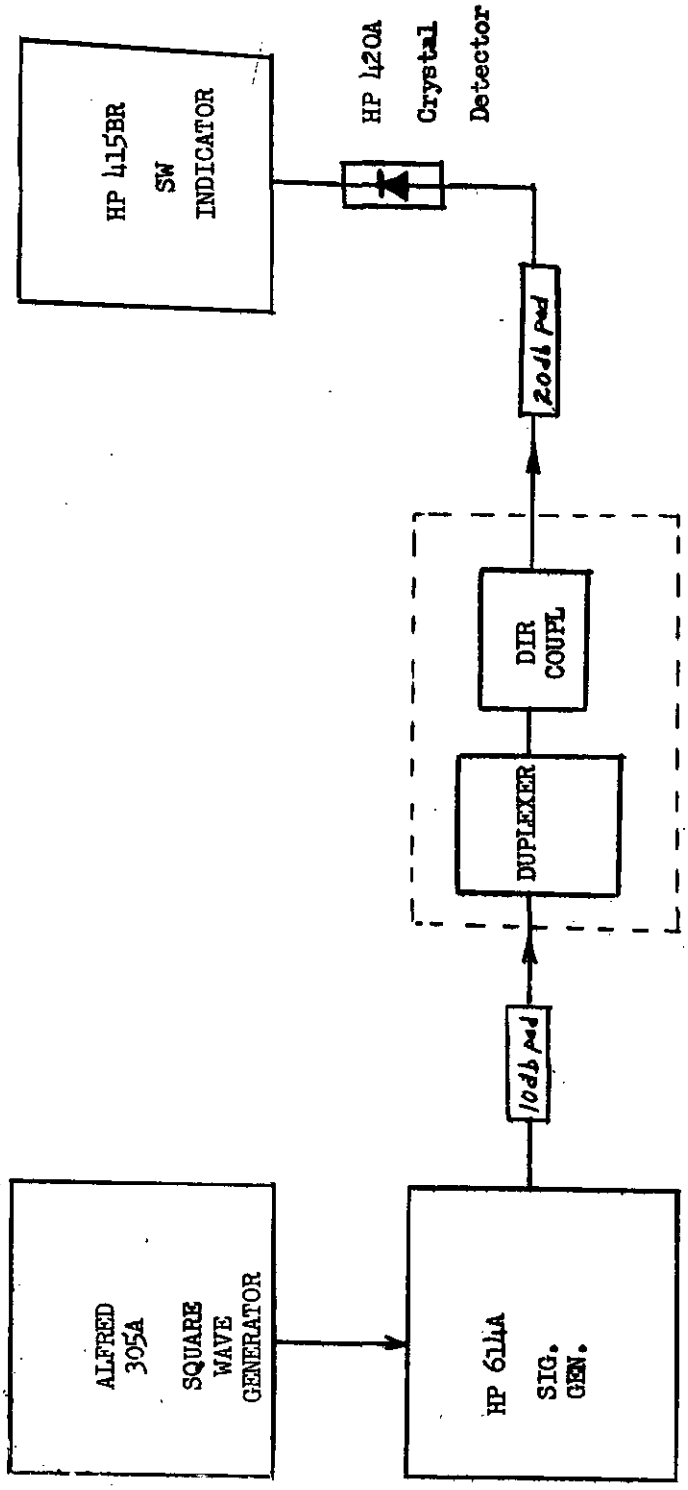


FIGURE 4

DUPLEXER INSERTION LOSS TEST

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

#20

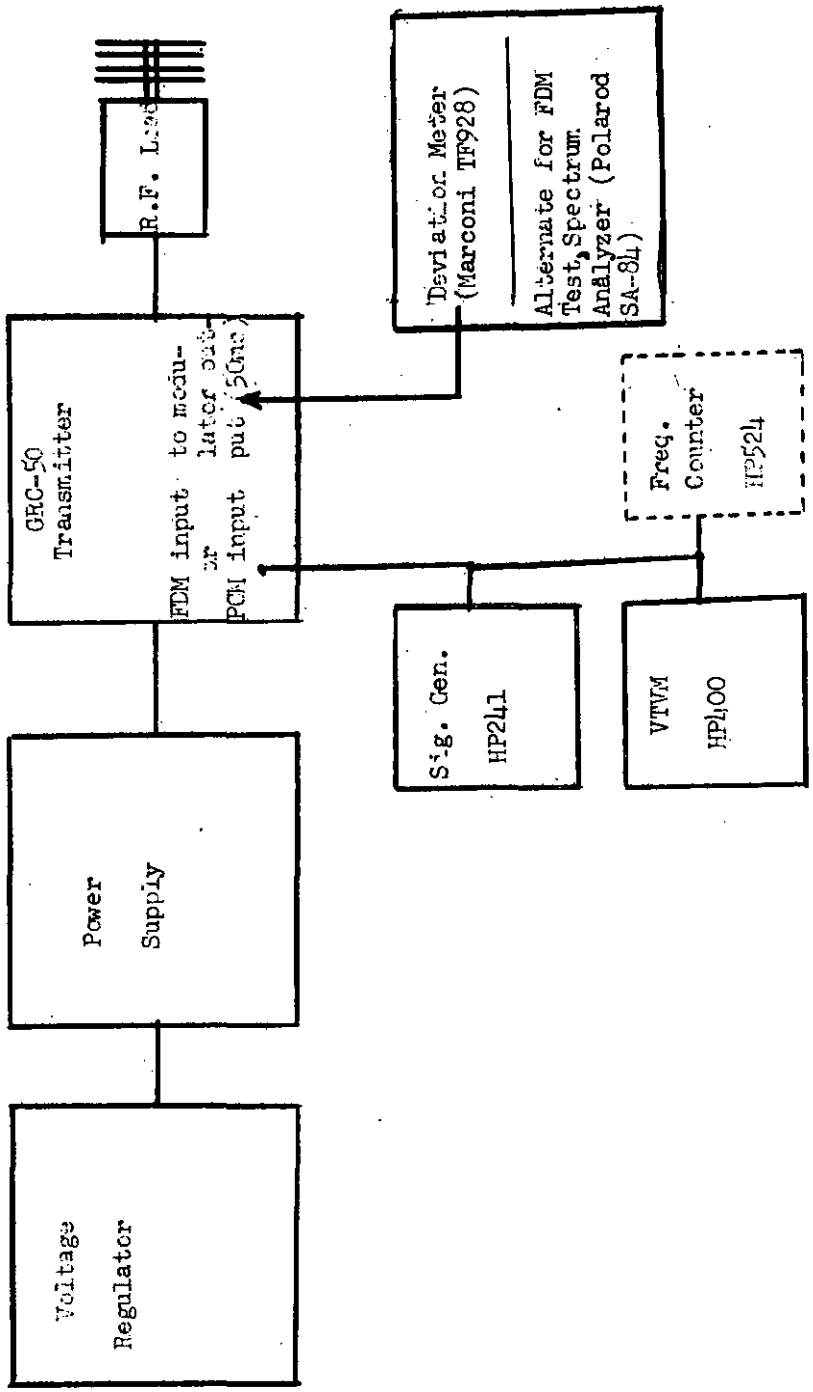


Figure 40 - Transmitter Deviation Sensitivity and Metering

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.3.7

Step 18. Depress switch which is a part of the extender cable and observe that the results of step 6 are again obtained.

NOTE 1: If head is removed from frame for a long period of time with no cooling air, the thermal relay may be activated.

Step 19. The requirements of 3.10.1.7 shall be met.

NOTE 2: After completion of test, remove extender cable and replace the transmitter head in the main frame.

4.16.3.8 Insertion Loss Test.- The equipment shall be set up as in figure 4. Standards test conditions shall apply. (See 4.16.1)

Step 1. Connect Alfred 605A output to HP 614A EXT MOD input.

Step 2. Set Alfred 605A Amplitude and Frequency controls to mid range.

Step 3. Set HP 614A Switch to EXT NEG.

Step 4. Set HP 415 BR input selector to 200 ohm Xtal, Range to 40 db, Gain to midrange and Meter Scale to NORMAL.

Step 5. The duplexer will be disconnected from the PA output and the Signal Generator will be connected to the input of the Duplexer (J-6) to which the Power Amp was connected. The ANT output (J-2) will be connected to the 20 db pad through a short cable.

Step 6. Set the Duplexer to the desired channel and tune the Signal Generator for max. on the HP 415B meter. Tune the Alfred Square Wave Generator for max. on HP-415B. Adjust the Signal Generator attenuator for a 2 dbm reference on the HP 415 meter.

Step 7. Remove cable from Duplexer input and connect it directly to the 20 db pad. The level on the HP-415B meter should not increase by more than 2 db. The requirements of 3.10.1.8 shall be met.

Step 8. Repeat steps 5, 6 & 7 with Signal generator connected to REC side and duplexer marked "To REC" (J-3).

4.16.3.9 Transmitter FDM Deviation Sensitivity Test.- With the equipment set up per Figure 4A.

Step 1. Tune the transmitter to the desired r.f. channel in accordance with 4.16.2.1.

Step 2. Calibrate the deviation meter at the deviation limits accordingly.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

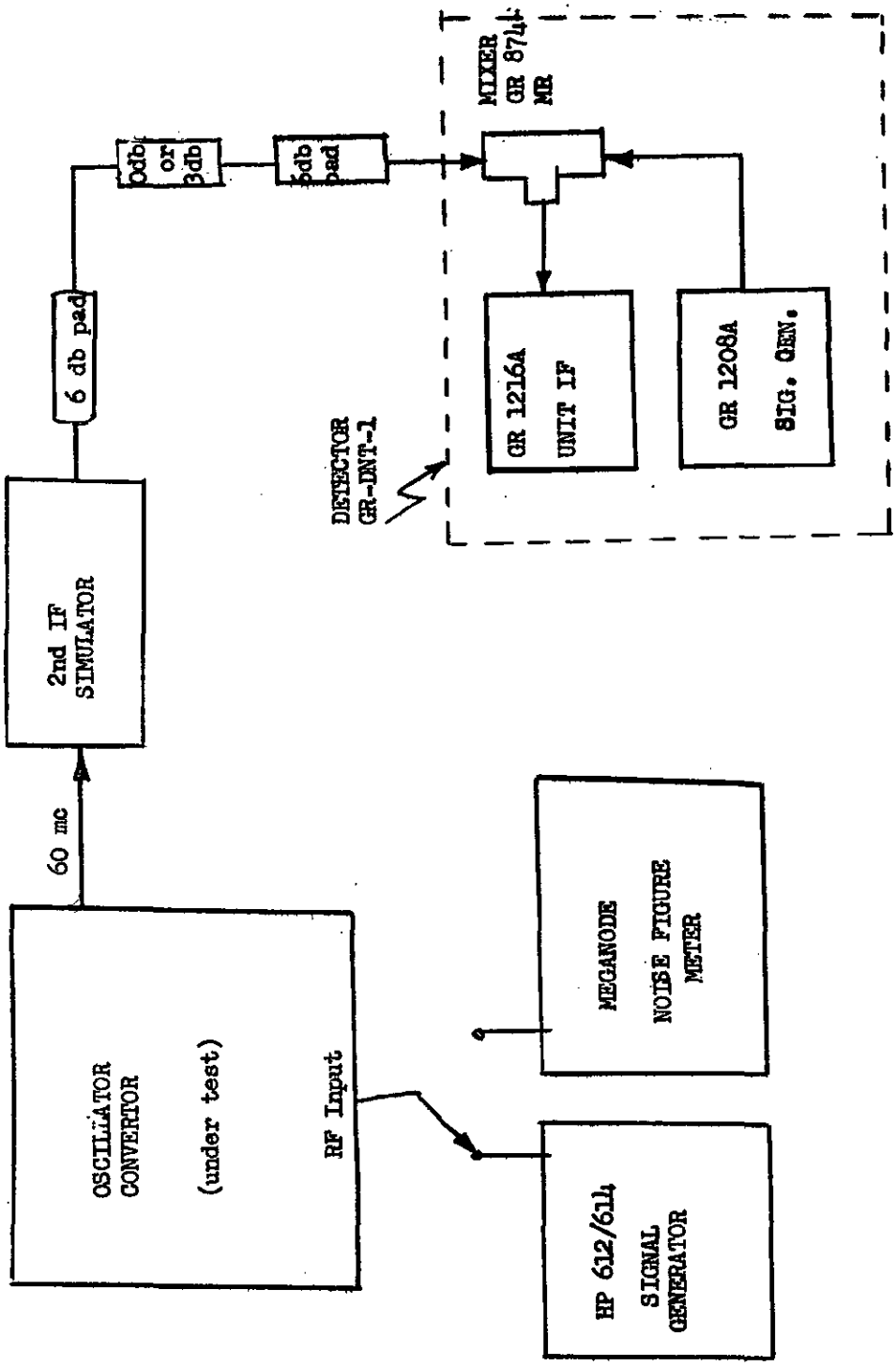


FIGURE 5

FREQUENCY, RANGE & NOISE FIGURE TEST

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.3.9 Step 2 (Contd)

Apply a 56 Kc signal to the FDM input terminal and adjust level to produce a 300 Kc deviation by using the 2nd Bessel Null method of observed on the spectrum analyzer. Calibrate the deviation meter by reading same. Correspondingly apply 85 Kc modulation and calibrate the deviation meter for 470 Kc deviation. Likewise apply 43.5 kc modulation for 240 Kc deviation and 68 Kc modulation for 375 Kc deviation. Record all deviation meter readings.

Step 3. Apply 72 Kc modulation at an input level of +13 dbm and determine if the deviation is within the limits of 300 Kc and 470 Kc as indicated on the deviation meter.

Step 4. The requirements of 3.10.1.9 shall be met.

Step 5. Adjust the input level to +11 dbm and similarly determine if the deviation is within the limits of 240 Kc and 300 Kc.

Step 6. The requirements of 3.10.1.9 shall be met.

4.16.3.10 Transmitter PCM Deviation Sensitivity.- With the equipment set up per Figure 4A.

Step 1. Tune the transmitter to the desired r.f. channel in accordance with 4.16.2.1.

Step 2. Calibrate the Marconi TF-928 deviation meter at the deviation limits of 300 and 470 Kc accordingly. Apply 56 Kc modulation to the input terminals and adjust the input level for 300 Kc deviation by using the spectrum analyzer to observe the Carrier Bessell Null. Measure and record the indicated reading on the deviation meter. Repeat the above for 85 Kc modulation and a 470 Kc deviation.

Step 3. Apply a 10 Kc sinusoidal signal to the PCM input terminals at a level of 0.53 volts rms (1.5 volts peak to peak) with the input PCM attenuator in mid position and measure the deviation with the deviation meter.

Step 4. The requirements of 3.10.1.10 shall be met.

4.16.3.11 AC Power interruption (Transmitter)- The equipment shall be set up as in figure 2. Standard test conditions shall apply (4.16.1)

Step 1. Tune the transmitter to the desired RF channel in accordance with 4.16.2.1.

Step 2. The output frequency of the channel selected shall be counted and recorded.

Step 3. Interrupt the AC power for a period of 2 seconds and turn it on again. NOTE: After a period of approximately 6 minutes, proceed with Step 4.

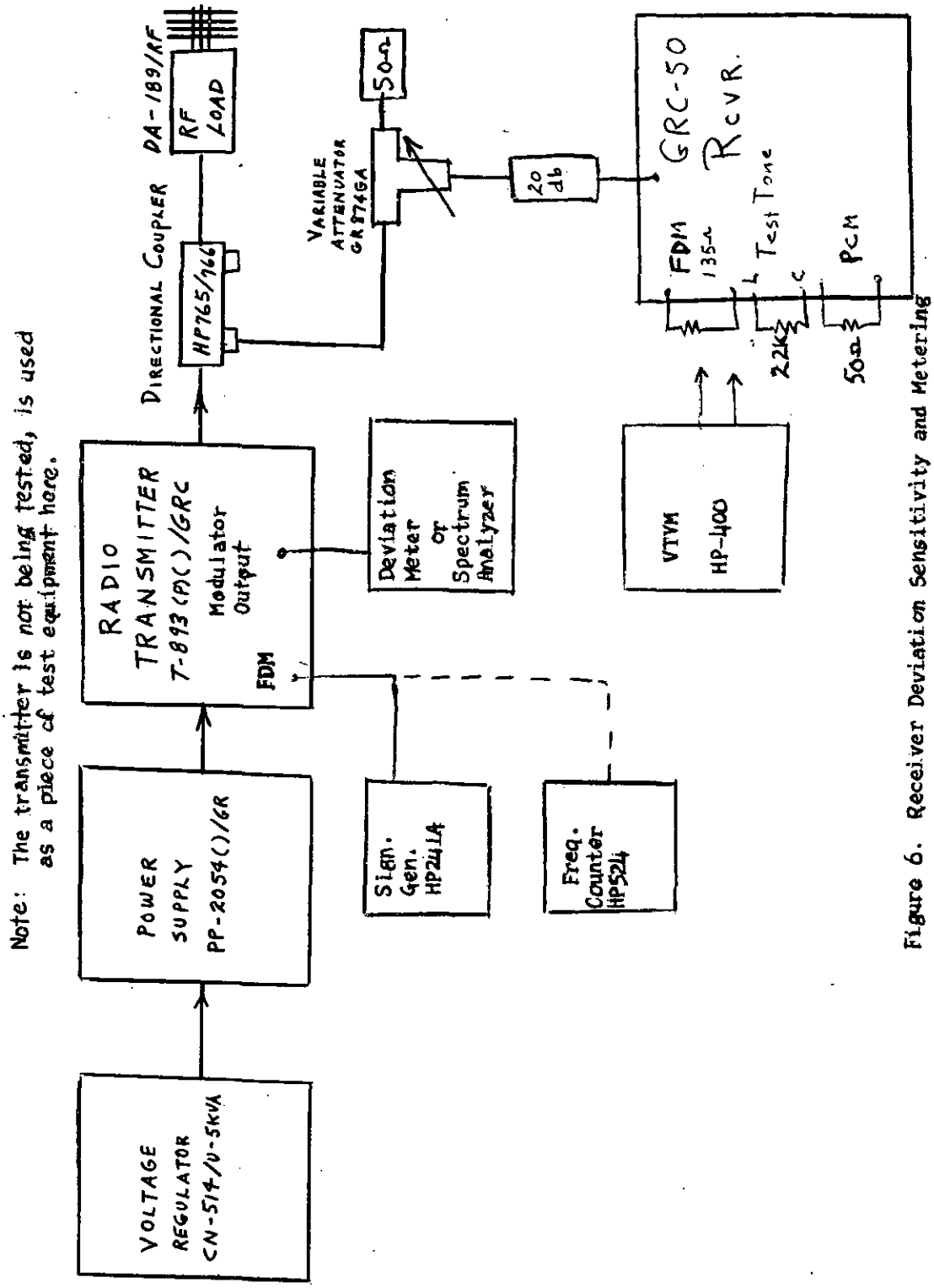
Step 4. The output frequency of the channel selected will again be counted and recorded. Requirements of 3.10.1.11 shall be met.

Step 5. Interrupt the AC power for a period of 90 seconds and turn it on again. Requirements of 3.10.1.11 shall be met.

4.16.4 Receiver tests (see 3.10.2).-

4.16.4.1 Frequency Range and Noise Figure Test.- The equipment shall be set up as in figure 5. Standard test conditions shall apply. (See 4.16.1)

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.



Note: The transmitter is not being tested, is used as a piece of test equipment here.

Figure 6. Receiver Deviation Sensitivity and Metering

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.1 Contd

Step 1. Tune the receiver to the desired channel in accordance with 4.16.2.2.

Step 2. Apply the signal generator to the receiver RF input.

Step 3. Set the oscillator and signal generator for the desired frequency.

Step 4. Tune the signal generator for a peak Indication on the Unit IF Amplifier (GR-1216A, or equal).

Step 5. Alternately tune the receiver and signal generator for peak indication on the unit if amplifier.

Step 6. Disconnect the receiver input from the signal generator and connect it to the noise generator.

Step 7. Change the signal generator frequency and set its attenuator for minimum output level.

Step 8. Note the indication on unit if amplifier with the noise generator cut-off.

Step 9. Adjust the output indicator control on the noise generator to add 5 db position.

Step 10. Increase the attenuation between the 2nd IF simulator and the detector by 3 db.

Step 11. Adjust the noise output adjustment on the noise generator for the indication noted in step 8 on the Unit IF meter.

Step 12. Record the noise generator indication and apply the transit time correction factor to determine noise figure.

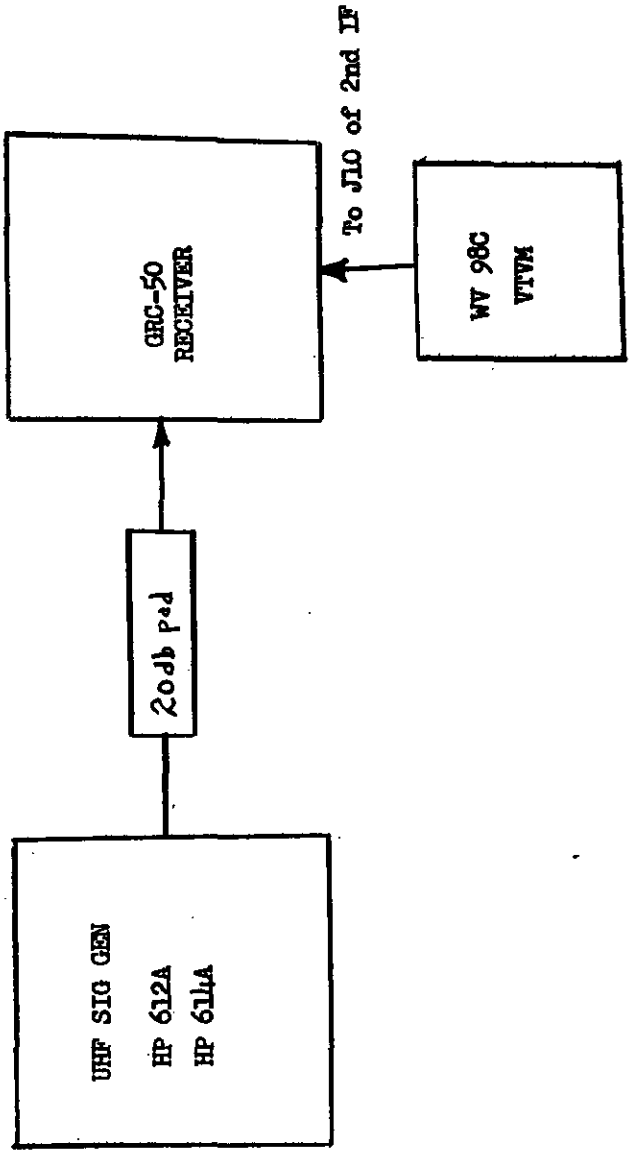
Step 13. Repeat steps 1 through 12 for rf channels 1, 250, 399, 400, 650, and 899, as applicable. The requirements of 3.10.2.1 shall be met.

4.16.4.2 Receiver FDM Deviation Sensitivity Test

The equipment shall be set up as in Figure 6. The standard test conditions shall apply (See 4.16.1)

Step 1. Tune the transmitter and receiver to the desired r.f. channel in accordance with 4.16.2.1 and 4.16.2.2 respectively.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



RECEIVER SPURIOUS RESPONSE TEST

FIGURE 7

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.2 Contd.

Step 2. Adjust the r.f. input to the receiver to from -70dbm to -50 dbm.

Step 3. Calibrate the Marconi TF-928 deviation meter for 375 and 300 Kc deviation accordingly. Apply 68 Kc modulation to the FDM input and adjust the input level for 375 Kc deviation by using the polar SA-84 spectrum analyzer to observe the carrier second Bessel Null. Measure and record the indicated reading on the deviation meter. Repeat the above procedure with 56 Kc modulation to obtain a calibration reading of the deviation meter for 300 Kc deviation.

Step 4. Apply 72 Kc modulation to the FDM input terminals and adjust the input level to obtain the same reading corresponding to 375 Kc as obtained in step (3) on the deviation meter.

Step 5. Measure the output level at the receiver FDM terminals.

Step 6. The requirements of 3.10.2.3 shall be met.

Step 7. Decrease the input level to obtain a 300 Kc deviation as per the deviation meter calibration.

Step 8. Measure the output level at the receiver FDM terminals.

Step 9. The requirements of 3.10.2.3 shall be met.

4.16.4.3 Receiver Spurious Response Test.- The equipment shall be set up as in figure 7. Standard test conditions shall apply. (4.16.1).

Step 1. The Signal Generator will be adjusted to the desired channel frequency, with a level of approximately -70 dbm (generator -50dbm, 20 db in the pad) applied to the receiver input.

Step 2. Tune the receiver, in accordance with the standard procedure, to the desired channel and then measure the DC voltage at J-10 of the 2nd IF for a reference.

Step 3. Remove 50 db from the Signal Generator Attenuator.

Step 4. While monitoring the voltage at J-10, tune the Signal Generator through the range of 600 to 1000 mc for the low band receiver and through the range of 1350 to 1850 mc for the high band receiver.

Step 5. The reference level established at J-10 at the desired channel frequency should not be exceeded for any spurious response.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.3(Contd)

Step 6. The test will be conducted with the receiver tuned to Channels 1, 250, 399, 400, 650 and 899 as applicable.

Step 7. The requirements of 3.10.2.4 shall be met.

4.16.4.4 Automatic frequency control test.- Automatic frequency control test for Receiver Radio R-1148(P) ( )/GRC with Amplifier Converter AM-1955( )/GRC or AM-1956 ( )/GRC. The equipment shall be set up as figure 8.

Step 1. After a 30 minute warm up period with the ambient temperature at a point between minus 20°F and plus 132°F and the relative humidity below 95%, tune the receiver to Channel 399 or 899 as applicable in accordance with 4.16.2.2.1

Step 2. Calibrate the discriminator error voltage characteristic by the following procedure:

a. Disconnect the input to J-2 of the second IF and apply a signal generator tuned to approximately 60 mc. at 20 mv output level.

b. Disable the secondary AFC. This is accomplished by inserting a jumper test cable between J7 and J5 of the second IF unit.

c. Adjust the generator for a zero discriminator voltage as measured at J6. Count and record the input frequency.

d. Change the frequency by plus (0.005 percent plus 90 kc) of step c and record the voltage J6.

e. Change the frequency by minus (0.005 percent plus 90 kc.) of step c, and record the voltage at J6.

f. Reconnect the normal input to J2 of the second IF. Restore the secondary AFC operation by removing the jumper cable.

Step 3. Apply a carrier input signal at an RF level between minus 80 dbm and minus 40 dbm at the assigned frequency to the input terminal of the receiver. Adjust the "REC SIG" control for a peak indication on the REC SIG meter. Set the generator output for a 40 microamp reading on the REC SIG meter. Count and record the input frequency.

Step 4. Count and record the 60 mc. frequency at J2. The algebraic difference between the error frequency on the input frequency (counter #1) and the error frequency of the 60 mc frequency (counter #2) shall be less than ± 0.005 percent ± 90 kc. of the assigned channel in order that the requirements of paragraph 3.10.2.5 shall be met.

Step 5. Measure the voltage at J6. It shall be within the limits established in steps 2d and 2e in order that the requirements of paragraph 3.10.2.5 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

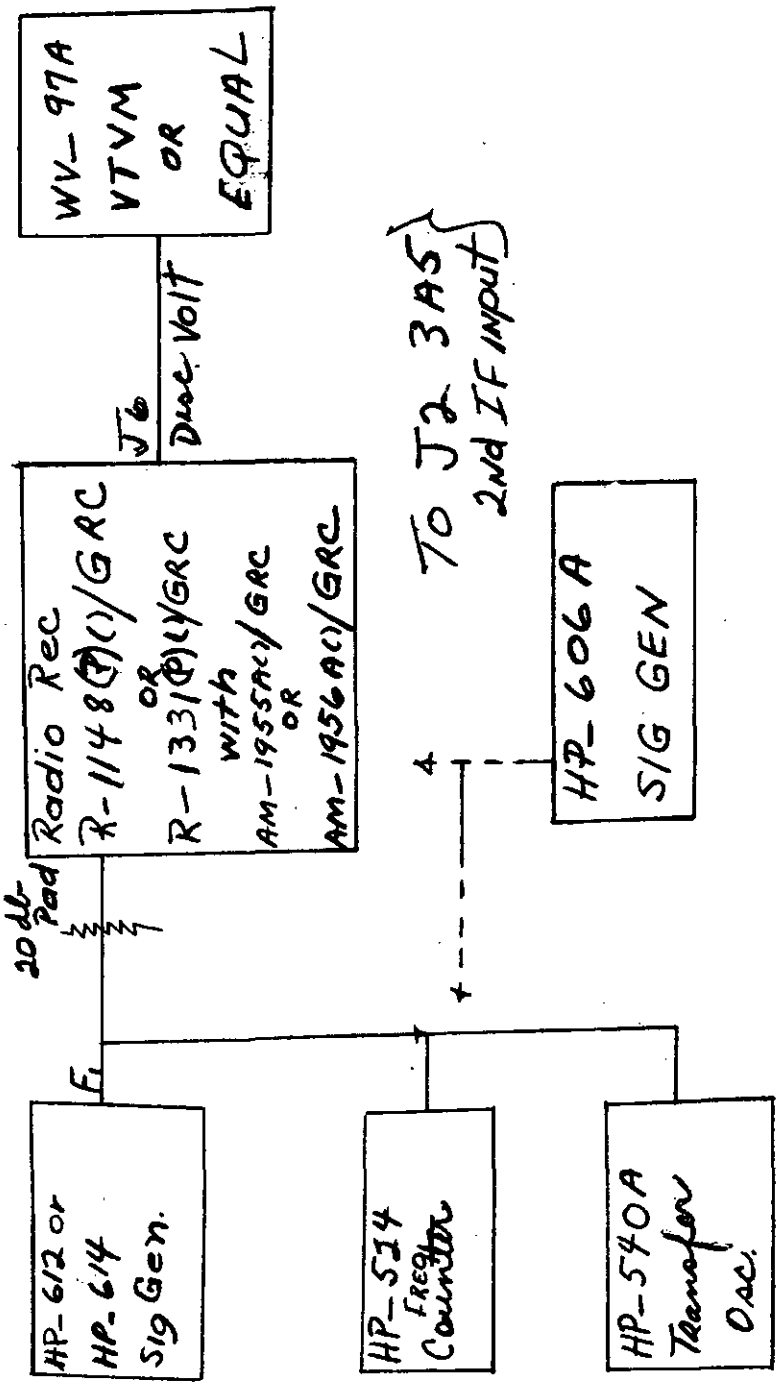


FIGURE 8

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

Step 6. Turn the AFC switch to the TUNE position, then return to ODD (or EVEN) whichever is applicable. The requirements of 3.10.25 shall be met.

Step 7. Repeat steps 1 through 6 at any time during an 8 hour period. Requirements of 3.10.2.5 shall be met.

Step 8. For preproduction tests, repeat steps 1 through 7 with the ambient temperature removed any amount up to 54 degrees F from the ambient temperature,at the time of initial setting of the receivers. (see 4.6) The requirements of 3.10.2.5 shall be met.

Step 9. For preproduction tests, continue the tests for 168 hours respeating steps 1 through 6 at any time. The requirements of 3.10.2.5 shall be met.

4.16.4.5 Frequency stability tests for receiver R-1331(P( )/GRC with Amplifier Converter AM-1955A( )/GRC or AM-1956A( )/GRC . The equipment shall be set up as in figure 8.

Step 1. After a 30 minute warmup period with the ambient temperature at a point between minus 20°F and plus 132°F and the relative humidity below 95%, tune the receiver to channel 399 or 899, as applicable, in accordance with 4.16.2.2.2.

Step 2. Apply a carrier input signal at an rf level between -80 dbm and -40 dbm at the assigned frequency to the input terminal of the receiver. Adjust the REC SIG control for a peak indication on the REC SIGNAL meter. Set the generator output for a 40 microamp reading on the REC SIGNAL meter. Count and record the input frequency.

Step 3. Count and record the 60 mc frequency at J2. The algebraic difference between the error frequency on the input frequency counter (#1) and the error frequency of the 60 mc. (counter #2), shall be less than ± 750 kc in order that the requirements of paragraph 3.10.2.6 shall be met.

Step 4. Repeat steps 1 through 3 at any time during an 8 hour period. Requirements of 3.10.2.6 shall be met.

Step 5. For preproduction tests, repeat steps 1 through 4 with the ambient temperature removed any amount up to 54 degrees F from the ambient temperature at the time of initial setting of the receiver (see 4.6). The requirements of 3.10.2.6 shall be met.

Step 6. For preproduction tests, continue the tests for 168 hours repeating steps 1 through 3 at any time. The requirements of 3.10.2.6 shall be met.

4.16.4.6 Secondary Automatic Frequency Control Tests for receiver with Amplifier Converter AM-1955A( )/GRC or AM-1956A( )/GRC. The equipment shall be set up as in figure 8.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.6 Contd

Step 1. After a 30 minute warmup period with the ambient temperature at a point between minus 20°F and plus 132° F and the relative humidity below 95%, tune the receiver to channel 399 or 899, as applicable, in accordance with 4.16.2.2.

Step 2. Calibrate the 60 KC discriminator-error voltage characteristic by the following procedure:

a. Disconnect the input to J2 of the 2nd IF and apply a signal generator tuned to approximately 60 mc @ 20 mv output level.

b. Disable the secondary AFC. This is accomplished in Receiver Radio R-1331(P) ( )/GRC by depressing the momentary AFC Disable Switch, and in Receiver Radio R-1148(P) ( )/GRC by inserting a jumper test cable between J-7 and J-5 of the second IF unit.

c. Adjust the generator for a zero discriminator voltage as measured at J6. Count and record the input frequency.

d. Change the frequency by plus 60 KC of step c and record the voltage at J6.

e. Change the frequency by minus 60 KC of step c and record voltage at J6.

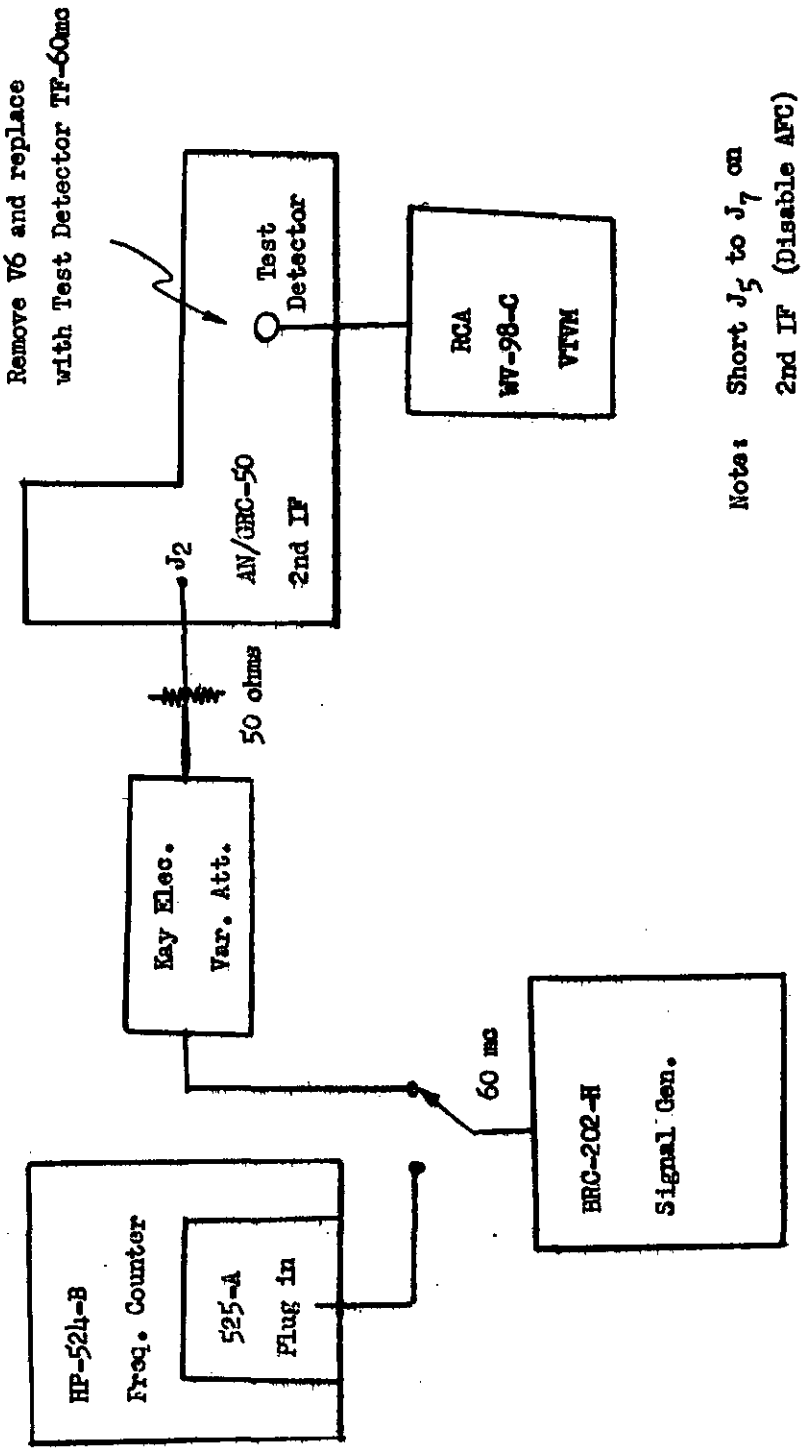
f. Reconnect the normal input to J2 of the 2nd IF. Restore the secondary AFC operation by releasing the AFC Disable Switch in R-1331(P) ( )/GRC or removeing the jumper cable in R-1148(P) ( )/GRC.

Step 3. Apply a carrier input signal at an rf level between -80 dbm and -40 dbm at the assigned frequency to the input terminal of the receiver. Adjust the REC SIG control for a peak indication on the REC SIGNAL meter. Set the generator output for a 40 microamp reading on the REC SIGNAL meter. Count and record the input frequency.

Step 4. Measure the voltage at J6. It shall be within the limits established in steps 2d and 2e in order that requirements of 3.10.2.7 shall be met.

Step 5. Repeat steps 1 through 4 at any time during an 8 hour period. Requirements of 3.10.2.7 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.



IF BANDWIDTH

FIGURE 9

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.6 Contd

Step 6. For preproduction tests, repeat steps 1 through 5 with the ambient temperature removed any amount up to 54 degrees F from the ambient temperature at the time of initial setting of the receiver. (see 4.6). The requirements of paragraph 3.10.2.7 shall be met.

Step 7. For preproduction tests, continue the tests for 168 hours repeating steps 1 through 4 at any time. The requirements or 3.10.2.7 shall be met.

4.16.4.7 IF Bandwidth .- The equipment shall be set-up as in Figure 9. Standard test conditions shall apply. (see 4.16.1)

Step 1. Set the Kay Variable Attenuator for -3DB.

Step 2. Adjust the BRC-202H frequency to approximately 60 MC.

Step 3. Turn the output level of the BRC-202H for minimum output, and note the reading on the WV-98-C VTVM. This is the noise level.

Step 4. Turn the output level of the BRC-202H up until an increase in voltage is noted on the WV-98-C-VTVM.

Step 5. Fine tune the BRC-202H frequency for a maximum indication on the WV-98-C. Then adjust the output level of the BRC-202H for a voltage of 0.10 volts above the noise level noted in Step 3.

Step 6. Remove the -3DB attenuation from the Kay Attenuator.

Step 7. Tune the frequency vernier of the BRC-202H to a higher frequency until the WV-98-C reads exactly the Step 5 value.

Step 8. Connect the BRC-202H output to the HP-524-B frequency counter input, increase the output level of the BRC-202-H sufficiently for the counter to operate.

Step 9. Measure the frequency of the BRC-202H and note this value.

Step 10. Insert -3DB attenuation on the Kay Attenuator.

Step 11. Reconnect the BRC-202-H output to the Kay Attenuator input.

Step 12. Repeat steps 5 through 9 tuning the BRC-202H to a lower value than the peak frequency.

Step 13. Subtract the two 3 db down frequency values obtained at Steps 9 and 12. Record this difference as the 3 db bandwidth.

Step 14. Repeat entire test through Step 13 using -60DB attenuation in the Kay Attenuator.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.7 IF Bandwidth Contd.

Step 15. The requirements of 3.10.2.8 shall be met.

4.16.4.8 Squelch and Alarm test.- The equipment shall be set up as in figure 7 for steps 1 through 10 below. Standard test conditions shall apply. (See 4.16.1)

Step 1. Select any one channel from each band.

Step 2. Tune the receiver to the desired RF channel in accordance with 4.16.2.2.

Step 3. Set the signal generator to an output of -70 dbm. Generator tuned to zero discriminator cross-over. (Twenty (20) db pad between generator and receiver gives a -90 dbm input to receiver.)

Step 4. Rotate the squelch control from the maximum clockwise position to the point where the NO SIGNAL lamp lights and buzzer operates.

Step 5. Rotate the squelch control to maximum clockwise and note that NO SIGNAL lamp goes out and buzzer quiets.

Step 6. Repeat step 4 and 5 with a generator output of -50 dbm (-70dbm to receiver.)

Step 7. Repeat step 4 with a generator output of -50 dbm (-70dbm to receiver.)

Step 8. Slowly increase the generator output level to the point where the buzzer stops. Note the generator output in dbm. The generator output shall be -47 dbm to -53 dbm.

Step 9. Slowly decrease the generator output level to the point where buzzer operates. This shall be within 3db of the level of the previous step.

Step 10. With the squelch operating(receiver muted-silence buzzer with BUZ OFF control), measure the output levels at the FDM,PCM, and ROW output terminals(terminated) with a flat weighted meter of detected range of 0.300 to 108 KC. Levels should be less than -45 dbm.

NOTE: Pull V2 in the Signalling Unit (with the set muted), to disable the 1600 cps oscillator and measure the noise level at the Remote Order Wire terminals in a band from 300 to 1800 cps. The requirements of 3.10.2.9shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

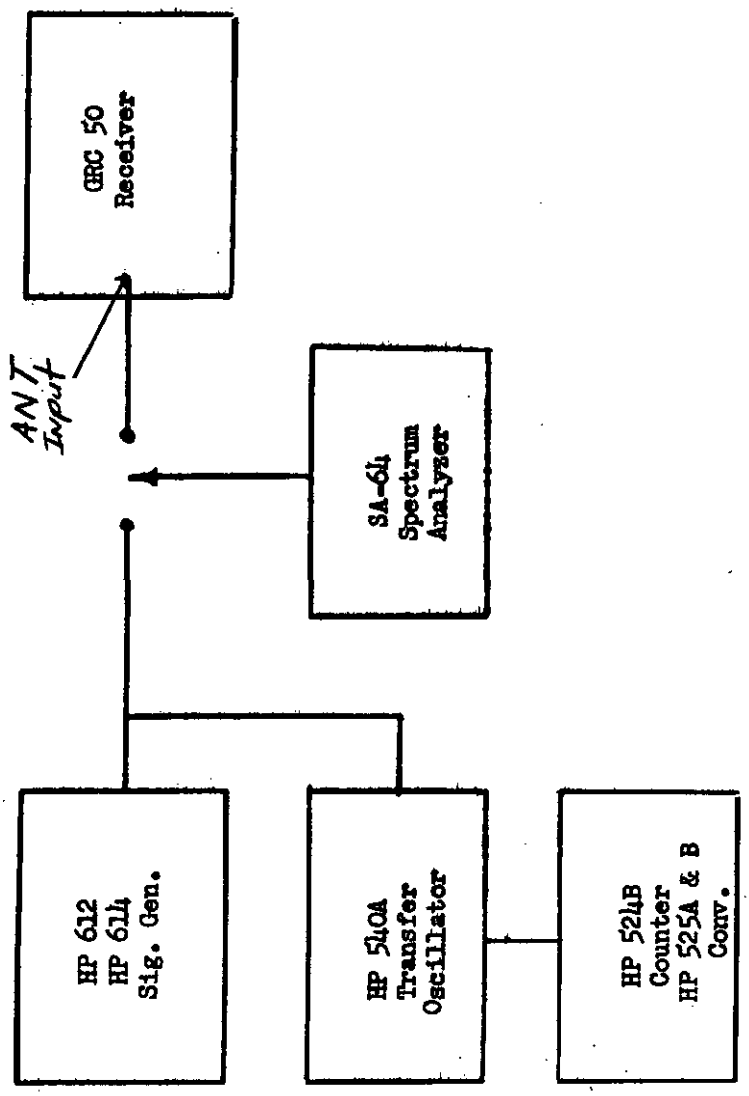


FIGURE 10

RECEIVER LOCAL OSCILLATOR RADIATION

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

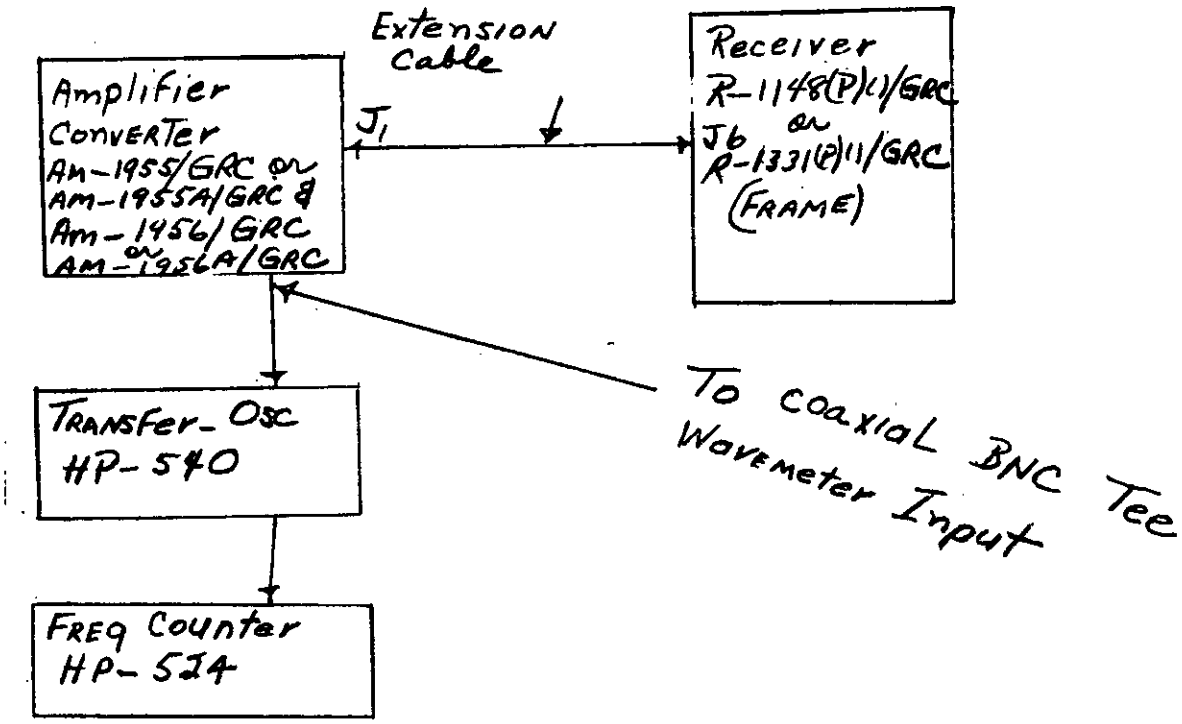


FIGURE 11

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.9 Receiver Local Oscillator Radiation Test.- The equipment shall be set up as in figure 10. Standard test conditions shall apply. (See 4.16.1).

Step 1. Tune the receiver to any desired channel in accordance with 4.16.2.2.

Step 2. Set the Signal Generator to the Receiver Local Oscillator frequency with the output level at 2500 (uv) microvolts when connected to the Spectrum Analyzer attenuator input. The attenuator must be set to at least 10 db to provide a 50 ohm input.

Step 3. Adjust the Spectrum Analyzer to the local oscillator frequency simulated by the HP614A (or the HP612A) and adjust the display so that the 2500 (uv) microvolts input produces 4 cm. of amplitude on the scope.

Step 4. Disconnect the Signal Generator and connect the receiver antenna input to the spectrum analyzer input. If local oscillator radiation can be detected it should be below the 4 cm level to meet the requirements of 3.10.2.10.

NOTE: A field intensity meter (NF-105 or equal) could be used in place of spectrum analyzer as the comparison device.

4.16.4.10 Receiver frequency settability test for Receiver, Radio R-1148(P)( )/GRC with Amplifier Converter AM-1955( )/GRC or AM-1956( )/GRC. The equipment shall be set up as in Figure 11.

Step 1. Remove the Amplifier Converter from the Receiver main frame and connect a cable extension between J-1 Converter Head and J-6, the Receiver main frame input.

Step 2. Connect a coax Tee to (3A2) J-6 WAVEMETER INPUT. Connect the cable marked A2 J-6 to one arm of the Tee and connect other arm of Tee through a coaxial cable to the counter and transfer oscillator.

Step 3. Tune the receiver to the desired rf channel in accordance with 4.16.2.2.1.

Step 4. Measure the local oscillator frequency.

Step 5. The difference between the measured local oscillator frequency and the nominal local oscillator frequency shall meet the requirements of 3.10.2.11.

Step 6. Repeat step 3 through 5 for channels 1, 250, 399, 400, 650 and 899 as applicable.

4.16.4.10.1 Receiver frequency settability. - Test for receivers with Amplifier Converter AM-1955( )/GRC or AM-1956A( )/GRC. The equipment shall be set up as in figure 11. Standard test conditions shall apply (4.16.1).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

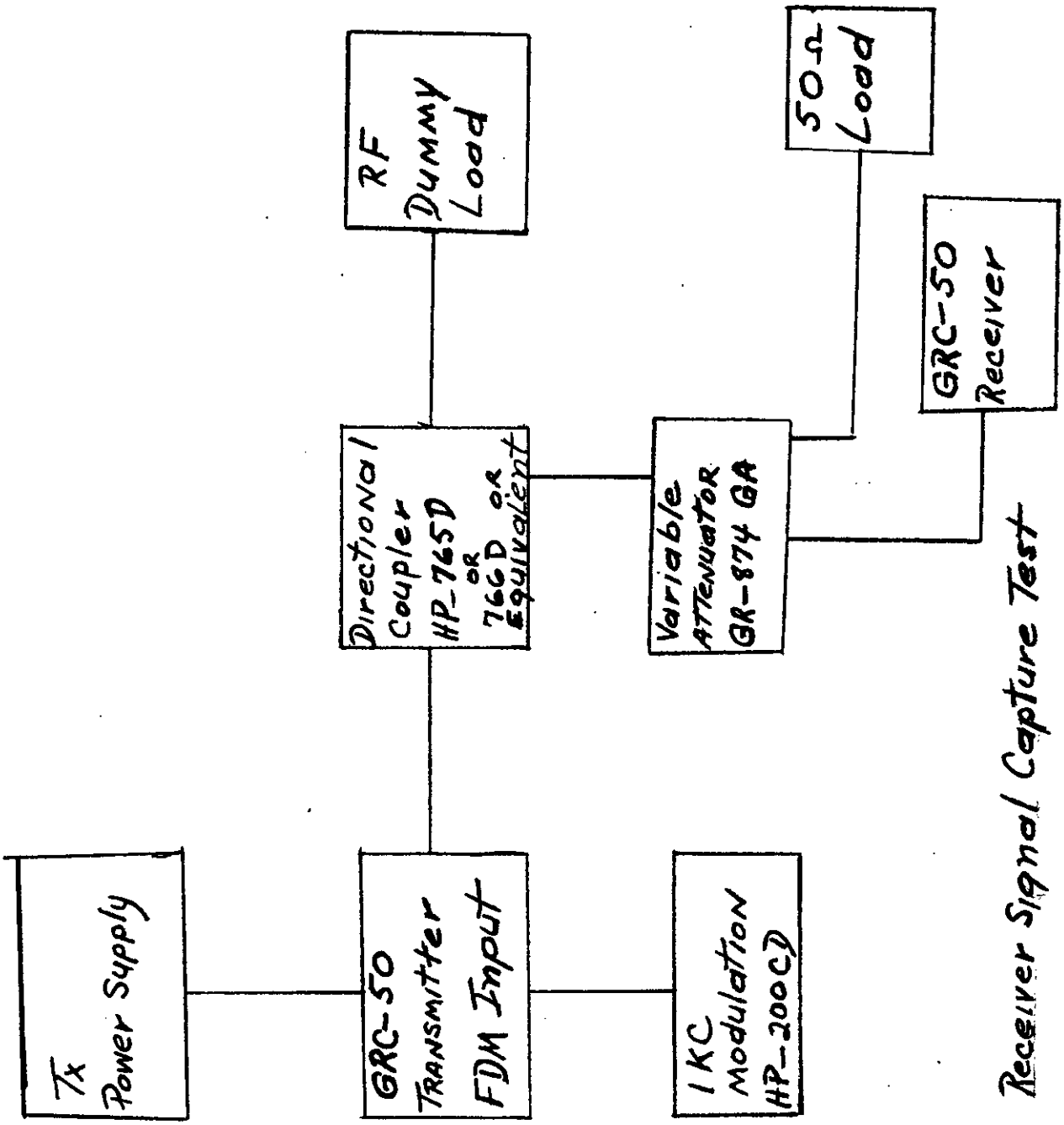


Figure 8a

Receiver Signal Capture Test

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.10.1 Contd.

Step 1. Remove the Amplifier Converter Head for the Receiver main frame and connect a cable extension between J-1 Converter Head and J-6 Receiver main frame input.

Step 2. Connect a coax Tee to (3A2) J-6 WAVEMETER INPUT. Connect the cable marked A-2 J-6 to one arm of the Tee and connect other arm of Tee through a coaxial cable to the counter and transfer oscillator.

Step 3. Tune the receiver to the rf channel in accordance with 4.16.2.2.2.

Step 4. Measure the local oscillator frequency.

Step 5. The difference between the measured local oscillator frequency and the nominal local oscillator frequency shall meet the requirements of 3.10.2.11.1.

Step 6. Repeat steps 3 through 5 for channels, 1, 250, 399, 400, 650 and 899 as applicable.

4.16.4.11 Receiver Signal Capture Test.- The equipment shall be set up as shown in figure 8A. Standard test conditions shall apply (4.16.1).

Step 1. After a 30 minute warm-up period, tune the receiver to any channel in accordance with 4.16.2.2.

Step 2. Calibrate the 60 kc discriminator-error voltage characteristic by the following procedure.

a. Disconnect the input to J2 of the second IF and apply a signal generator tuned to approximately 60 mc at 20 mv output level.

b. Disable the secondary AFC. This is accomplished in Receiver, Radio R-1331(P) ( )/GRC by depressing the momentary AFC disable switch and in Receiver,Radio R-1148(P) ( )/GRC by inserting a jumper test cable between J-7 and J-5 of the second IF unit.

c. Adjust the generator for a zero discriminator voltage as measured at J6. Count and record the input frequency.

d. Change the frequency by plus 60 kc of step c and record the voltage at J6.

e. Change the frequency by minus 60 kc of step c and record the voltage at J6.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.11 Contd.

f. Reconnect the normal input to J2 of the second IF. Restore the secondary AFC operation by releasing the AFC Disable Switch in R-1331(P) ( )/GRC or removing the jumper cable in R-1148(P) ( )/GRC.

Step 3. Apply a carrier input signal at an RF level between minus 80 dbm and minus 40 dbm at the assigned frequency to the input terminal of the receiver. Adjust the REC SIG control for a peak indication on the REC SIGNAL meter. Set the generator output for a 40 microamp reading on the REC SIGNAL meter. Count and record the input frequency.

Step 4. Measure the voltage at J6. It shall be within the limits established in steps 2d and 2e.

Step 5. Adjust the signal generator (connected to J2) for a 750 kc. error from the assigned channel.

Step 6. Repeat step 4 above.

Step 7. Lower the generator level below 125 dbm.

Step 8. Increase the RF level to minus 87 dbm.

Step 9. Repeat step 4 above. Requirements of 3.10.2.12 shall be met.

4.16.4.12 AC Power Interruption (Receiver).-

4.16.4.12.1 AC Power Interruption for Receiver R-1148(P) ( )/GRC with Amplifier Converter AM-1955( )/GRC or AM-1956( )/GRC. - The equipment shall be set up as in figure 8. Standard test conditions shall apply (4.16.1).

Step 1. After a 30 minute warm up period, tune the receiver to the desired RF channel in accordance with 4.16.2.2.1.

Step 2. Apply a carrier input signal at an RF level between minus 80 dbm and minus 40 dbm at the assigned frequency to the input terminal of the receiver. Adjust the REC SIG control for a peak indication on the REC SIGNAL meter. Set the generator output for a 40 microamp reading on the REC SIG meter. Count and record the input frequency.

Step 3. Count and record the 60 mc. frequency at J2. The algebraic difference between the error frequency on the input frequency (counter #1) and the error frequency of the 60 mc frequency (counter #2) shall be less than ±0.005 percent ± 90 kc of the assigned channel.

Step 4. Interrupt the AC power for a period of 2 seconds and turn it on again. (Note: After a period of approximately 3 minutes, proceed with step 5.)

Step 5. Repeat steps 2 and 3. Requirements of 3.10.2.13.1 shall be met.

Source: https://assist.dla.mil -- Downloaded: 20261O-26T04:14Z
Check the source to verify that this is the current version before use.

4.6.4.12.1 Contd.

Step 6. Interrupt the AC power for a period of 90 seconds and turn it on again. Repeat steps 2 and 3. Requirements of 3.10.2.13.1 shall be met.

4.16.4.12.2 AC Power Interruption for receivers with Amplifier Converter AM-1955A( )/GRC or AM-1956A( )/GRC.- The equipment shall be set up as in figure 8. Standard test conditions shall apply (4.16.1).

Step 1. After a 30 minute warm up period, tune the receiver to the desired RF Channel in accordance with 4.16.2.2.

Step 2. Calibrate the 60 kc discriminator error voltage characteristic by the following procedure:

a. Disconnect the input to J2 of the second IF and apply a signal generator tuned to approximately 60 mc. at 20 mv output level.

b. Disable the secondary AFC. This is accomplished in Receiver, R-1331(P) ( )/GRC by depressing the momentary AFC Disable switch, and in Receiver R-1148(P) ( )/GRC by inserting a jumper test cable between J7 and J5 of the second IF unit.

c. Adjust the generator for a zero discriminator voltage as measured at J6. Count and record the input frequency.

d. Change the frequency by plus 60 kc of step c and record the voltage at J6.

e. Change the frequency by minus 60 kc of step c and record the voltage at J6.

f. Reconnect the normal input to J2 of the second IF unit. Restore the secondary AFC operation by releasing the AFC Disable switch in R-1331(P) ( )/GRC or removing the jumper cable in R-1148(P) ( )/GRC.

Step 3. Apply a carrier input signal at an RF level between minus 80 dbm and minus 40 dbm at the assigned frequency to the input terminal of the receiver. Adjust the REC SIG control for a peak indication on the REC SIGNAL meter. Set the generator output for a 40 microamp reading on the REC SIGNAL meter. Count and record the input frequency.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

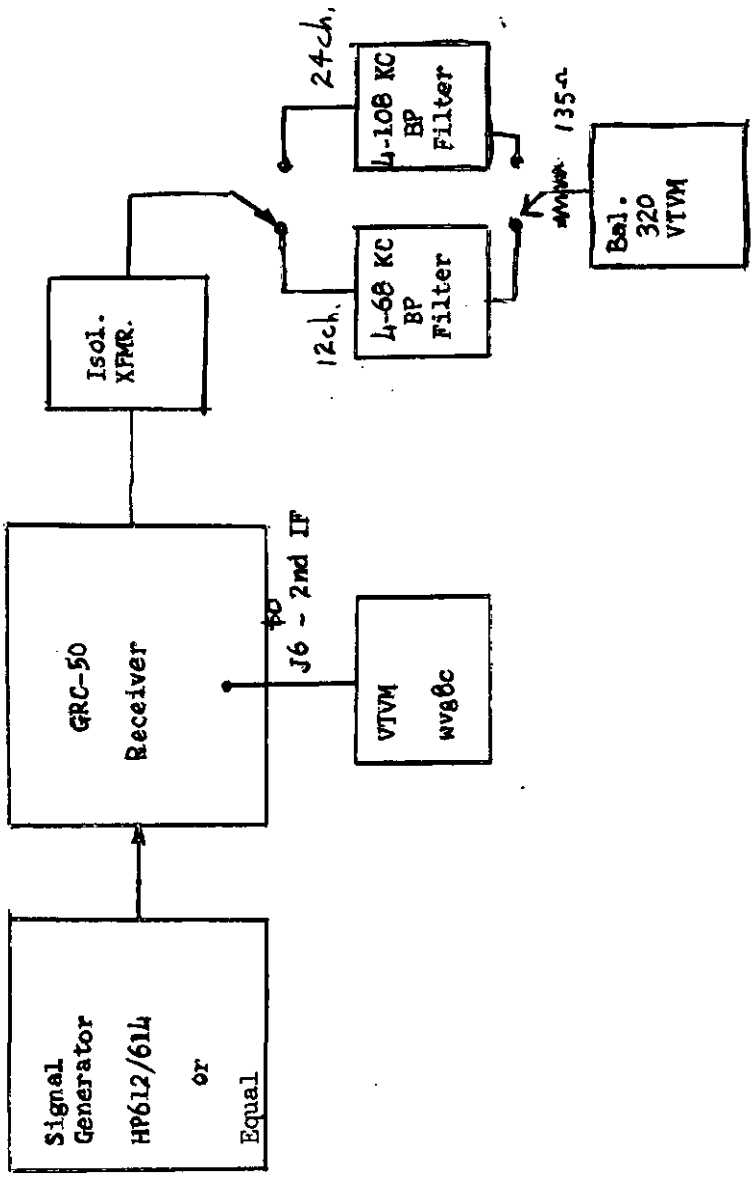


Figure 64. Receiver S+N/N Test

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.12.2 Contd

Step 4. Measure the voltage at J6. It shall be within the limits established in steps 2d and 2e.

Step 5. Interrupt the AC power for a period of 2 seconds and turn it on again. (Note: After a period of approximately 3 minutes, proceed with step 6.)

Step 6. Repeat steps 3 and 4. Requirements of 3.10.2.13.2 shall be met.

Step 7. Interrupt the AC power for a period of 90 seconds and turn it on again. Repeat steps 3 and 4. Requirements of 3.10.2.13.2 shall be met.

4.16.4.13 Receiver PCM Deviation Sensitivity Test. - The equipment shall be set up as in Figure 6. Standard test conditions shall apply (See 4.16.1).

Step 1. Tune the transmitter and receiver to the desired r.f. channel in accordance with 4.16.2.1 and 4.16.2.2 respectively.

Step 2. Adjust the r.f. input to the receiver to any level from -70 to -50 dbm.

Step 3. Calibrate the deviation meter for 375 Kc accordingly. Apply 68 Kc modulation to the PCM input terminals and adjust level to obtain 375 Kc deviation by observing RF input signal on spectrum analyzer for 2nd Bessel Null of carrier. Measure and record indication reading of same for the deviation meter.

Step 4. Apply 10 Kc modulation at PCM input terminals and adjust level to obtain same indicaton on the deviation meter as obtained in the step (3) calibration.

Step 5. Measure the PCM output level.

Step 6. The requirement of 3.10.2.14 shall be met.

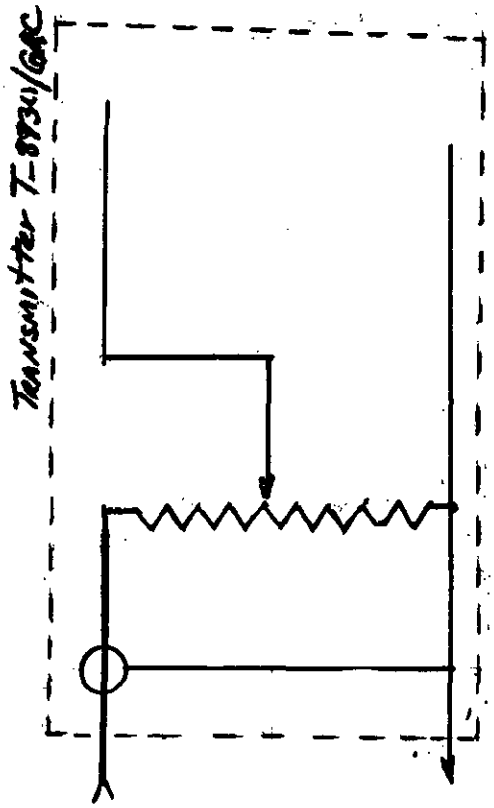
4.16.4.14 Receiver FDM S+N/N Test .- This test shall be performed on receivers only after they have met the requirements of 3.10.2.3. The equipment shall be set up as in Figure 6A. Standard test conditions shall apply (4.16.1).

Step 1. Tune the receiver to the desired channel in accordance with 4.16.2.2.

Step 2. With the AFC DISABLE button depressed or 2nd IF test points J-5 and J-7 shorted to each other, tune the signed generator for zero discriminator voltage as monitored at 2nd IF test point J-6.

Step 3. Release AFC DISABLE or remove J-5 to J-7 short as applicable.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



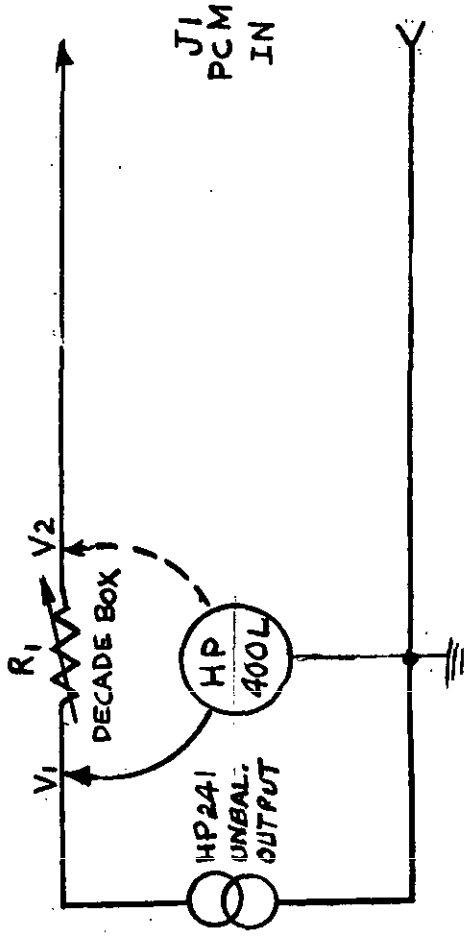


FIG 12

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

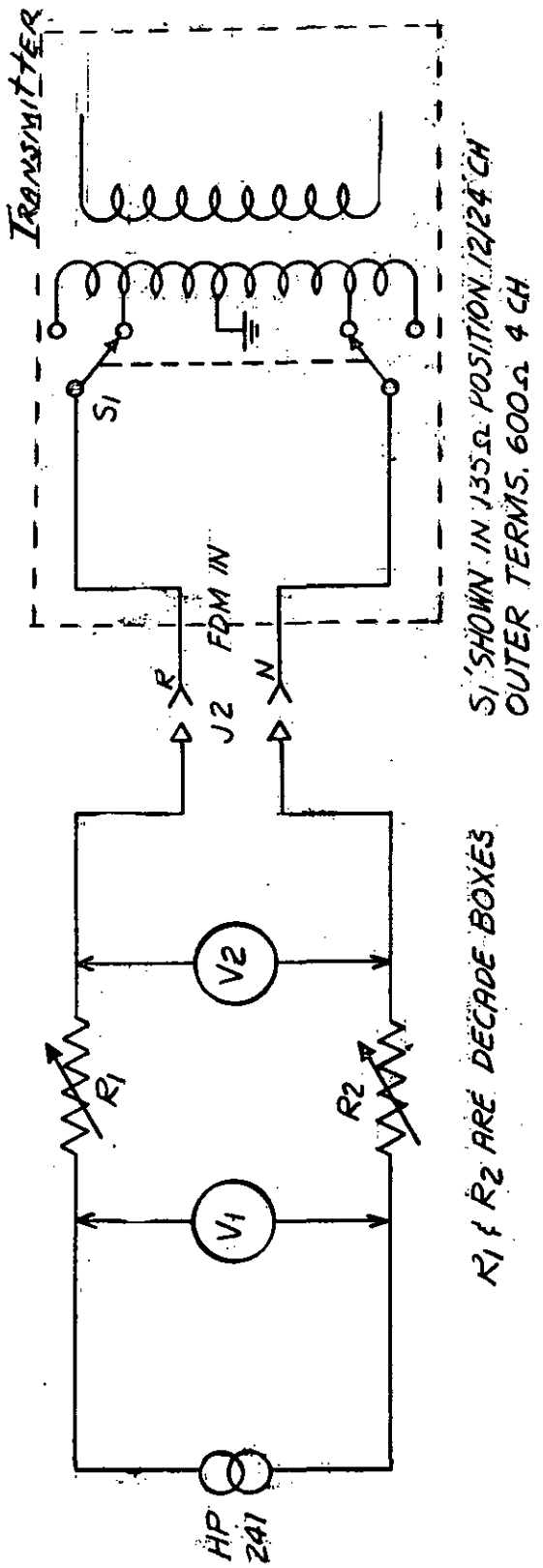


FIG. 12a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.4.14 Contd

Step 4. Make noise measurements at bandpass filter output, at r.f. input levels of -60 dbm and -80 dbm using both the 12 and 24 channel filters. Record noise levels.

Step 5. Compute the signal-plus noise to noise ratio (S+N/N). Use the output level measured in the test of 4.16.4.2.1 for the (S+N) for 24 channel and 2 db less for 12 channel. The noise value shall be the value measured in step 4 corrected for the in-band attenuation of the isolation transformers and filters used.

Step 6. The requirements of 3.10.2.15 shall be met.

4.16.5 Broadband modulation tests (see 3.10.3).-

4.16.5.1 PCM Input Impedance Test. - The equipment shall be connected to the transmitter as illustrated in figure 12.

Step 1. Set the decade box to approximately 50 ohms.

Step 2. Set the PCM attenuator to mid position.

Step 3. Adjust HP 241 oscillator output for 1 volt at 10 KC.

Step 4. Adjust the decade box so that the voltage V2 at PCM IN is exactly one-half (1/2) of generator output V1.

Step 5. Read the decade box resistance as the input impedance.

Step 6. Repeat the test at 0.3 KC and 680 KC.

Step 7. The requirements of 3.10.3.1 (a) shall be met.

4.16.5.2 FDM Input Impedance and Balance Test

(a) Input Impedance - 135 ohm terminals - The equipment shall be connected to the transmitter as illustrated in 12a.

Step 1. Set S-1 to 12/24 position on transmitter

Step 2. Set each decade box to 67 ohms.

Step 3. Set HP 241 to 0.3 KC and adjust its output to 0.74 volts at V-1.

Step 4. Adjust the decade boxes R-1 and R-2 equally and incrementally until the voltage at V-2 is exactly one-half (1/2) the voltage at V-1.

Step 5. Read the sum (R-1 + R-2) as the input impedance.

Step 6. Repeat the test at 108 KC.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

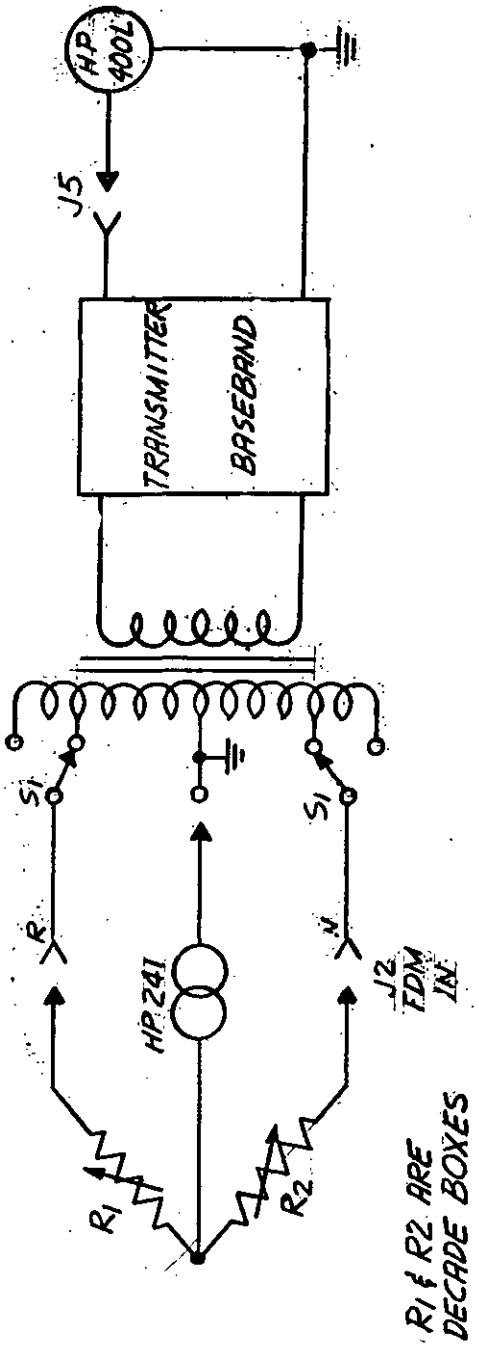


FIG. 12b

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.5.2 Contd

Step 7. The 135 ohms ± 15%, resistive, requirement of 3.10.3.1 (b) shall be met.

(b) Input Impedance -600 ohm terminals -The equipment shall be connected as in 4.16.5.2 (a).

Step 1. Set S-1 to 4 channel.

Step 2. Set each decade box to 300 ohms.

Step 3. Set HP 241 to 0.3 KC and output V-1 to 1.55 volts.

Step 4. Adjust R1 and R2 until voltage at V-2 is exactly one-half (1/2) the voltage at V-1.

Step 5. The sum (R1 + R2) is the input impedance.

Step 6. Repeat the test at 20 KC.

Step 7. The 600 ohms ± 15%, resistive, requirements of 3.10.3.1 (b) shall be met.

(c) Balance - 135 ohm terminals- The equipment shall be connected to the transmitter as illustrated in 12 b.

Step 1. Set S-1 on transmitter to 12/24 channel position.

Step 2. Set FDM attenuator to mid position.

Step 3. Set each decade box to 67 ohms.

Step 4. Adjust HP 241 for maximum output at 0.3 KC.

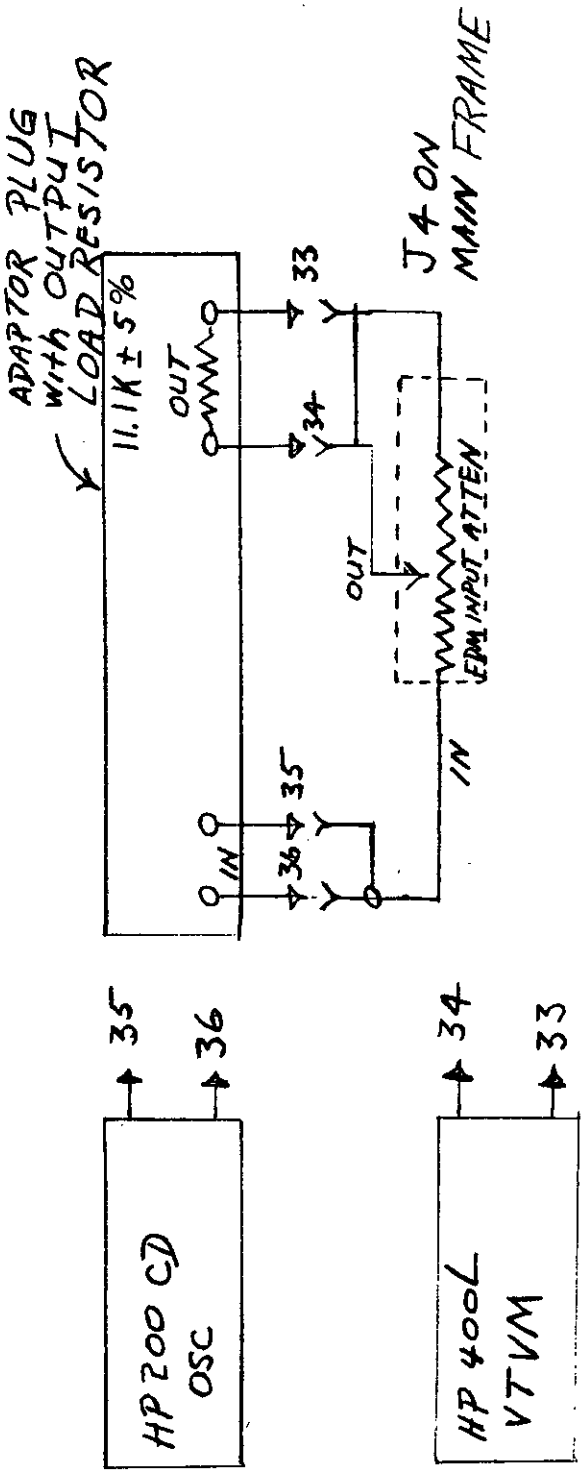
Step 5. Adjust decade box R-1 or R-2 above or below 67 ohms while observing a null in voltage at J-5.

Step 6. Calculate degree of balance by substituting value of R-1 and R-2 decade boxes in the following formula:

$$\frac{R_{smaller}}{R_{larger} - R_{smaller}} = 22.4$$

$(20 \log_{10} 22.4 = 27 \text{ db})$

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



Input and output ATTENUATOR Tests

Figure 13

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.5.2 Contd

Step 7. Repeat the test for 108 KC.

Step 8. The 135 ohm circuit balance requirements of 3.10.3.1 (b) shall be met.

(d) Balance-600 ohm terminals - The equipment shall be connected as in 4.16.5.2 (b).

Step 1. Set S-1 on transmitter to 4 channel position.

Step 2. Set each decade box to 300 ohms.

Step 3. Set HP 241 to 0.3 KC.

Step 4. Adjust decade boxes incrementally while observing a null at J-5.

Step 5. Record the values of R-1 and R-2 and compute degree of balance as in 4.16.5.2 (c) step 6.

Step 6. Repeat test at 20 KC.

Step 7. The 600 ohm circuit balance requirements of 3.10.3.1 (b) shall be met.

4.16.5.3 Input and Output Attenuator Tests.- Standard test conditions shall apply. (See 4.16.1). The equipment shall be set up as in figure 13.

(a) FDM Input Attenuator:

Step 1. No power is applied to the transmitter in this test.

Step 2. The transmitter base band (2A3) will be removed from the main frame.

Step 3. Use Adaptor plug to make connection through J-4 on main frame to the FDM Input Attenuator.

Step 4. Rotate FDM attenuator fully clockwise.

Step 5. Set HP 200 CD frequency to 1 KC and level at atten. "IN" for + 2 dbm (.975V).

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the currentversion before use.

4.16.5.3 Contd

Step 6 While maintaining input constant at +2db, vary the FDM attenuator clockwise noting that level changes 1db ± 0.2 at each step. The total attenuation at any step should be within ± 0.5 db of the indicated position.

Step 7. Repeat the test at 10 and 108 KC. The requirements of 3.10.3.2 (a) shall be met.

(b) FDM Output Attenuator:- The equipment shall be set up as in figure 13a .

Step 1. No power is applied to the Receiver in this test.

Step 2. Disconnect plug P4 which mates with J3 on 2nd IF chassis.
Disconnect plug P6 which mates with J3 on Rec. Baseband.
Disconnect plug P9 which mates with J2 on Rec. Baseband.
Remove the Receiver Baseband 3A3 from the mainframe.

Step 3. Connect plug P6 to a BNC jack with a 4.4 K resistor in series to the HP 200 CD Oscillator. Connect plug P9 to BNC jack with the 220 K output load resistor.

Step 4. Rotate FDM attenuator fully clockwise.

Step 5. Adjust HP 200 CD for +12 dbm (3.08V) 1 KC to "IN" (P6).

Step 6. Rotate FDM attenuator counter clockwise noting level change of "OUT" P9 at each step of 1 db ± 0.2 db while maintaining +12 db at "IN" P6. The total attenuation at any step should be within ± 0.5 db instead of the indicated position. (The input level must be increased to +14 db to test the last two steps).

Step 7. Repeat the test at 10, and 108 KC. The requirements of 3.10.3.2 (b) shall be met.

(c) PCM Input Attenuator: The equipment shall be set as in figure 13b.

Step 1. No power is applied to the transmitter in this test.

Step 2. Disconnect P1 on mainframe from J-2 on the baseband.

Step 3. Connect a BNC jack with a 580 ohm load resistor to P1 on mainframe.

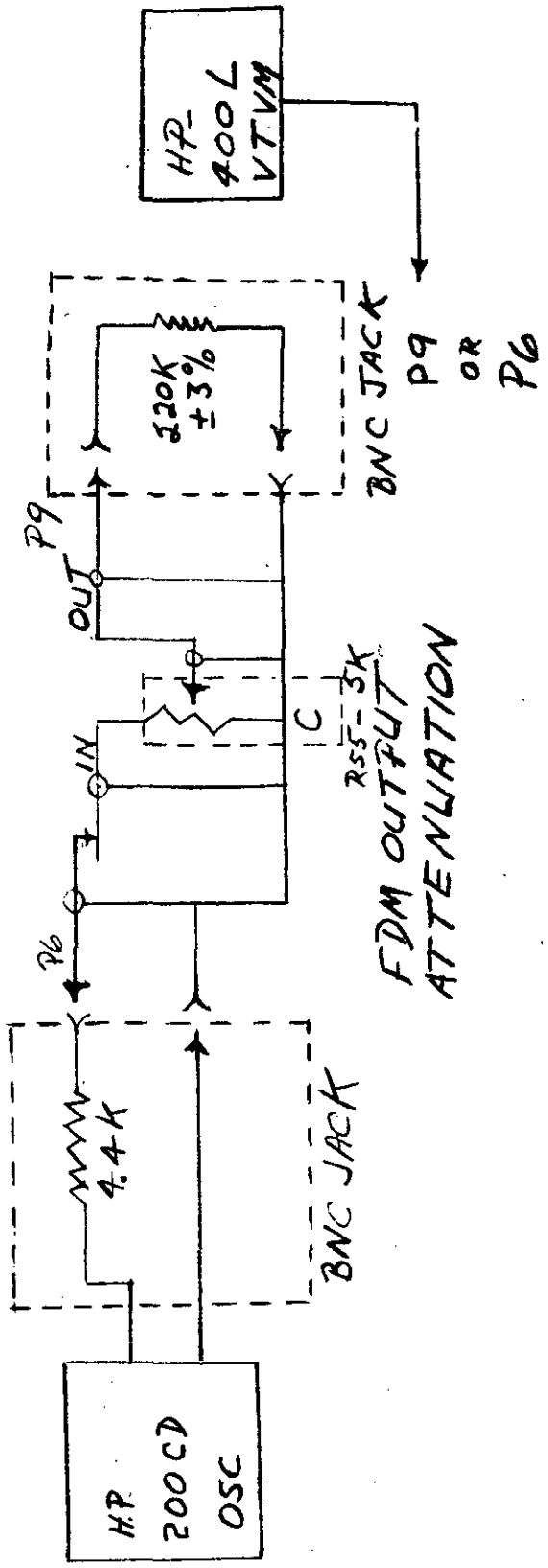
Step 4. Rotate PCM attenuator fully clockwise.

Step 5. Adjust HP 200 CD OSC to 10 KC and set level to + 2 dbm (0.975v) on HP 400 L at PCM IN and measure at P1 PCM OUT.

Step 6. Rotate PCM attenuator counter clockwise noting a 1 ± .2 db change at each step. The total attenuation at any step should be within ± 0.5 db of the indicated position. In all these checks keep the input level constant.

Step 7. The test is made at 10 KC only. The requirements of 3.10.3.2 (c) shall be met.

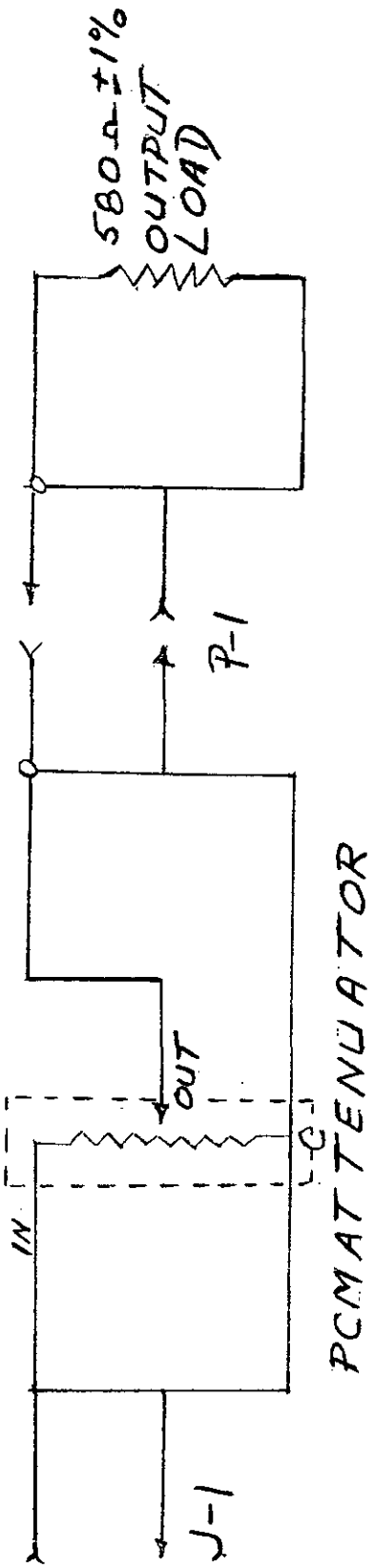
Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



FDM OUTPUT ATTENUATOR

Figure 13a

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



PCM INPUT ATTENUATOR

Figure 136

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

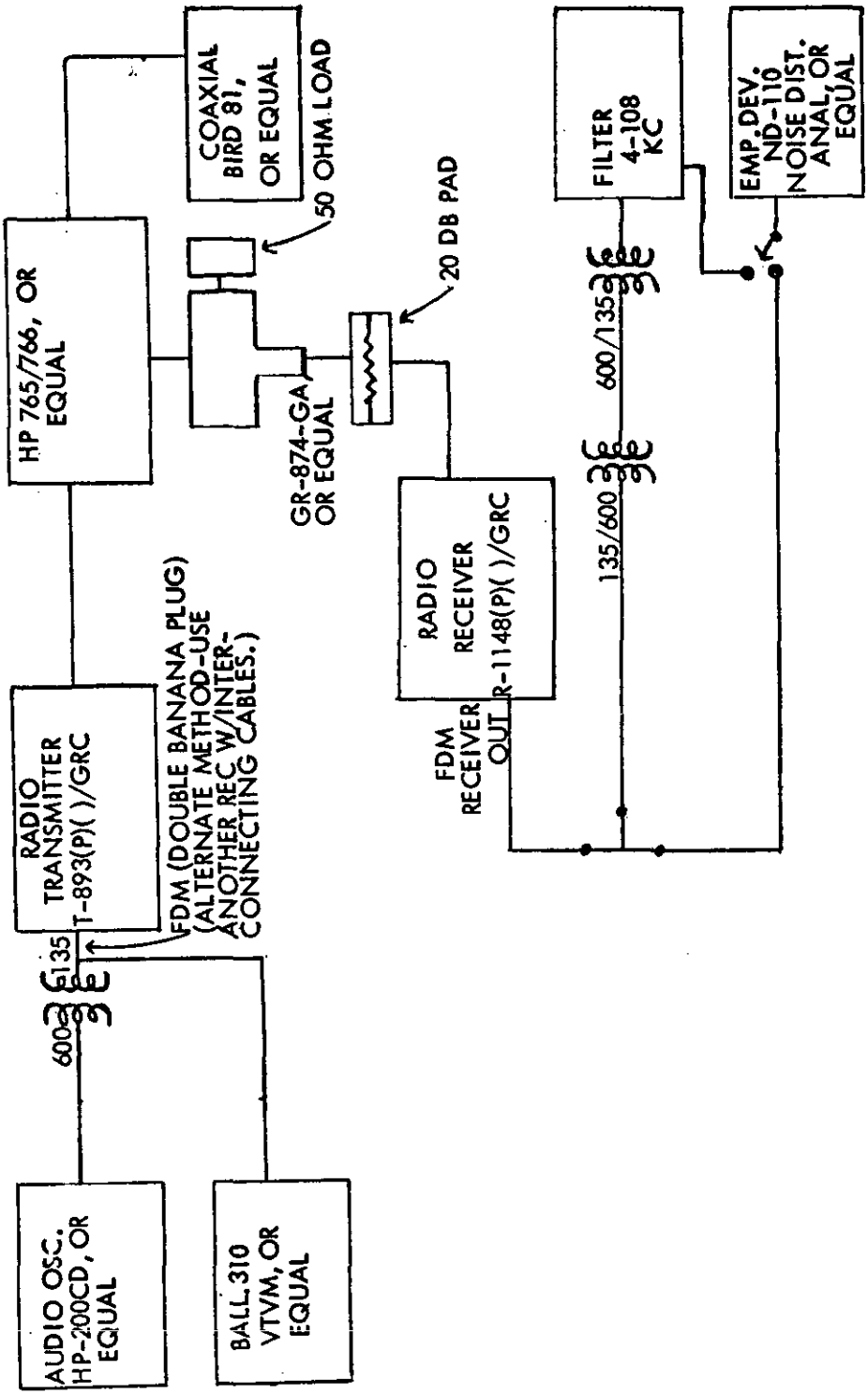


FIGURE 14

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.5.4 FDM signal-plus-noise to noise test. - The equipment shall be set up as in Figure 14. Standard test conditions shall apply. (See 4.16.1)

Step 1. Tune the transmitter and receiver to the desired channel 399 or 899 as applicable in accordance with 4.16.2.1 and 4.16.2.2, respectively. (See step 14 below).

Step 2. Calibrate receiver signal level meter for -73 and -60 dbm rf inputs with signal generator and record these values.

Step 3. Disconnect the rf generator and connect the GR-874 Attenuator, or equal, output to receiver and adjust for signal level of -73 dbm.

Step 4. Set 24/12-4 channel switch to 24/12 position.

Step 5. Adjust audio oscillator for 72 kc and to read input of 1.64 volts with vtvm(135 ohm side of transformer).

Step 6. Connect ND-110, or equal, to FDM output and set ND-110, or equal for 135 ohms balanced input on wideband. With transmitter and receiver FDM Attenuator on mid-position, reading should be +13.0 ± 2.db. If not, adjust the receiver attenuator to obtain same. The system will not be adjusted for 0 db net gain.

Step 7. Disconnect audio oscillator signal.

Step 8. Switch ND-110, or equal, to narrow band and read the slot noise as specified.

Step 9. Switch ND-110, or equal, to calibrate and turn attenuators to maximum.

Step 10. Connect the 135/600 transformers and filter between he FDM output and the ND-110, or equal, being careful to observe the terminating conditions of the FDM output and the filter, i.e., maintaining a balanced condition for the FDM output and the proper ND-110 (or equal), input conditions.

Step 11. Adjust the GR-874 attenuator (or equal) to produce the receiver level indication of -60 dbm.

Step 12. Switch ND-110, (Or equal), to wideband (-30 to -80dbm ranges) and measure noise.

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the currentversion before use.

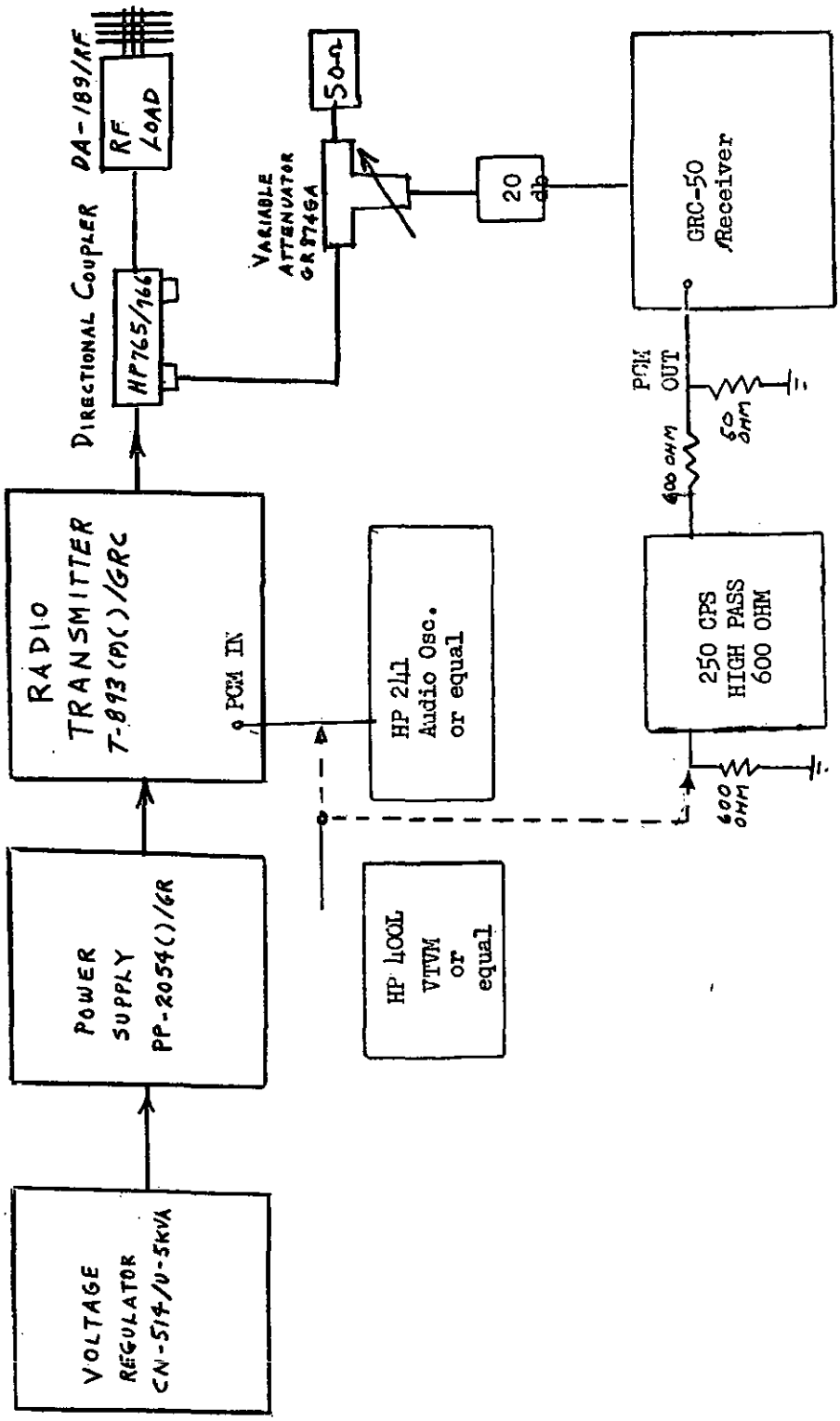


FIGURE 15

PCM S+N/N

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.5.4 Contd

Step 13. Return attenuators to maximum and input switch to calibrate.

Step 14. Repeat steps 1 through 13 for channels 1, 250, 400 and 650, as applicable in preproduction tests only. The requirements of 3.10.3.3(a) and (b) shall be met for all channels.

4.16.5.5 PCM signal-plus-noise to noise test.- The equipment shall be set up as in Figure 15. Standard test conditions shall apply. (See 4.16.1).

Step 1. Select any one channel from each band.

Step 2. Tune the transmitter and receiver to the desired rf channel in accordance with 4.16.2.1.

Step 3. Adjust the attenuator (GR-874), or equal) for -80 dbm input.

Step 4. Set the HP-200CD generator (or equal) for 10 KC with PCM input of 0.53 volts.

Step 5. Set the PCM Transmitter input attenuator to mid-position and read Receiver PCM output voltage, should be -15.5 dbm ± 2 db.

Step 6. Record the value as the signal.

Step 7. Remove generator from input and read HP-400 (or equal) PCM output voltage.

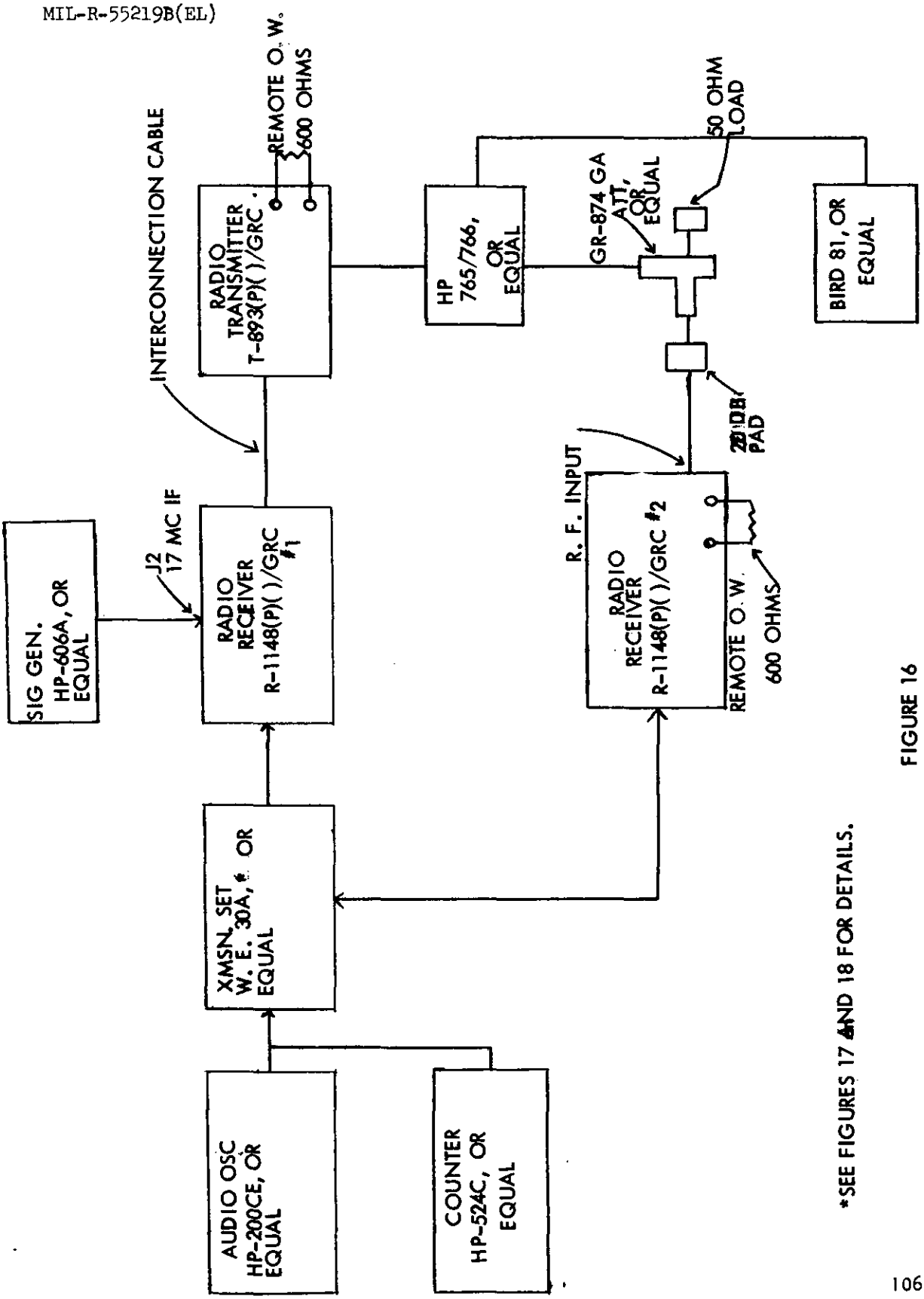
Step 8. Record the value as noise.

Step 9. The ratio of the voltage of Step 6 to the voltage in Step 8 is the (S+N)/N.

Step 10. Readjust the attenuator for receiver input level of -73 dbm.

Step 11. Read the pcm output voltage and record as noise.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



*SEE FIGURES 17 AND 18 FOR DETAILS.

FIGURE 16

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

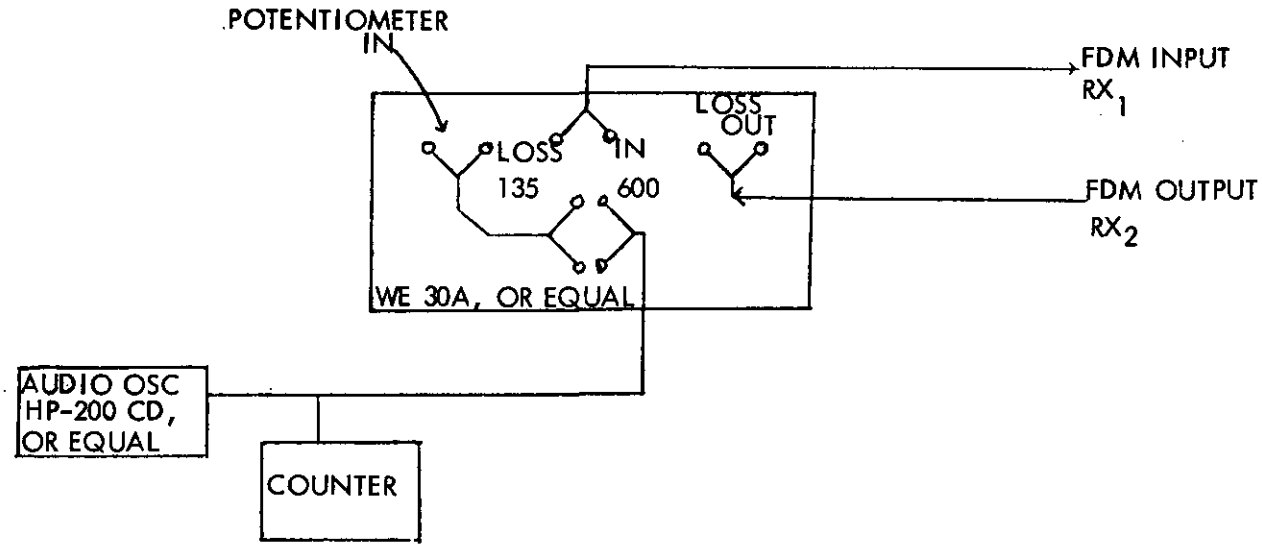


FIGURE 17. 135-OHM SET-UP.

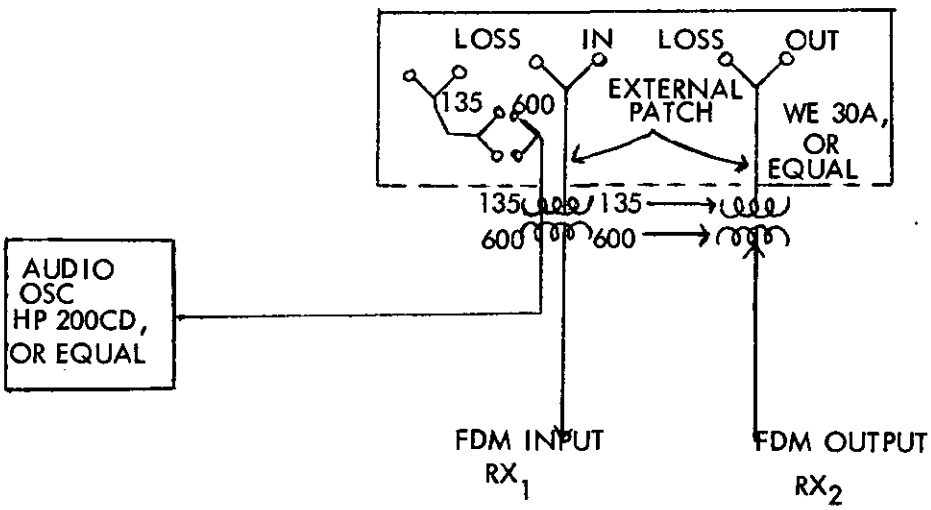


FIGURE 18. 600-OHM SET-UP.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.5.4 Contd

Step 12. The ratio of the voltage of Step 6 to the voltage in Step 11 is the (S+N)/N. The requirements of 3.10.3.4(a) and (b) shall be met.

4.16.6 Frequency response tests (see 3.10.4).

4.16.6.1 FDM frequency response test.- The equipment shall be set up as in Figure 16, and 17 or 18, as applicable. Standard test conditions shall apply. (See 4.16.1).

Step 1. Select any one channel from each band.

Step 2. Tune the transmitter and receiver to the desired rf channel in accordance with 4.16.2.1 and 4.16.2.2 respectively.

Step 3. Terminate the Remote orderwire at both the send and receive ends with 600 ohms.

Step 4. Adjust the rf input signal from the transmitter with the variable attenuator for -53 dbm.

Step 5. Adjust the HP 606 Signal Generator (or equal) to approximately 60 mc for maximum signal level on RC 1 panel meter and adjust level for a signal level of 40 ua.

Step 6. Set the 12/24-4 channel switch to 12/24 position at both receiver and transmitter. Set the FDM input and output attenuators in mid-position.

Step 7. Calibrate the WE-30(or equal) transmission set as per instruction book.

Step 8. Set the level and frequency of the audio oscillator to 0.30, 0.4, 0.7, 1, 2, 4, 7, 10, 20,40,68,100 and 108 kc respectively, using the WE-30(or equal) as per instruction book.

Step 9. Measure and record system net loss versus frequency starting at 1 kc. (If system net loss exceeds ± 1 db adjust FDM RC2 and TX attenuators for a maximum of ± 1 db each to obtain a net loss of less than ± 1 db. Adjust FDM RC2 and TX attenuators at 1 KC only.)

Step 10. Remove the 600-ohm termination from the remote ow at both send and receive ends and repeat Step 8, and 9 without adjustment.

Step 11. Replace 600-ohm termination on remote ow.

Step 12. Set the 12/24-4 channel switch to 4 channel position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

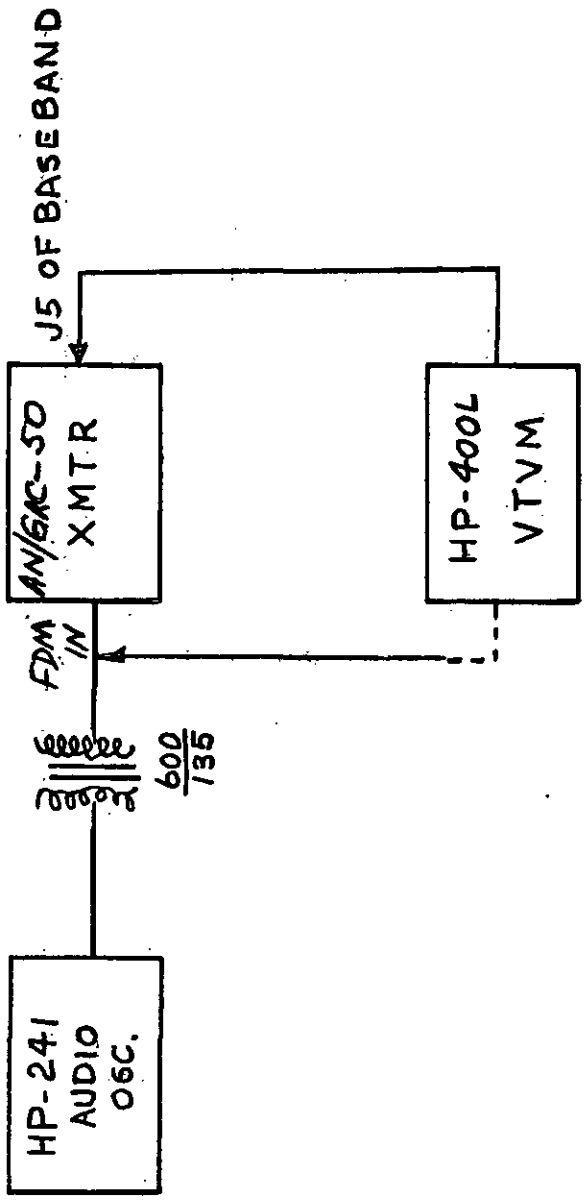
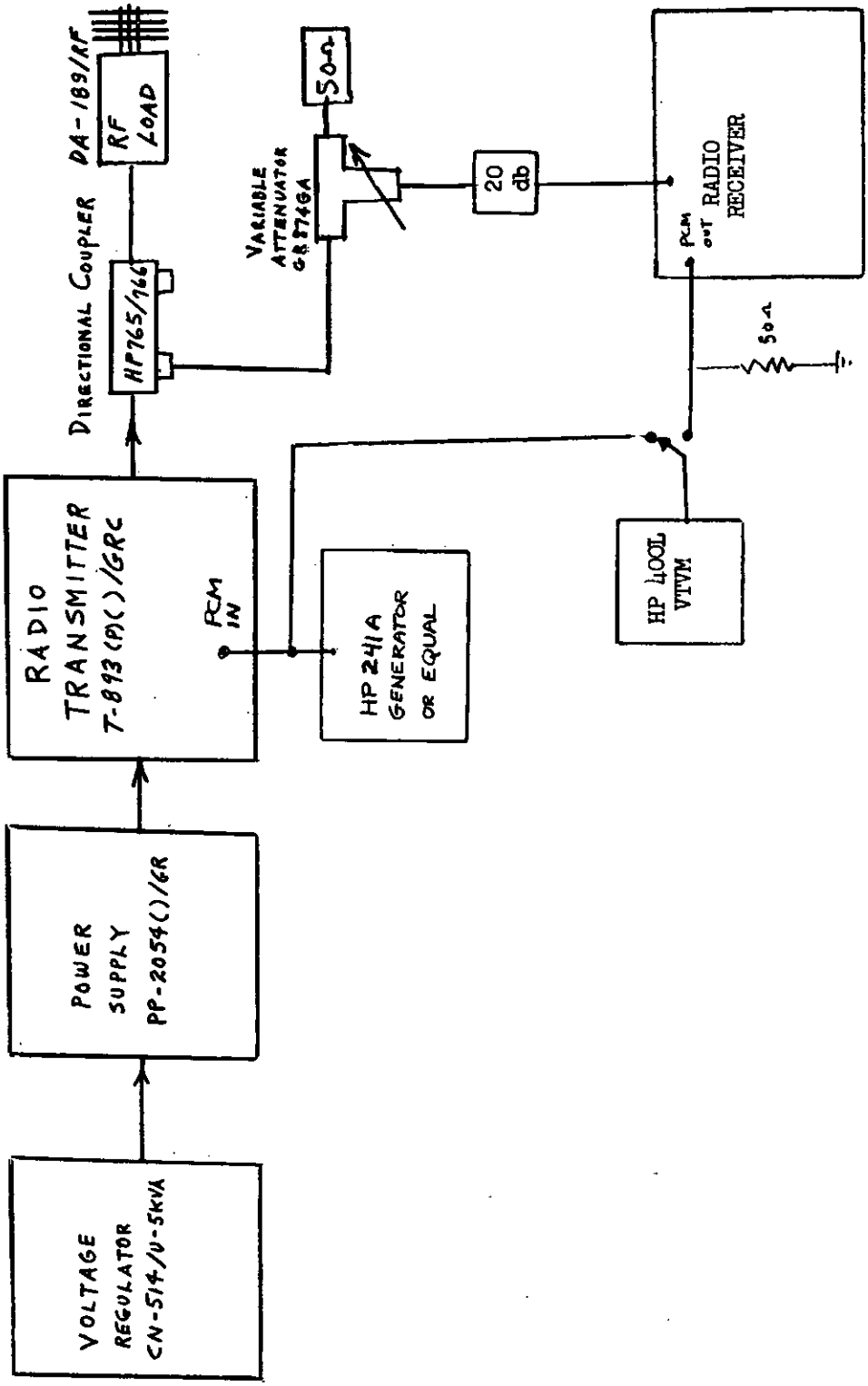


FIG 19

F.D.M. TRANSMITTER FREQUENCY RESPONSE

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



PCM FREQUENCY RESPONSE

FIGURE 20

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.6.1 Contd

Step 13. Repeat measurements taken in Step 9 for frequencies 0.30, 0.4, 0.7, 1, 2, 4, 7, 10 and 20 kc.

Step 14. Remove 600-ohm termination at remote ow and repeat Step 13 without adjustments. The requirements of 3.10.4.1 (a) and (b) shall be met.

4.16.6.2 Transmitter FDM Frequency Response Test.- The equipment shall be connected as illustrated in figure 19.

Step 1. Place S-1 on transmitter in 12/24 channel position.

Step 2. Place FDM attenuator in the mid position.

Step 3. Adjust the HP 241 generator or equal for 0.37 volts (0 dbm) at 1 KC to FDM input (as read on the HP 400L).

Step 4. Read the output on the HP 400L switched to read the output at J-5 of the transmitter baseband. This is the reference.

Step 5. Maintaining the input constant at 0.37 volts, measure the output at J-5 for the following frequencies 0.3 kc, 0.4 kc, 0.7 kc, 4 kc, 7 kc, 10 kc, 15 kc, 20 kc, 50 kc, 70 kc, 100 kc, and 108 kc.

Step 6. Place the S-1 channel switch to the 4 channel position .

Step 7. Remove the 600/135 matching transformer and connect the HP 241 directly to the FDM input terminals.

Step 8. Adjust the HP 241 generator for 0.775 volts (0 dbm) at 1 KC to the FDM input.

Step 9. Read the output on the HP 400 L switched to read the output at J-5 of the transmitter baseband. This is the reference.

Step 10. Maintaining the input constant at 0.775 volts measure the output at J-5 for the following frequencies: 0.3 kc, 0.4 kc, 0.7 kc, 4 kc, 7 kc, 10 kc, 15 kc, and 20 kc.

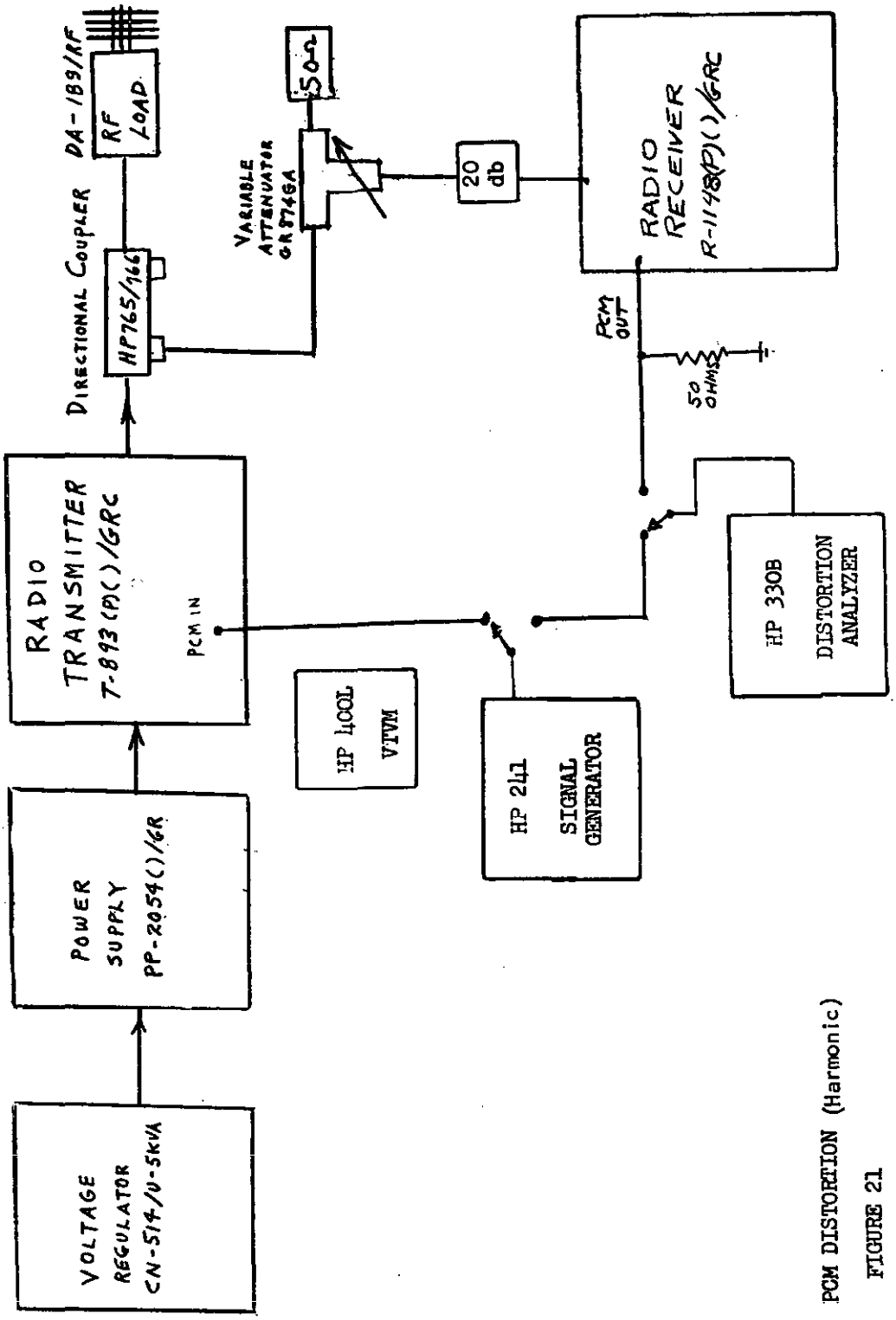
Step 11. The requirements of 3.10.4.2 shall be met.

4.16.6.3 PCM frequency response test.- The equipment shall be set up as in figure 20. Standard test conditions shall apply. (See 4.16.1).

Step 1. Select any one channel from each band.

Step 2. Tune the transmitter and receiver to the desired channel in accordance with 4.16.2.1 and 4.16.2.2 respectively.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.



PCM DISTORTION (Harmonic)

FIGURE 21

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.6.3 Contd.

Step 3. Adjust receiver input lead for -53 dbm.

Step 4. Connect HP-241 (or equal) audio generator, with 56 ohms shunted across terminals, to the PCM input and set for an output level of 0.53 volts at 10 kc.

Step 5. With the PCM transmitter input attenuator set in mid-position, measure the PCM output voltage at the receiver and vtvm (Ballantine 310, or equal). The level should be 0.265 ± 2 db. (Record this value as the zero reference level.)

Step 6. With input voltage constant at 0.53, measure and record the output voltage for 0.18, 0.5, 1.0, 2.0, 5.0, 10, 20, 50, 70, 138, 196, 240, 340, 480, and 680 kc. The requirements of 3.10.4.3 shall be met.

4.16.6.4 PCM Harmonic distortion test.- The equipment shall be set up as in figure 21. Standard test procedure shall apply. (See 4.16.1.1)

Step 1. Select any one channel from each band.

Step 2. Tune the transmitter and receiver to the desired rf channel in accordance with 4.16.2.1 and 4.16.2.2 respectively.

Step 3. RF input level to receiver shall be adjusted for -53 dbm (Record receiver REC SIGNAL indication).

Step 4. Adjust the audio oscillator for 1 kc for a level of 0.53 volts.

Step 5. Connect the distortion analyzer across the audio oscillator and measure the input distortion.

Step 6. Record as 1 kc residual.

Step 7. Repeat steps 4 through 6 with oscillator set at 10 kc and 20 kc, respectively.

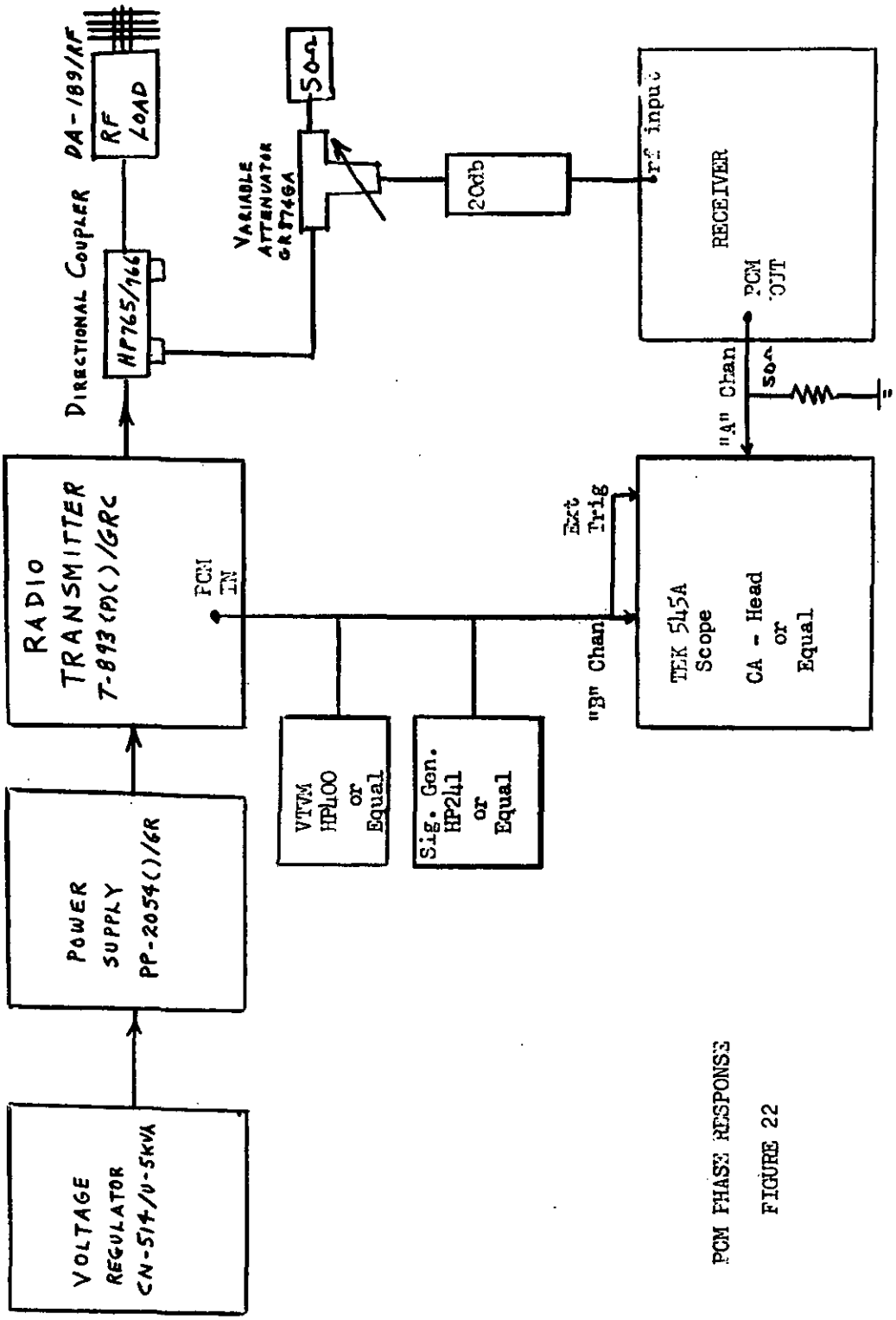
Step 8. Connect the distortion analyzer to the PCM output terminal and record the 20 kc distortion (terminate with 50 ohms).

Step 9. Change the oscillator to 10 kc at 0.53 volts and record 10 kc distortion.

Step 10. Change the oscillator to 1kc at 0.53 volts and record 1 kc distortion.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

#28



PCM PHASE RESPONSE

FIGURE 22

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.6.4 Contd.

Step 11. Subtract the residual distortion from the total distortion to obtain the net distortion. The requirements of 3.10.4.4 shall be met.

4.16.6.5 PCM PHASE RESPONSE test.- The equipment shall be set up as in figure 22. Standard test conditions shall apply. (See 4.16.1).

Step 1. Select any one channel from each band .

Step 2. Tune the transmitter and receiver for desired rf channel in accordance with 4.16.2.1 and 4.16.2.2, respectively.

Step 3. Set the HP-200CD audio generator (or equal) for an output of 0.53 v at 1 kc.

Step 4. With the oscilloscope connections shown in the test set-up, adjust the vertical amplifier gain of channel A and B independently so that each waveform fills the entire grid area. (See NOTE A).

Step 5. Adjust the centering controls so that the two waveforms fall precisely on top of one another.

Step 6. Expand the time basis axis so that the difference in time between the corresponding zero crossover points is displayed over approximately 5 to 10 centimeters (cm) on the trace.

Step 7. Adjust the relative delay so that the first crossover appears at the extreme left side of the trace on a grid intersection point.

Step 8. Record the time in microseconds (usec) between corresponding crossover points.

Step 9. Calculate the phase delay in degrees from the following: $\phi 1000 = 0.36 \times t$, where t is the delay time in microseconds.

Step 10. Record the calculated phase angle for the 1 kc.

Step 11. Reset the HP-200CD audio generator (or equal) to 300 cps at a level of 0.53 volts.

Step 12. Readjust the oscilloscope as in Steps 4 through 7.

Step 13. Record the delay in usec.

Step 14. Calculate the phase delay for the 300 cps as follows:

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

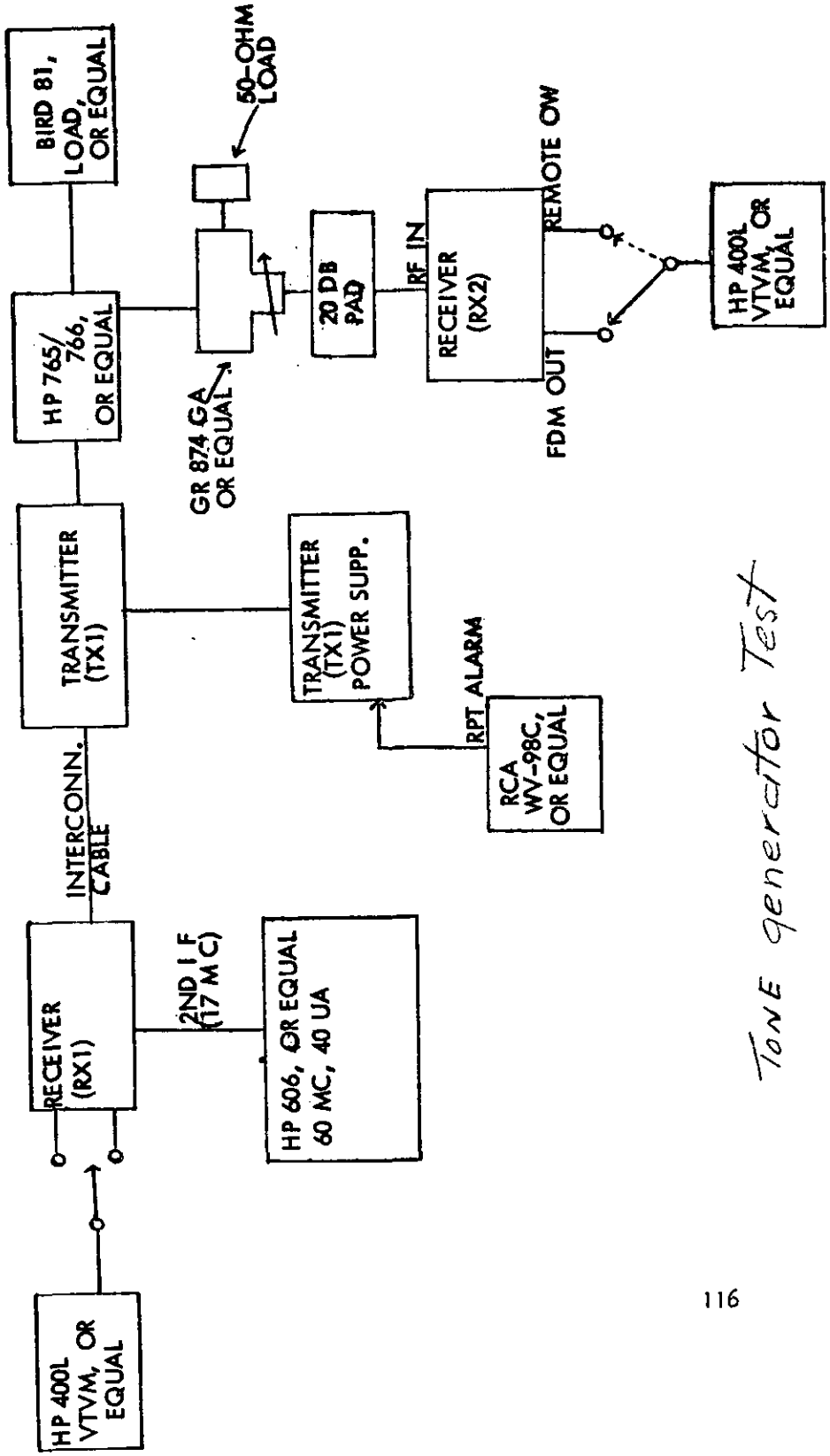


FIGURE 23

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.6.5 (Step 14 Contd)

ϕ 300 = 0.108 x t (usec).

Step 15. Calculate and record the phase difference.

ϕ300 - ϕ1000 - ϕt

The requirements of 3.10.4.5 shall be met.

NOTE A. Conditions for preliminary set-up of TEK-545 dual trace scope:

TIME A	TIME B
AC Mode Stability & Level Clockwise	Trigger Mode Automatic External Adjust Level & Stability
HORIZONTAL DISPLAY	DUAL HEAD
"A" Delayed by "B"	Chopped A & B Polarity (positive)

4.16.7 Orderwire tests (see 3.10.5).-

4.16.7.1 Remote alarm test.- The equipment shall be set up as in figure 23. Standard test conditions shall apply. (See 4.16.1)

Step 1. Select any one channel from each band.

Step 2. Tune transmitter (X1) and receiver (RX1) to the desired rf channel in accordance with 4.16.2.1 and 4.16.2.2, respectively.

Step 3. Tune receiver RX2 50 mc away, and quiet with a -53 dbm signal level.

Step 4. Turn on squelch on RX 2 and adjust until buzzer and light operates, then slowly back-off until indications go off.

Step 5. Adjust the TX1 low power alarm until buzzer and light operates, then slowly adjust until indications go off.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

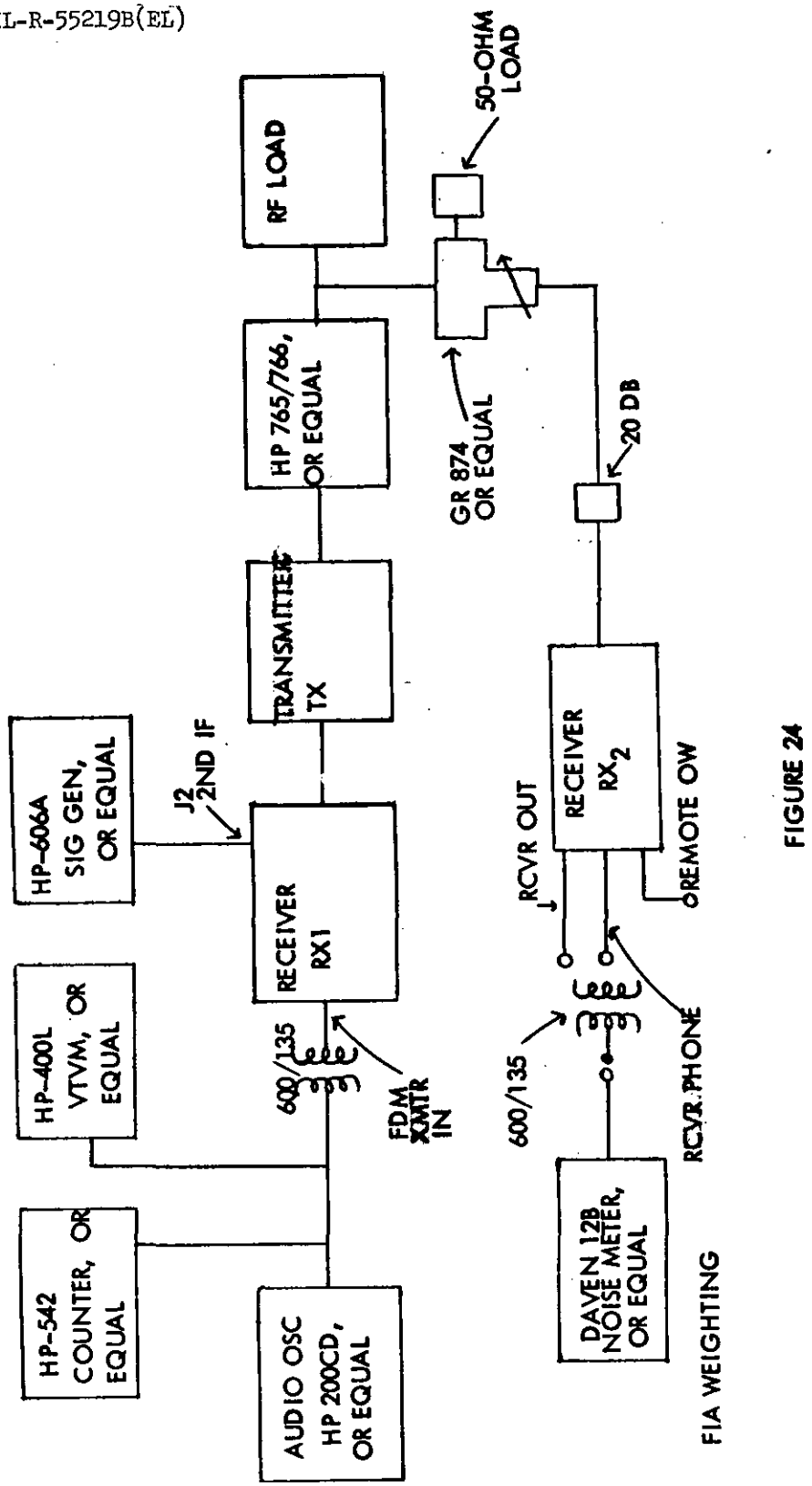


FIGURE 24

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.7.1 Contd

Step 6. Terminate the send and receiver end, phone and remote o.w. circuits, with 135/600 ohms.

Step 7. Detune the TX1 power output control and note the start of operation of the buzzer and alarm lights on TX1, RX1, and RX2.

Step 8. Retune the transmitter for power output and note that the alarm indications have stopped.

Step 9. Adjust the squelch control until buzzer and squelch of RX1 come on.

Step 10. Turn off squelch control The requirments of 3.10.5.1 shall be met.

4.16.7.2 ORDERWIRE signal-plus-noise to noise test.- The equipment shall be set up as in figure 24. Standard test conditions shall apply. (See 4.16.1).

Step 1. Select any one channel for each band.

Step 2. Tune the transmitter and receiver to the desired rf channel in accordance with 4.16.2.1and 4.16.2.2 respectively.

Step 3. Set RF input to RX 2 at -87 dbm.

Step 4. Adjust audio oscillator for 1 kc at a level of 0 dbm (0.775v).

Step 5. Set the 24/12-4 channel switch to 12/24 channel position.

Step 6. Calibrate the Daven 12B, or equal, as per instruction book.

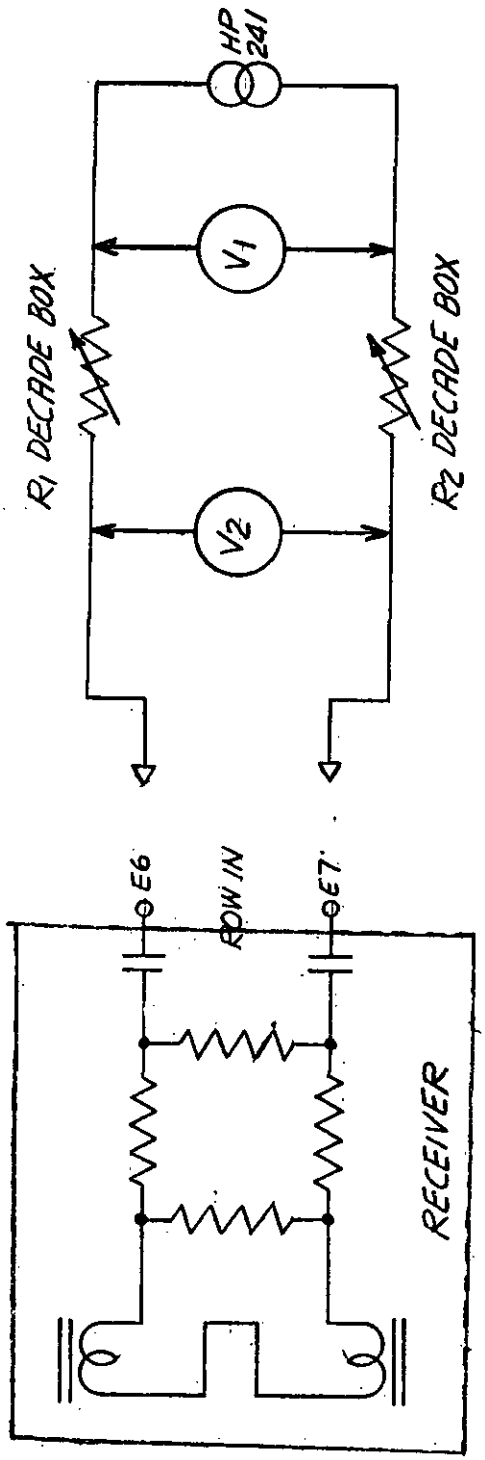
Step 7. Record the signal outputs of FDM, receiver phone, and remote orderwire.

Step 8. Remove the audio oscillator signal.

Step 9. Record the noise outputs of FDM, receiver phone, and remote O.W.

Step 10. The difference between the readings taken in Step 8 and Step 9 is the (S + N)/N ratio. The requirements of 3.10.5.2 shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



REMOTE ORDERWIRE (2W) IMPEDANCE TEST
Figure 25

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.7.3 Remote Orderwire (2W) Impedance Test. - The equipment shall be connected to the Receiver as illustrated in Figure 25.

Step 1. Set R1 and R2 to 300 ohms each.

Step 2. Adjust generator output (V1) at 1 KC to 1.55 V (+6 dbm).

Step 3. Adjust R1 and R2 incrementally and equally until the V2 voltage across ROW input is exactly one half the generator output voltage V1.

Step 4. The sum of resistances R1 + R2 is the input impedance.

Step 5. Repeat the test at 300 and 1800 cps.

Step 6. The requirements of 3.10.5.3 shall be met.

4.16.7.4 Peak Limiting Test. -The equipment shall be set up as in figure 26.

Step 1. Tune the transmitter to the desired RF channel in accordance with 4.16.2.1.

Step 2. Quiet the receiver as follows:

a. Tune the generator for 60 mc. and apply into V2 of the second IF (3 A5).

b. Adjust the generator for an output level such that a 40 microamp reading is obtained on the REC SIGNAL meter with the meter selector switch in the REC SIG level position.

Step 3. Set the HP 241 to 1 kc, and apply this signal at a level of 0.078 volts (minus 10 dbm) to the mike input.

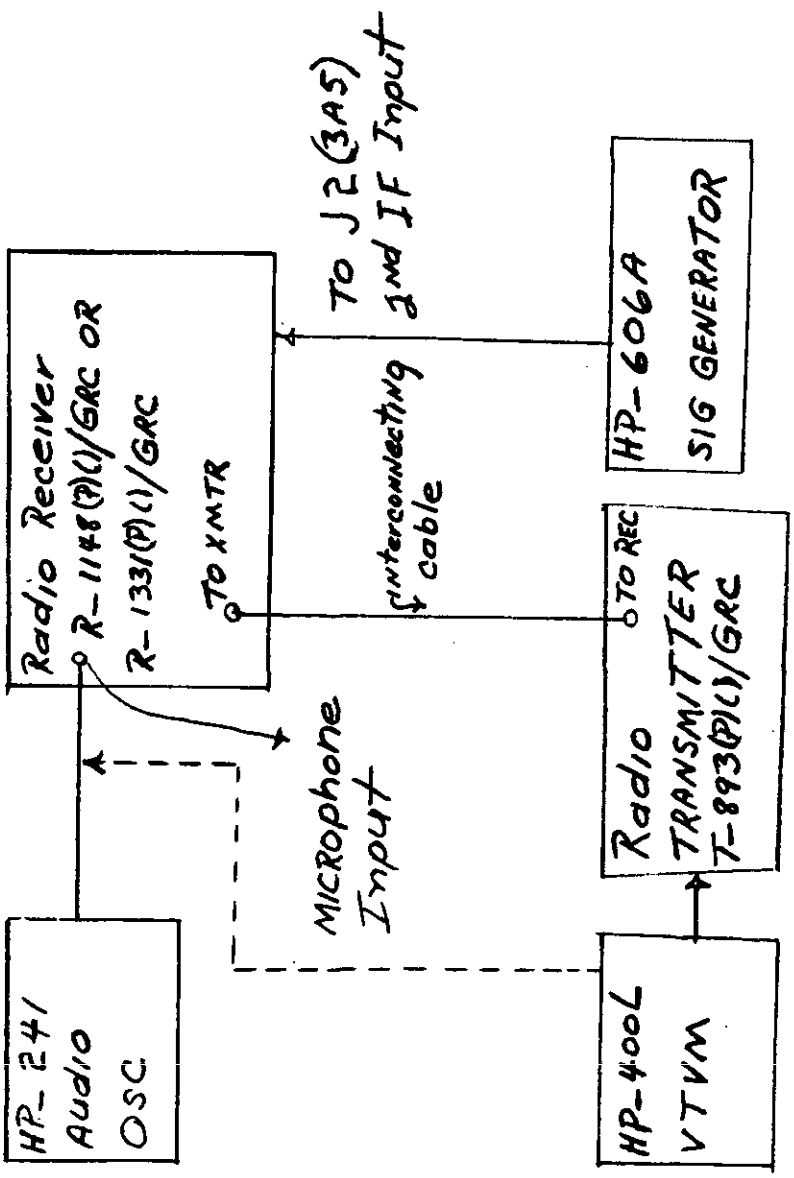
Step 4. Measure the level at J5 of the transmitter baseband chassis (2a3). This is the reference level.

Step 5. Repeat steps 3 and 4 at an input level of 0.245(0dbm). The level at J5 shall be from 9.6 db to 10 db of the referenced level reading.

Step 6. Repeat steps 3 and 4 at an input level of 1.38 volts (+ 15 dbm). The levels at J5 shall be from 12 dbm to 18 db of the referenced level reading.

Step 7. Requirements of 3.10.5.4 shall be met.

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:14Z
Check the source to verify that this is the current version before use.



PEAK LIMITING TEST

FIGURE 26

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.7.5 Orderwire frequency response test.- The equipment shall be set up as in figure 27. Standard test conditions shall apply.

Step 1. Select any one channel from each band.

Step 2. Tune the transmitter and receiver to the desired rf channel in accordance with 4.16.2.1 and 4.16.2.2, respectively.

Step 3. RF Input to RX 2 at -53 dbm.

Step 4. Terminate send end and receive end with the handset, a TA-312/PR, or equivalent.

Step 5. Terminate receive and remote ow with 600 ohms.

Step 6. Adjust the input signal from the audio generator to 0 dbm (0.775v) at 1 kc.

Step 7. Read and record output at the receive and remote ow for 0 db reference level.

Step 8. Repeat step 6 for each of the following frequencies: 300, 500, 1000, 1,500 and 1,800, cps (Record outputs noted at the receive end remote ow.)

Step 9. Remove the 600 -ohm termination from receiver remote ow.

Step 10. Repeat Steps 6 through 8.

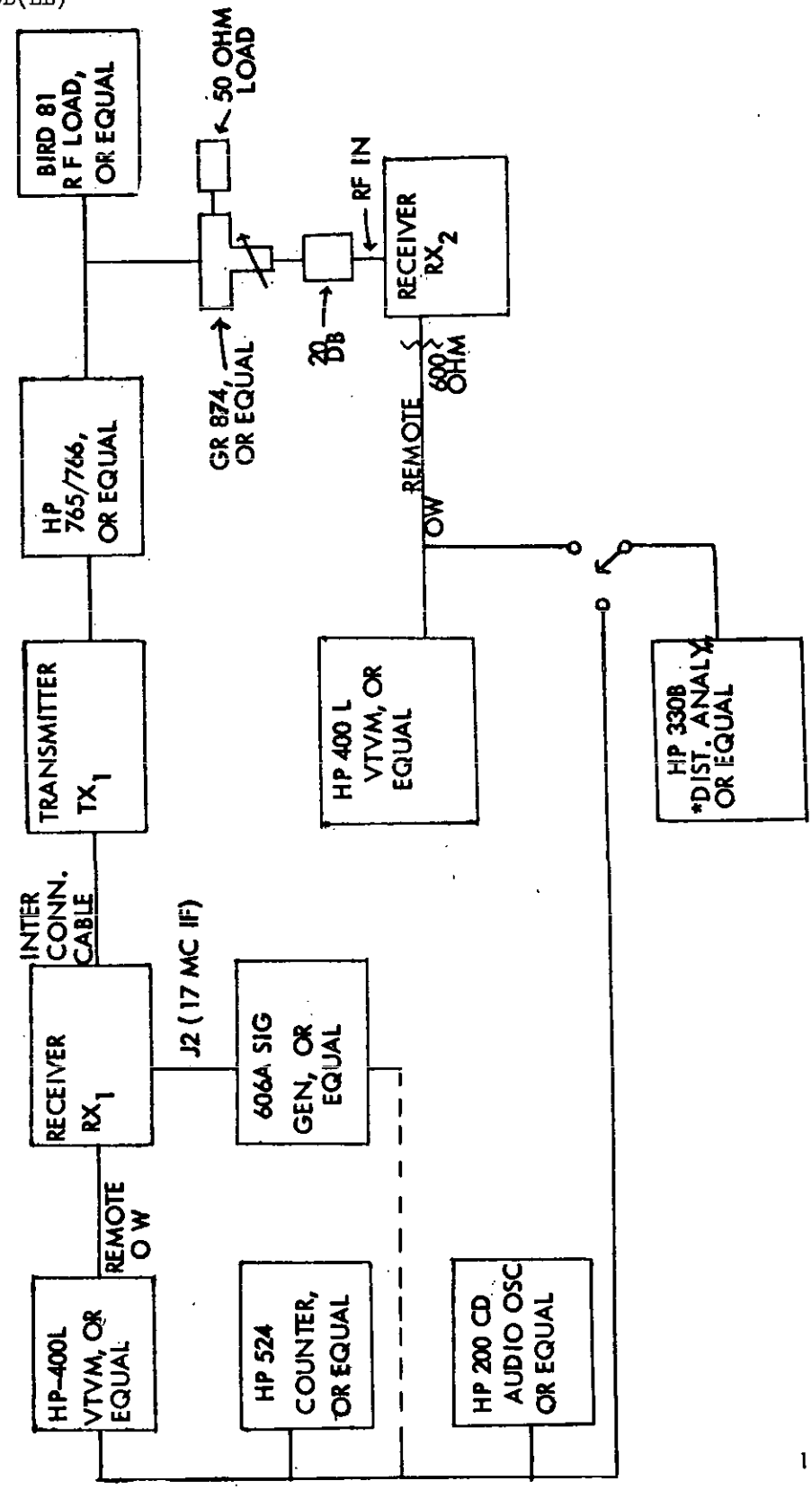
Step 11. Repeat above test for the local orderwire. The requirements of 3.10.5.5(a) and (b) shall be met.

4.16.7.6 Orderwire Distortion test.- The equipment shall be set up as in figure 27. Standard test conditions shall apply. (See 4.16.1)

Step 1. Select any one channel from each band.

Step 2. Tune the transmitter and receiver to the desired RF channel in accordance with 4.16.2.1 and 4.16.2.2, respectively.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.



*NOTE: GROUND DISTORTION ANALYZER CASE TO RX2.

FIGURE 27

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.7.6 Contd.

Step 3. Connect the distortion analyzer across the audio generator.

Step 4. Measure and record the residual distortion at 300, 600, and 900 cps for a 0 dbm input.

Step 5. Connect the distortion analyzer across the receive end remote ow terminal (600 ohms terminated).

Step 6. Measure and record the 300, 600, and 900 cps distortion for an 0 dbm input.

Step 7. Subtract the residual distortion of Step 4 from Step 6. The requirements of 3.10.5.6 shall be met.

4.16.7.7 PCM Combiner Input Impedance Test.

Step 1. The equipment will be connected to the transmitter as illustrated in figure 28. Standard test conditions shall apply.

Step 2. Set the Decade Box resistance to approximately 600 ohms.

Step 3. Adjust HP 241 Osc output (V1) for 1.55 volts (+6 dbm) at 1 KC.

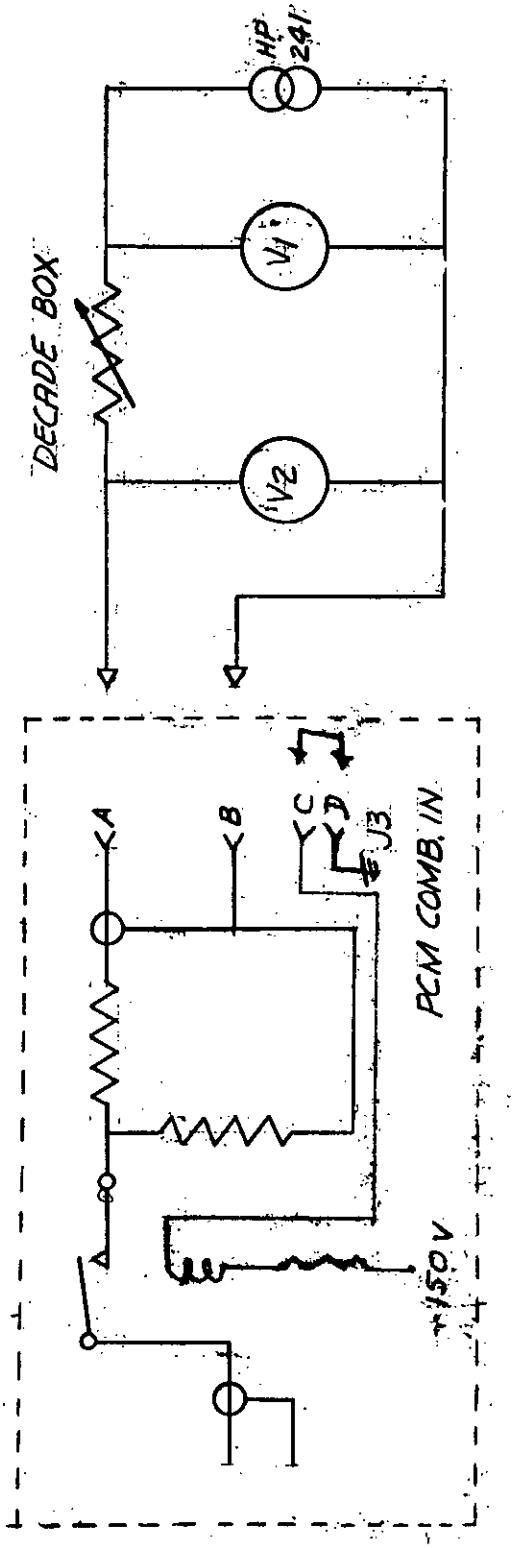
Step 4. Adjust the Decade Box until the voltage V-2 at the J3 input is half of the voltage at V-1.

Step 5. Read the Decade Box resistance as the input impedance.

Step 6. Repeat the test at 0.3 and 1.8 KC.

Step 7. The requirements of 3.10.5.7 shall be met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.



PCM Combiner Input Impedance Test
Fig. 28

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

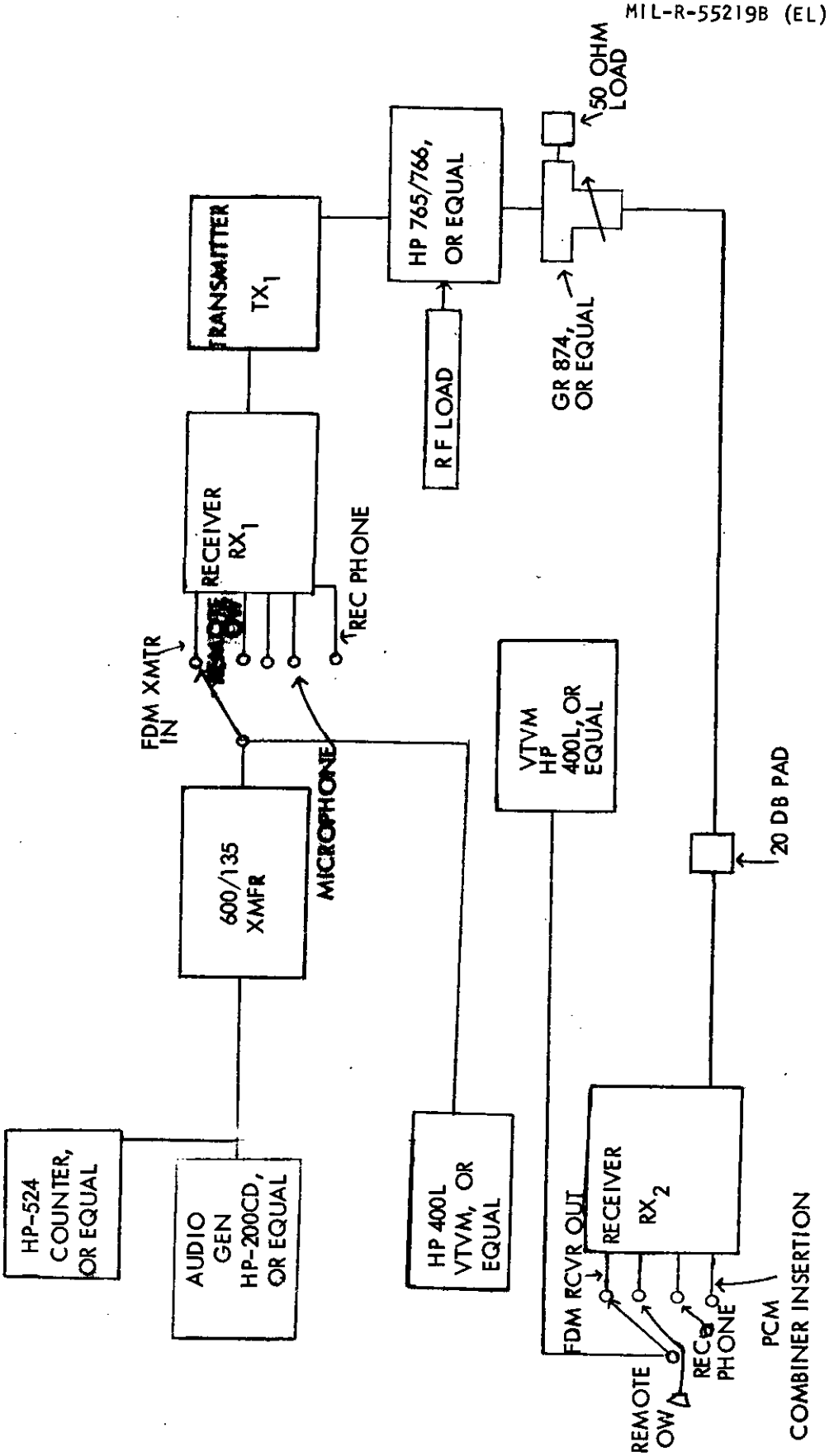


FIGURE 2a

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.8 System tests (see 3.10.6)-

4.16.8.1 Sidetone and Net Loss Test - The equipment shall be set up as in figure 29. Standard test conditions shall apply. (See 4.16.1)

Step 1. Select any one channel from each band.

Step 2. Tune the transmitter and receivers to the desired rf channel in accordance with 4.16.2.1 and 4.16.2.2, respectively.

Step 3. Adjust receiver rf input signal to -53 dbm.

Step 4. Terminate inputs and outputs on both send and receive ends.

Step 5. Switch 12/24-4 channel switches to 12/24 position.

Step 6. Connect a 600/135 ohm matching transformer between the HP-200CD (or equal) audio generator and the FDM input.

Step 7. Set the audio generator (HP-200, or equal) to 1 KC with on 0 dbm output (0.37 volts) measured on the 135 ohm side.

Step 8. Measure and record the outputs to:

(a) Remote ow - send and receive end.

(b) Receive phone - send and receive end.

(c) FDM output - receive end (135 ohms).

Step 9. Remove the 600/135 ohm matching transformer and connect the audio generator directly to the send end remote ow terminals.

Step 10. Set the level at 1 kc to 0 dbm (0.775v).

Step 11. Measure and record the outputs to:

(a) FDM Output send and receive end (135 ohm).

(b) Receive phone send and receive end.

(c) Remote ow receive end.

Step 12. Remove the audio generator from the ow and connect into the microphone input.

Step 13. Set the audio generator for 1 kc at 0.245 volts across the microphone input.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

Step 14. Measure and record the outputs to:

(a) FDM Output-send and receive end.

(b) Remote ow-send and receive end.

(c) Receive phone-send end.

(d) Receive phone-receive end.

Step 15. Disconnect the audio generator from the microphone input and connect it to the receive end (RC2) FCM combiner input directly.

Step 16. Set the level to -4 dbm (0.49v).

Step 17. Measure and record the outputs to:

(a) FDM Output receive end.

(b) Remote ow receive end.

(c) Receive phone receive end.

The requirements of 3.10.6.1 shall be met.

4.16.8.2 Level Adjust and Metering - The equipment shall set up as in Figure 30. Standard test conditions shall apply. (See 4.16.1.)

Step 1. Tune the transmitter and receiver to the desired rf channel in accordance with 4.16.2.1 and 4.16.2.2, respectively. Observe and record the following indications when tuning the transmitter: a) Osc. b) Main Tune, c) Mixer, d) Amp, e) Power Out, f) Out Frequency. Observe and record the following level indications when tuning the receiver: a) Osc., b) AFC Level (when applicable).

Step 2. Adjust the receiver rf input to -80 dbm and record the receiver signal level indication in position REC SIG. Shall be at least 5 ua.

Step 3. Adjust the receiver rf input to -53 dbm.

Step 4. Disconnect the interconnecting cable between the 1st and 2nd if (AJ52) and connect a Marconi Deviation Meter, or equal, to the 1st if cable. (Use low level input, Range 2).

Step 5. Set 12/24-4 channel switches to 12/24 position and terminate all outputs.

68 KC Metering

Step 6. Apply a 68 kc tone to the FDM input terminals (RX1) and adjust the level for a 25 ua meter deflection of the Tx multimeter in position "68 KC IN."

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:14Z
Check the source to verify that this is the current version before use.

Step 7. Measure the input level and record. Shall be 0 dbm ± 2 db (.29 - .465 volts).

Step 8. Adjust the 68 kc input for a 25 ua meter deflection of the Tx multimeter in position "68 kc MOD."

Step 9. Measure the deviation - shall be 84 kc ± 2 db (67 - 106 kc).

Step 10. Adjust the 68 kc input for a 25 ua meter deflection of the R x 2 multimeter in position "68 kc OUT."

Step 11. Measure the FDM output level at R x 2. Shall be 0 dbm ± 2 db (.29 - .465 volts).

1 KC Metering

Step 12. Apply a 1 kc tone to the FDM input terminals (Rx1) and adjust for a 25 ua meter deflection of the Tx multimeter in position "1 KC IN."

Step 13. Measure the input level and record. Shall be 0 dbm ± 2 db (.29 - .465 volts).

Step 14. Adjust the 1 kc input for a 25 ua meter deflection of the Tx multimeter in position "1 kc MOD."

Step 15. Measure the deviation. Shall be 59 kc ± 2 db (47 - 74 kc).

Step 16. Adjust the 1 kc input for a 25 ua meter deflection of the Rx2 multimeter in position "1 kc OUT."

Step 17. Measure and record the FDM output level at Rx2. Shall be 0 dbm ± 2 db (.29 - .465 volts).

Step 18. Remove the input signal and turn on the TEST TONE switch of Rx1. Adjust test tone level for a 25 ua meter deflection in position TEST TONE CAL.

Step 19. Measure and record the deviation. Shall be 59 kc ± 2 db (.47 - 74 kc).

PCM Metering

Step 20. Turn off the test tone and apply 10 kc to the PCM input (Tx). Adjust the level for a 25 ua meter indication on position "PCM IN."

Step 21. Measure and record level at PCM IN. Shall be .53 volts ± 2 db (.42 - 67 volts).

Step 22. Adjust the level for a 25 ua meter indication at Rx2 on position "PCM OUT."

Step 23. Measure and record level at PCM OUT. (Loaded with 50 ohms). Shall be .265 volts ± 2 db (.21 - .335 volts).

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

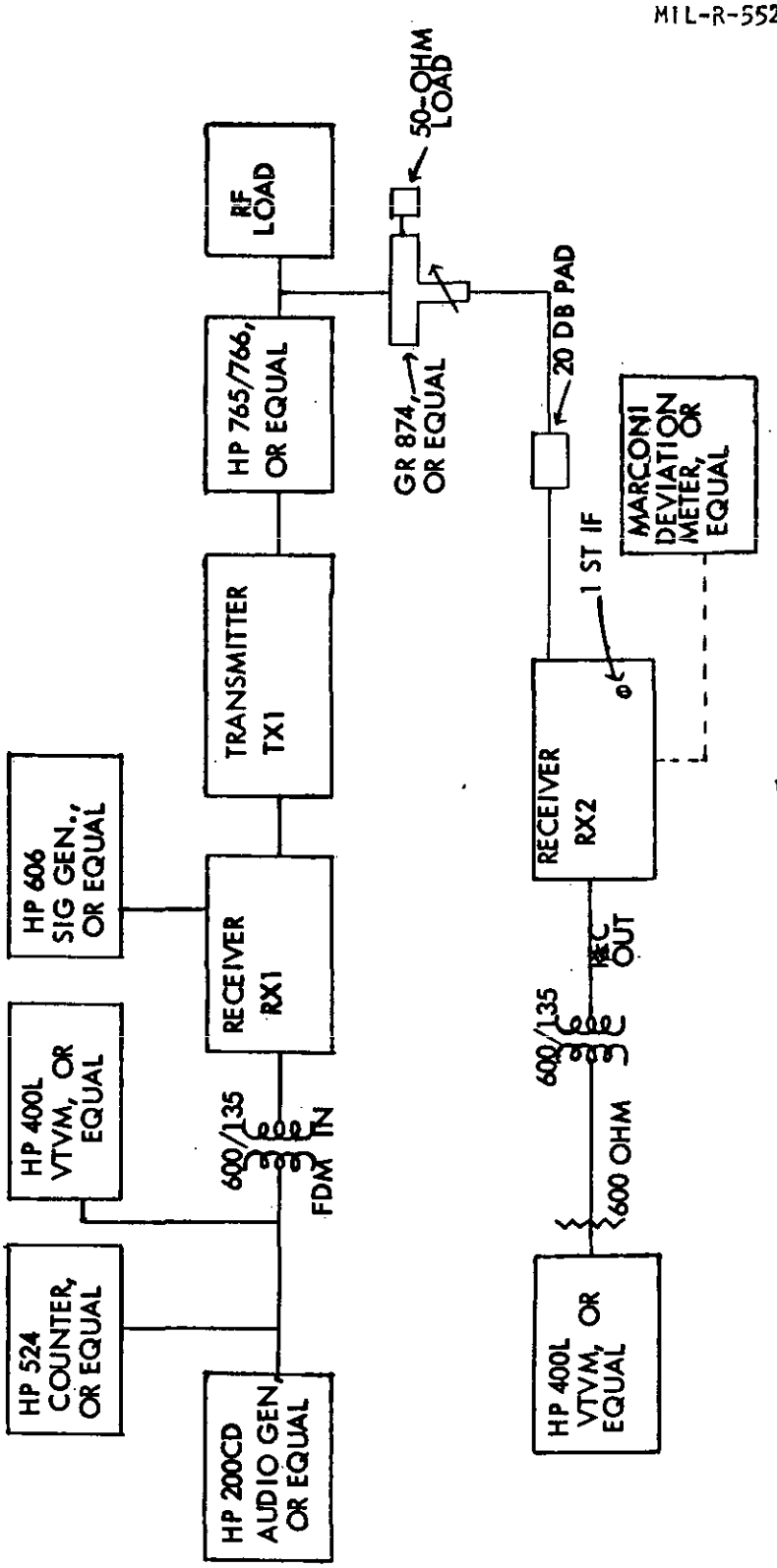


FIGURE 30

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the currentversion before use.

Step 24. The requirements of 3.10.6.2 shall be met.

REFLECTED POWER METERING TEST

Step 25. With the transmitter rf output terminated with a dummy load, tune the transmitter to any channel in accordance with 4.16.2.2.

Step 26. Set the multimeter to "REFL. PWR" and observe the meter reading.

Step 27. Terminate the transmitter in a load that will produce a VSWR of 2:1 (Two 50 ohm loads in parallel).

Step 28. Observe that the REFL PWR indication has increased by at least 2.5 ua from that of Step 2.

Step 29. The requirements of 3.10.6.2 shall be met.

4.16.8.3 Duplexer operation test - The equipment shall be set up as in Figure 31 except that the receiver will temporarily be connected directly thru a 20 db pad to the signal generator.

Step 1. Tune the transmitter to any selected channel in accordance with 4.16.2.1. Place the transmitter in "standby."

Step 2. Preset the "Rec Sig 2" dial on the transmitter, 16 channels higher or lower than the transmitter operating channel selected.

Step 3. Tune the receiver, to 16 channels higher or lower than the transmitter operating channel previously selected and in agreement with the Rec Sig 2 channel, in accordance with 4.16.2.2.

Step 4. Set the signal generator to the frequency corresponding to the channel selected for receiver operation. Set the signal generator attenuator to minus 40 dbm output. Tune generator slowly for zero discriminator voltage at J-6 and peak Rec Sig 1 dial with receiver metering switch in Rec Signal position. Adjust the signal generator to minus 73 dbm and note "REC - SIG Level."

Step 5. Disconnect the signal generator and 20 db pad from the receiver and connect them to the 20 db arm of the directional coupler. Connect the receiver to the transmitter jack marked "to Rec." Tune "REC SIG" on transmitter (peak).

Step 6. Adjust the signal generator output until the rec signal level meter indicates the same level as that which was obtained for -73 dbm input.

Step 7. Place the transmitter into operation. Modulate the transmitter with plus 3 dbm of white noise in the 12 KC to 60 KC band. Measure the 34 KC "slot noise" of the receiver with the ND - 110. The requirements 3.10.3.3(a) of FDM Signal plus noise to noise at the 34 KC slot shall be met.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

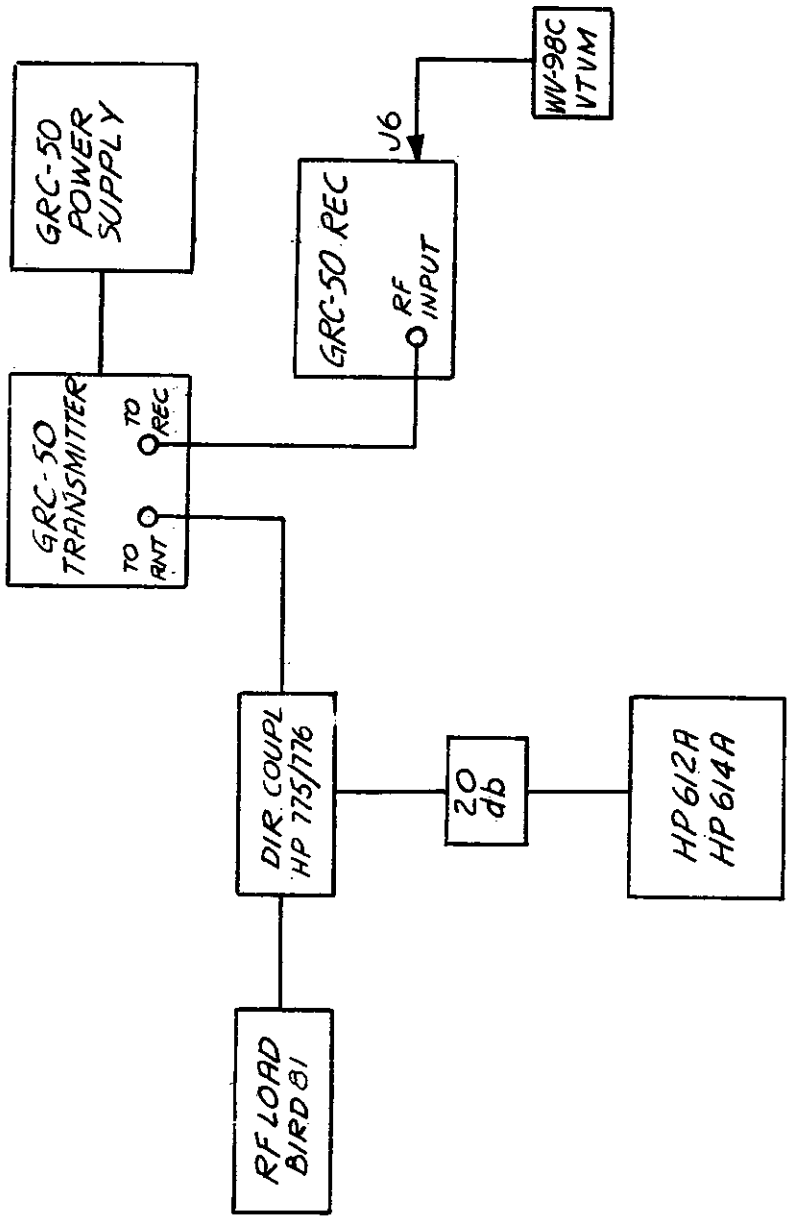


FIG. 31

Duplexer Operation Test

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

NOTE: If the slot noise exceeds the level specified in para 3.10.3.3(a), the degradation may be the result of spurious outputs from the transmitter which are within the passband of the receiver duplexer response region and which cannot be eliminated by the duplexer. To determine if this is the case, slowly tune the "OSC" control of the Amplifier Oscillators AM-1957( )/GRC, AM-1958( )/GRC and AM-1958A( )/GRC, as applicable while simultaneously observing the slot noise level for a change. An abrupt increase or decrease in the slot noise indicates a spurious transmitter output is present. Continue to tune the "OSC" for a minimum noise and then measure and record the noise level. The requirements of 3.10.3.3(a) shall be met.

Step 8. The requirements of 3.10.6.3 shall be met.

4.16.8.4 Level stability test.- The equipment shall be set up as in figure 32.

Step 1. Tune the transmitter and receiver to the desired rf channel in accordance with 4.16.2.1 and 4.16.2.2, respectively.

Step 2. Apply a modulated signal consisting of a uniform random noise spectrum from 16 to 56 kc of mean power +3 dbm and a 68 kc tone at 0 dbm, to the 135-ohm input terminals of the transmitter.

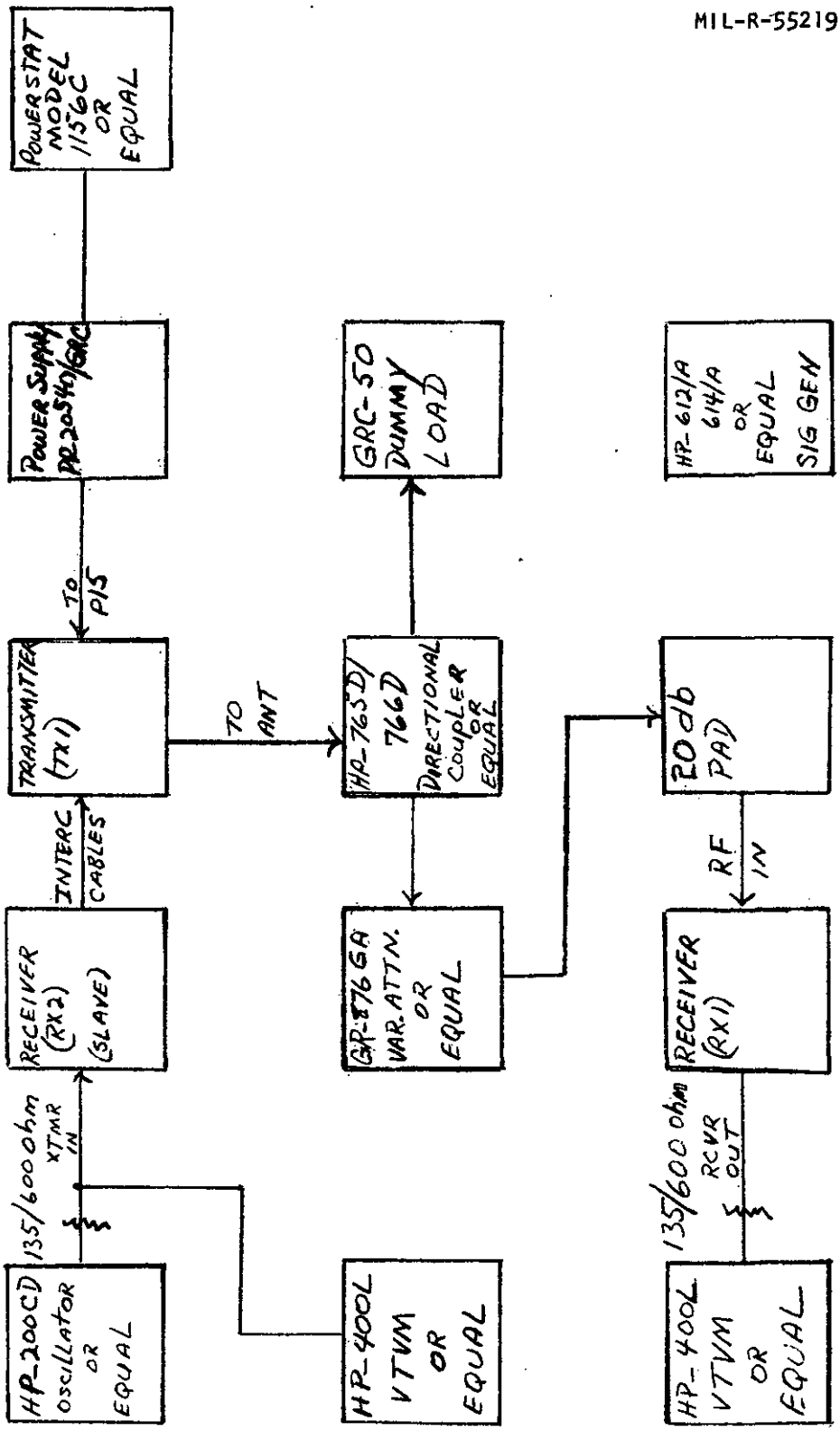
Step 3. Adjust the system for a -60 dbm signal level at the receiver input terminals.

Step 4. Adjust the overall system for a zero net gain.

Step 5. With the above conditions vary the receiver signal to -40 dbm and then -80 dbm. Record gain measurement.

Step 6. The following variables shall be supplied to the 135 ohms system in accordance with the table below. The system gain will be measured and compared to the reference.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.



Level stability Test

FIGURE 32

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

REC SIG LEVEL	60 CYCLE LINE VOLTAGE	GAIN MEASURE AND LIMITS (REFERENCES)
-60 dbm	115	
-60 dbm	109	within ± 1 db of ref.
-60 dbm	121	within ± 1 db of ref.
-40 dbm	121	within ± 1 db of ref.
-40 dbm	109	within ± 1 db of ref.
-40 dbm	115	within 1 db of ref.
-80 dbm	115	within ± 1 db of ref.
-80 dbm	109	within ± 1 db of ref.
-80 dbm	121	within ± 1 db of ref.

Step 7. The requirements of 3.10.6.4 shall be met.

4.16.8.5 Crosstalk Test - The equipment shall be set up as in figure 33. Standard test condition shall apply. (See 4.16.1)

Step 1. Select any one channel from each band.

Step 2. Tune the transmitter and receiver to the desired rf channel in accordance with 4.16.2.1 and 4.16.2.2, respectively.

Step 3. Adjust the rf input to the receiver to -53 dbm. (Calibrate REC SIG indication using HP 612/614).

Step 4. Set 12/24-4 channel switch to 12/24 position.

Step 5. Terminate all outputs not under test with specified loads.

Step 6. Calibrate ND-110, or equal, on Wide Band and Narrow Band.

Step 7. Switch the noise loading to the 12-channel filter (16-56 KC).

Step 8. Connect the noise loading rack to the ND-110 (135 ohm balanced).

Step 9. Switch the HP 512 oscillator to 0.05v position and adjust the noise generator for a reading /3.0 dbm on the ND-110.

Step 10. Disconnect the unit amplifier from the input to the filter.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

Step 11. Adjust the 512 oscillator to 68 kc at a level of 0 dbm on the ND-110.

Step 12. Reconnect the noise and read the total of 68 kc plus noise. (should be ⁄4.7 dbm)

Step 13. Connect the noise loading rack to the FDM at the transmitter directly through a test plug adapter or via a normally connected receiver. Connect the ND-110 to the FDM out on the receiver.

Step 14. With the transmitter and receiver FDM attenuators in mid-position, read the wide band noise with the ND-110 Noise & Distortion Analyzer Input switch to 135 ohm balanced. Adjust FDM attenuators until it is 4.7 dbm ± 1 db.

Step 15. Switch the ND-110 to Narrow Band and measure the outputs for the 14 KC, 34 KC and 60 KC slots with ND-110 and record. Requirements of para 3.10.6.5(a) shall be met.

Step 16. Adjust the GR attenuator (or equal) for -73 dbm input.

Step 17. Calibrate the Daven 12B (or equal).

Step 18. Record the noise at the remote ow terminals with the Daven 12B, or equal (FIA weighting). Requirements of para 3.10.6.5(b) shall be met.

Step 19. Readjust the GR attenuator for -53 dbm.

Step 20. Switch the noise input filter switch on the noise loading rack to the 24-channel position.

Step 21. Connect the noise loading rack to the ND-110.

Step 22. Remove the 68 kc tone by switching to the 0.05v range on the 512 oscillator.

Step 23. Adjust the noise generator to read ⁄5 dbm on the ND-110.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

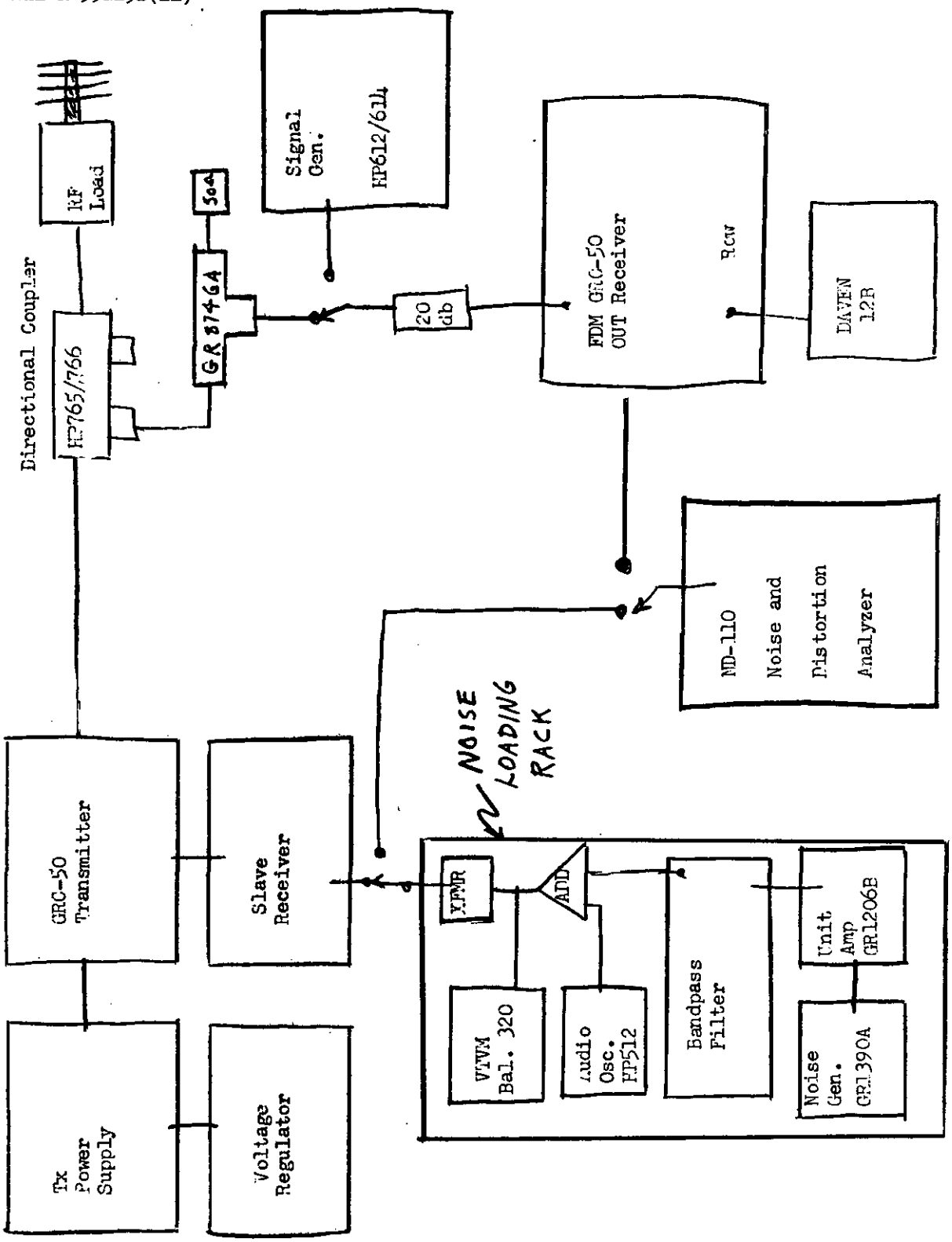


FIG. 33

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

Step 24. Disconnect the unit amplifier at the filter input and recheck the 68 kc tone level for 0 dbm.

Step 25. Reconnect the noise and check the total input: ⁄6.2 dbm.

Step 26. Reconnect per step 13.

Step 27. Switch ND-110 to wideband and record wideband noise; this level should be 6.2 dbm (± 1 dbm).

Step 28. Switch the ND-110 to narrow band.

Step 29. Record the output at the 6 KC, 34 KC, and 106 KC slots. Requirements of para 3.10.6.5(c) shall be met.

Step 30. Remove the noise input cable from the FDM terminals.

4.16.8.6 Tone generator test.- The equipment shall be set up as in figure 23. Standard test conditions shall apply. (See 4.16.1.)

Step 1. Select any one channel from each band.

Step 2. Tune the transmitter (TX1) and receiver (RX1) to the desired rf channel in accordance with 4.16.2.1 and 4.16.2.2, respectively.

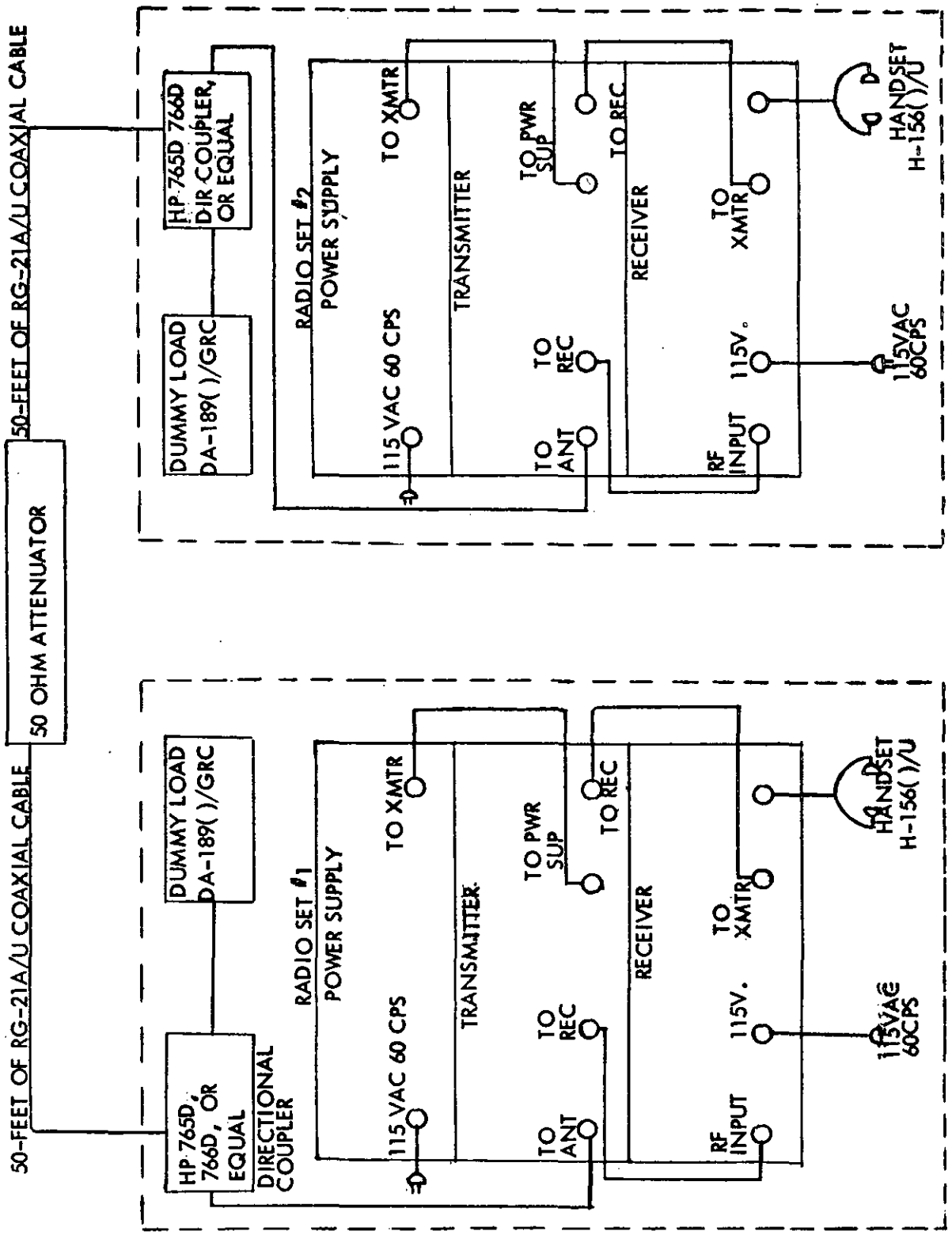
Step 3. Tune receiver (RX2) 50 MC away.

Step 4. Quiet receiver (RX2) under test with a signal level of -53 dbm.

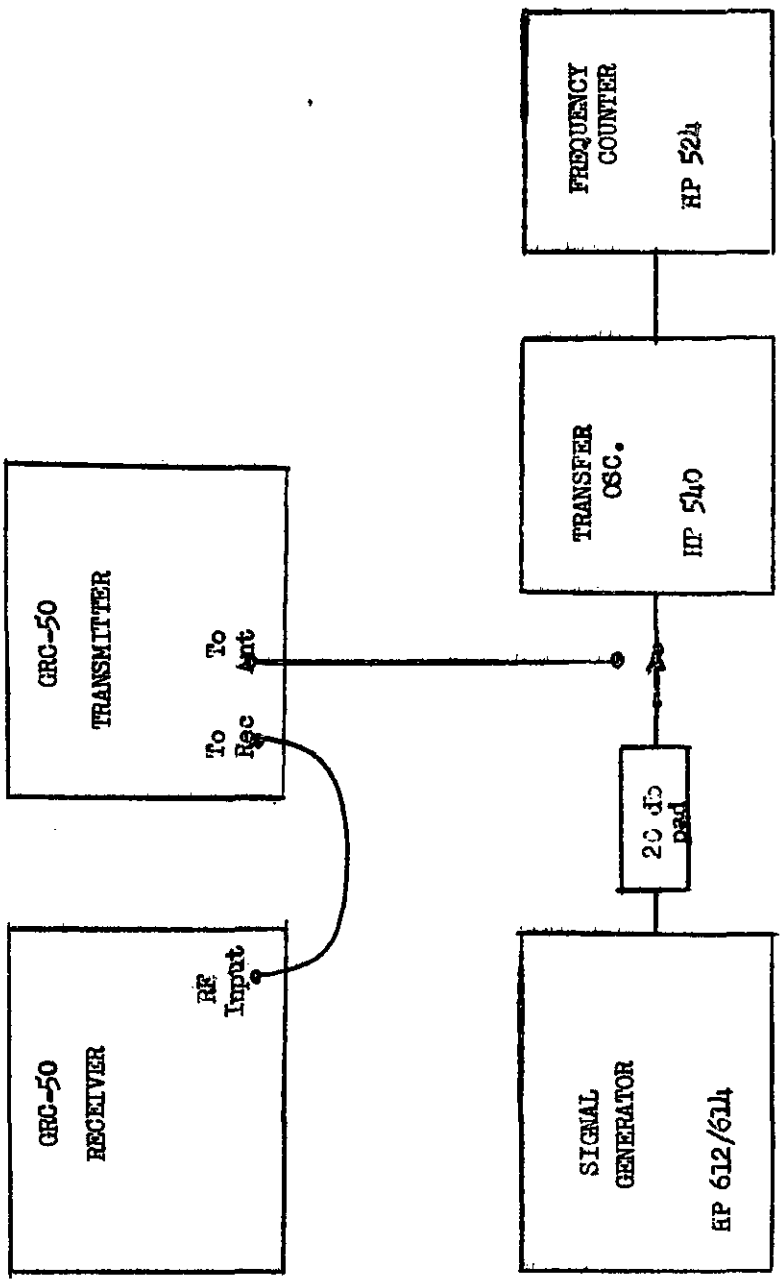
Step 5. Depress the ring button on RX1 and note indication on RX2. The requirements of 3.10.6.6 shall be met. (Both receiver headsets shall emit a clear, steady 1600 cycle tone.

Step 6. Repeat tests for receiver RX2. Requirements of 3.10.6.6 shall be met.

Source: https://assist.dla.mil -- Downloaded: 20260-26T04:14Z
Check the source to verify that this is the current version before use.



Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



DIAL ACCURACY TEST

FIGURE 35

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.8.7 Operational test.- The equipment shall be set up as in figure 34. This test shall be performed using both high and low band heads.

Step 1. Tune one transmitter and corresponding receiver to the desired RF channel in accordance with 4.16.2.1 and 4.16.2.2, respectively.

Step 2. Tune the other transmitter and receiver to the desired RF channel, separated by minimum 15 mcs in accordance with 4.16.2.1 and 4.16.2.2

NOTE: Insert attenuation as necessary in the transmission line between the two systems for receiver REC Signal panal meter readings of 40 microamperes maximum.

Step 3. Using Headsets H-156( )/U, communicate back and forth between the two radio sets.

Step 4. The requirements of 3.11 shall be met.

4.16.8.8 Preselector Dial Accuracy Test.- The receiver and transmitter shall be connected as illustrated in figure 35. No power is applied to the transmitter. No signal input to the transmitter "ANT" jack.

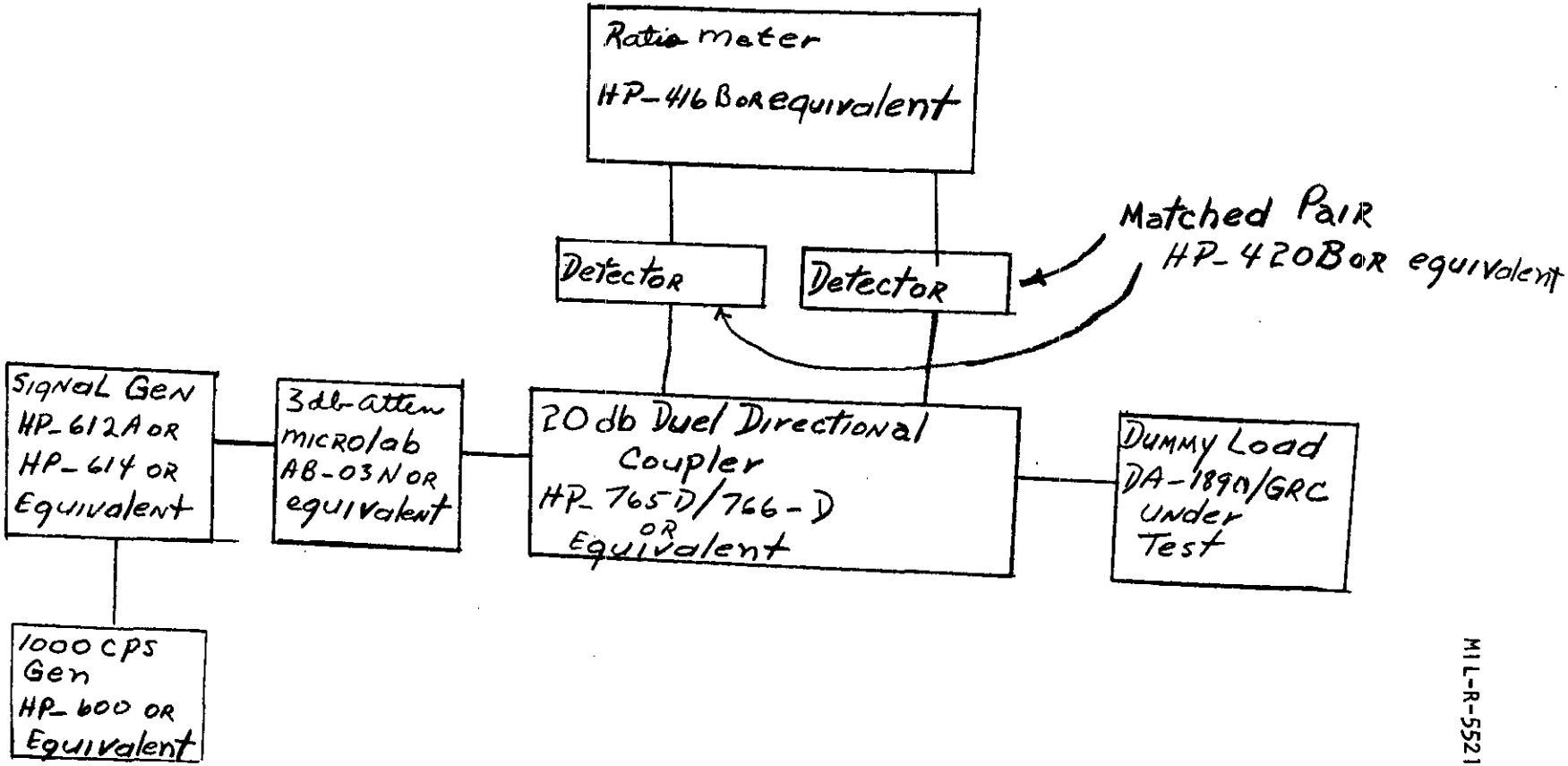
Step 1. Preset the receiver preselector dial "Rec Sig 1" precisely to the channel number desired.

Step 2. Preset the transmitter duplexer dial, "Rec Sig 2" precisely to the channel number desired.

Step 3. Tune the receiver to the desired channel in accordance with the standard procedure (paragraph 4.16.2.2) for setting the local oscillator, using the wavemeter provided.

Step 4. Adjust the signal generator HP614A or HP612A to the exact channel frequency using the transfer oscillator and frequency counter. Set the generator output level to -60 dbm when loaded with the 20 db pad.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.



Test set up for Dummy Load DA-1890/GRC VSWR Test

Figure 36

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

Step 5. Place the rec metering switch to Rec Sig 1 position and note reading.

Step 6. Apply the signal generator output through the 20 db pad to the transmitter "Ant" jack. The rec sig meter should indicate at least a 2 us change. Requirements of 3.10.6.7 shall be met.

4.16.9 Dummy Load, Electrical DA-189( )/GRC tests.- VSWR and calibration accuracy shall be checked from 600 MC to 1000 MC and from 1350 MC to 1850 MC as stated below:

4.16.9.1 VSWR test.- The equipment shall be set up as in figure 36.

Step 1. Measure the reflection coefficient at 100 MC increments from 600 MC to 1000 MC and from 1350 MC to 1850 MC and record same.

Step 2. Calculate the VSWR from the following relationship:

$$\text{VSWR} = \frac{1 + \text{reflection coeff}}{1 - \text{reflection coeff}}$$

Step 3. The requirements of 3.10.7.1 shall be met.

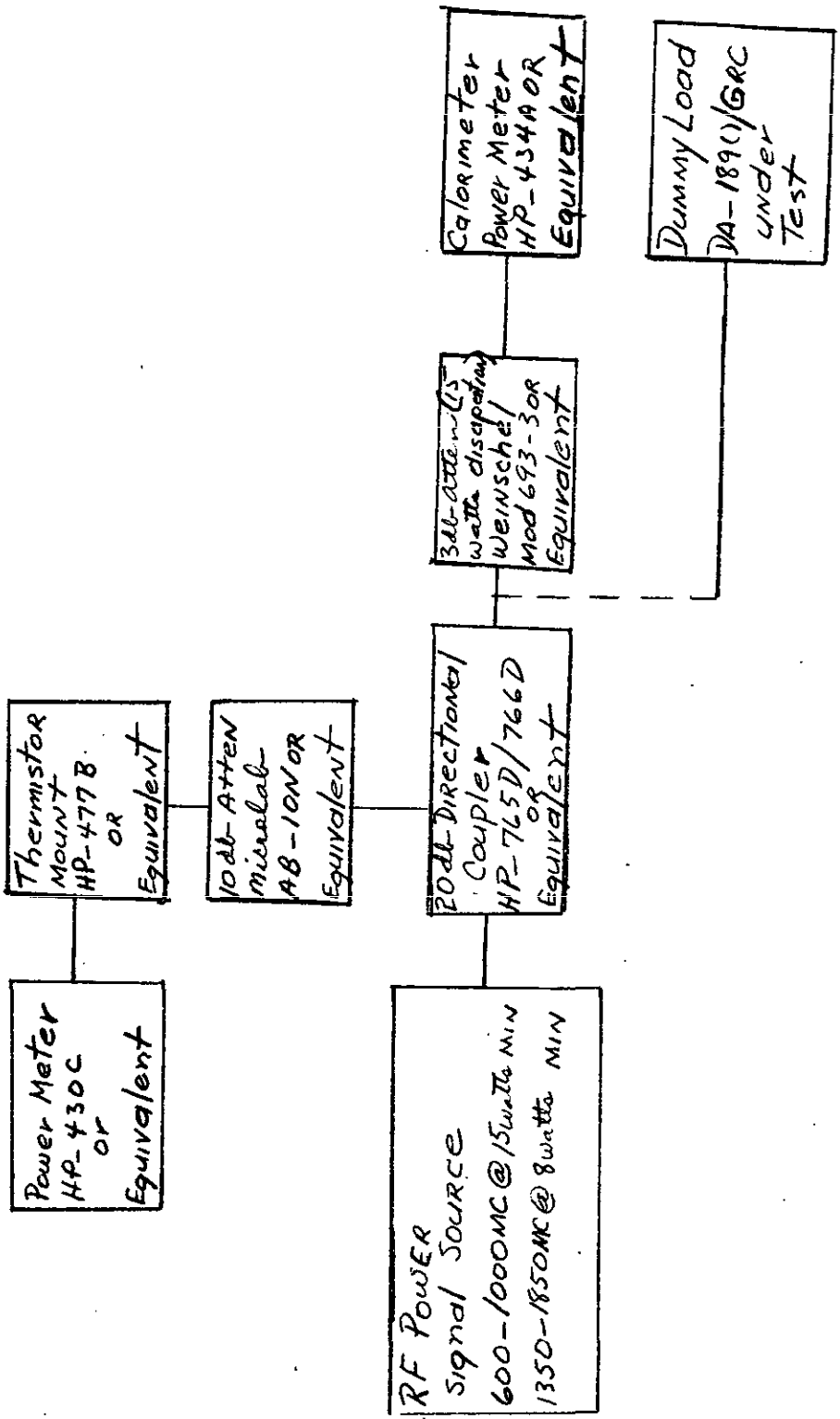
4.16.9.2 Power Calibration Accuracy Tests.- The equipment shall be set up as in figure 37.

Step 1. With the calorimeter wattmeter and 3 db attenuator as the load, adjust the rf power source level at 600 MC for 15 watts into the load (± 11.76 dbw). This is accomplished when the calorimeter reading =(11.76 dbw - the calibrated attenuation of the 3 db attenuator) which will vary with the specific attenuator but which will give a reading of approximately ⁄8.76 dbw.

Step 2. For the conditions of (1) read the power indicated on the HP-430C and record same.

Step 3. Remove the 3 db attenuator and calorimeter load and replace with Dummy Load DA-189( )/GRC.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.



Test Set up for Dummy Load DA-1891/GRC
POWER ACCURACY TEST

Figure 37

Source: https://assist.dla.mil -- Downloaded: 202640-26T04:14Z
Check the source to verify that this is the currentversion before use.

4.16.9.2 (Contd)

Step 4. Adjust the power source if necessary to obtain the same reading on HP-430C meter as read in Step 2.

Step 5. Read the power indicator of the Dummy Load DA-189( )/GRC.

Step 6. The requirements of 3.10.7 .2 shall be met.

Step 7. Repeat the above for 700, 800, 900 and 1000 MCS.

Step 8. The requirements of 3.10.7.2 shall be met.

Step 9. Repeat the above at 1350 MC with a total power into load of 8 watts (+ 9.0 dbw). Calorimeter will read (9.0-3.0 db attenuator) = + 6.0 dbw.

Step 10. The requirements of 3.10.7.2 shall be met.

Step 11. Repeat for 1450, 1550, 1650, 1750 and 1850 MCS.

Step 12. The requirements of 3.10.7.2 shall be met.

4.16.10 Power Supply, PP-2054( )/GRC (see 3.10.8)

4.16.10.1 Relay Operation Test. - The equipment shall be connected as shown in Figure 38. Standard Test conditions shall apply.

1. Set the Standby-Operate switch to "Operate."
2. Set the Circuit Breaker switch (CB1) to "ON".
3. The requirements of paragraph 3.10.8.1 shall be met.

4.16.10.2 High Voltage & Ripple Test - The equipment shall be connected as shown in Figure 38. Standard Test Conditions shall apply.

1. Set input Voltage (Variac) for 115 VAC.
2. Set dummy load to 800V (R) position.
3. Record the D.C. output and AC ripple readings.
4. Set dummy load to 650V(W) position.
5. Record the D.C. output and A.C. ripple readings.
6. Set dummy load to 645V(M) position.
7. Record D.C. output & AC ripple readings.
8. The requirements of paragraph 3.10.8.2 shall be met.

4.16.10.3 Low Voltage & Ripple test.- The equipment shall be connected as shown in Figure 38. Standard test conditions shall apply.

1. Set input voltage (Variac) for 115 Vac.
2. Set dummy load to 360V(K) position.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.16.10.3 (Contd)

3. Record D.C. output and AC ripple readings.
4. Set dummy load to 250V (V) position.
5. Record D.C. Output and AC ripple readings.
6. Adjust input voltage for 109V.
7. Record D.C. Output and AC ripple readings.
8. Adjust input voltage for 121V.
9. Record D.C. output and AC ripple readings.
10. Adjust input voltage for 115 V
11. Set dummy load to 150V (T) position.
12. Record the D.C. output and AC ripple readings.
13. Adjust the input voltage for 109V.
14. Record the D.C. output and AC ripple readings.
15. Adjust the input voltage for 121V.
16. Record the D.C. output and AC ripple readings.
17. Adjust the input voltage for 115V.
18. Set dummy load to 108V(A) position.
19. Record the D.C. output and AC ripple voltage.
20. Adjust the input voltage for 109V.
21 Record the D.C. output and AC ripple readings.
22. Adjust the input voltage for 121V.
23. Repeat Step 21.
24. The requirements of paragraph 3.10.8.3. shall be met.

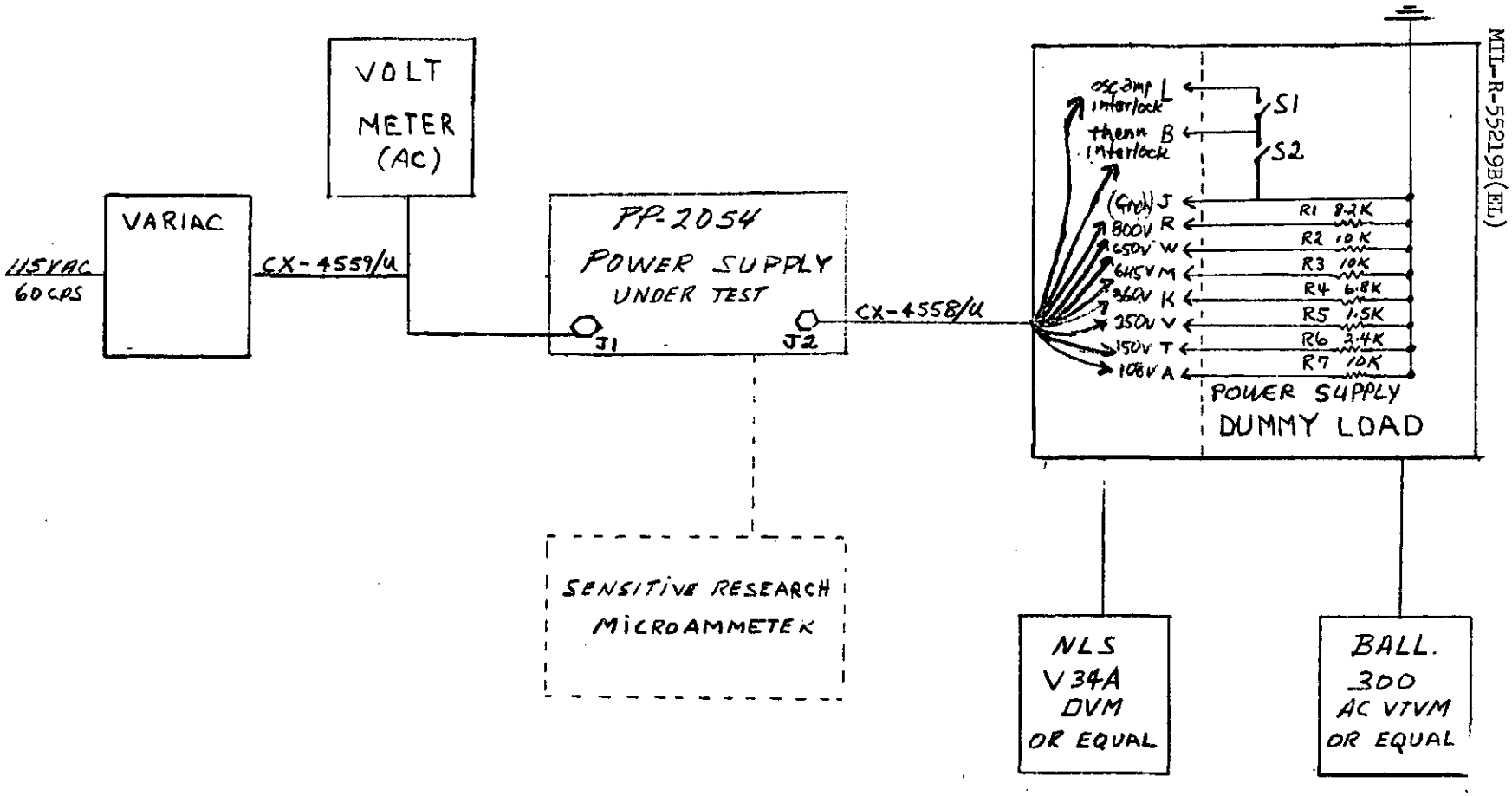
4.16.10.4 Current Test- The equipment shall be connected as shown in Figure 38. Standard test conditions shall apply.

1. Adjust the input voltage (Variac) for 115 V.
2. Connect the ground cable from the microammeter to J10 of power supply under test.
3. Connect the lead with a built-in 750 ohm resistor from the microammeter to each of the following terminals: J3, J4, J5, J6, J7, J8, J9 and record the readings. The requirements of paragraph 3.10.8.4 shall be met.

4.16.10.5 Interlock Test - The equipment shall be connected as shown in Figure 38. Standard test conditions shall apply.

1. Adjust the input voltage (Variac) for 115V.
2. Open switch S2 on dummy load for 10 seconds and then close. The requirements of paragraph 3.10.8.5 (a) shall be met.
3. Open switch S1 on dummy load for 5 seconds and then close. The requirements of paragraph 3.10.8.5 (b) shall be met.
4. Open switch S2 on dummy load for 10 seconds and then close. The requirements of paragraph 3.10.8.5 (c) shall be met.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.



Note: Power Supply under test should not be operated without the P.S. Dummy Load connected to J2, output connector.

Figure 38

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.17 Inspection for Interchangeability.- The electrical performance for the transmitter and the receiver listed in Table IV-Group A inspection shall be measured and the mechanical dimensions listed in 4.17.1 shall be gaged to determine conformance to 3.5.18. When a performance requirement or listed dimension is not within specified or design limits, it shall be considered a major defect.

4.17.1 Mechanical Dimensions.- The dimensions listed below shall be gaged or measured:

(a) External and internal dimensions of cases, covers, and insertable assemblies, when such dimensions affect mating of parts.

(b) Dimensions of cavities, when such dimensions affect insertion of items.

(c) Location of hinges and fasteners on separable parts or assemblies which must mate, such as cases, covers, and mountings.

(d) Location of connectors, locking pins, fasteners, slides, and mountings which receive mating parts of plug-in assemblies and major units; and location of the mating parts on the plug-in assembly or major unit.

(e) Size and form of special threads.

(f) Tuning cavities that require close fitting.

(g) Tube cavities for 4037A, and similar type tubes.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

4.17.2 Electrical performance.- Prior to re-assembly into the equipment, the plug-in modules and sub-assemblies listed below shall be interchanged among the Group B samples selected and tested to determine compliance with the above reference Table:

PP-2054
AM-1955 or 1955A
AM-1956 or 1956A
AM-1957
AM-1958 or 1958A
Baseband assy (3A3)
Second IF assy (3A5)
Signaling Unit (3A6)
Baseband Assy (2A3)
AFC assy (2A4 or 3A4)
Modulator assy (2A5)

4.18 Visual and mechanical inspection.- Parts and equipment shall be examined for the defects listed in Standard MIL-STD-252.

4.19 Quality conformance inspection of preparation for delivery.- Preparation for delivery shall be inspected in accordance with Specification MIL-P-116 to determine conformance to the requirements of Section 5.

4.20 Electromagnetic Compatibility Tests.- The preproduction model equipped for the reduction of radio interference in accordance with paragraph 3.7, shall be tested as described in MIL-E-55301(EL) by the contractor or his private testing organization. The contractor shall furnish the contracting officer, for approval, the Electromagnetic Compatibility test report required in MIL-E-55301(EL). The test report will include a block diagram of the test set-up, a description of the interference reduction system applied and a list of the test equipment utilized as described in MIL-E-55301(EL). Upon approval by the contracting officer and provided all other requirements of the specification are met, the preproduction model shall be used as a model for all production units.

5. PREPARATION FOR DELIVERY

5.1. Preservation and packaging.-(Packaging Levels, see section 6).

5.1.1 Level A - Radio Set AN/GRC-50 . - Packaging shall be developed in accordance with the methods listed below per MIL-P-116, employing kinds and types of materials which when applied will afford adequate protection against corrosion, deterioration and damage during world-wide shipment, handling and open storage.

ITEM	METHOD
Antenna AT-903 ( )/G	111
Case CY-2582/GRC	111
Case CY-2583/GRC	111
Mast Base AB-577 ( )/GRC	111
Amplifier Oscillator AM-1957( )/GRC	1C-5

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

5.1.1 (Contd)

ITEM	METHOD
Amplifier Oscillator AM-1958 ( )/GRC	1C-5
Transmitter Oscillator T-983 ( )/GRC	1C-5
Amplifier Converter AM-1955( )/GRC	1C-5
Amplifier Converter AM-1956 ( )/GRC	1C-5
Receiver Radio ( )/GRC	1C-5
Dummy Load	1C-5

5.1.1.1 Cushioning for Method 1C-5 shall be fabricated of material equal to the water-resistant requirements of the unit package container.

5.1.1.2 Technical literature shall be packaged Method 1C-1.

5.1.1.3 Running spares. - Running spare parts as required, shall be packaged as follows:

5.1.1.4 Tubes (spares). - Spare tubes shall be individually packaged in accordance with the applicable provisions of MIL-E-75.

5.1.1.5 Vibrators, Fuses, Fuselinks, Lamps, Insulators, and Antenna Sections.- Each item shall be packaged individually Method 1I1 as follows: Cushion each item by wrapping in Type II wrapping, cushioning paperboard, conforming to PPP-P-291. Secure the wrap with Class 1, gummed paper tape conforming to UU-T-111.

5.1.1.6 Cords and Cable. - Each cord and cable shall be packaged individually Method 1I1 as follows: Wind each item into a coil of proportionate dimensions. Tie in three places with cotton twine, Type 1, 6 ply, conforming to T-T-871 secure the ends with the ties.

5.1.1.7 Carbon Brushes, Connectors, Relays, Telephone Indicators, Protectors and Coils. - Each item shall be individually packaged Method 1C-6 as follows: Cushion each item by wrapping in Type II wrapping cushioning paperboard, conforming to PPP-P-291. Secure the wrap with Class 1, gummed paper tape conforming to UU-T-111, prior to placing in bag.

5.1.1.8 All Other Items. All other items shall be packaged in accordance with MIL-E-17555.

5.1.1.9 Stowage within compartments.- When space is provided in the equipment for stowage of running spares, the applicable items packaged as specified above will be stowed within the compartments provided. Overflow packages, or if no space is provided in the equipment will be consolidated as specified below for consolidated package.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

5.1.1.10 Consolidated Package. Consolidate the items packaged as specified above within a close-fitting, RSC fiber box, W5c conforming to PPP-B-636. Fill all voids with Type II, cushioning wrapping paperboard conforming to PPP-P-291, to prevent movement. Box closure shall be as specified in the appendix of the box specification.

5.1.2 Level C. - Each AN/GRC-50 shall be preserved and packaged in a manner that will afford adequate protection against corrosion, deterioration and damage during shipment from the supply source to the first receiving activity.

5.2 Packaging.- Radio Set AN/GRC-50.

5.2.1 Level A.- Selection of the container(s) shall be made from those cited below based upon the most economical container required to provide minimum, adequate protection for the packaged item(s) for this packing level of protection:

CONTAINER REQUIREMENTS

Container	Specification	Type, Class, Style	Max. Gross Wt. Limitation (lbs)
Box, Nailed Wood	PPP-B-621	Style 4, Class 2 Style 7, Class 2	200 (Note A) 1000 (Note A)
Box, Wirebound	PPP-B-585	Style 2 or 3 Class 2	200
Box, Wood Cleated Fiberboard	PPP-B-591	Class II	200 (Note A)
Box, Cleated Plywood	PPP-B-601	Overseas Type	200 (Note A)
Boxes, Fiberboard, Corrugated, Triple Wall	PPP-B-640	Class 2 Style E	
Box, Fiberboard	PPP-B-636	Type CF Class Weather-resistant	Table II (Special Requirements)
Box, Fiberboard, Corrugated, Double Wall, Weather Resistant	MIL-B-43291	Type CF	Table II

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

NOTE A. When the weight of a single item together with the shipping container exceeds 200 pounds, the maximum gross weight limitation of the container specification shall apply. In addition, skids, or pallet bases shall be applied in accordance with the requirements of the container specification, or if not specified in the specification, in a manner which will adequately support the item and facilitate the use of material handling equipment.

5.2.1.1 Container waterproofing.- If the container(s) being placed directly into the shipping container is not waterproofed, waterproofing shall be effected by use of an individual wrap or a Grade B caseliner conforming to MIL-L-10547 and appendix thereto, prior to packing. Dunnage not inclosed within the waterproof caseliner, bag or wrap shall consist of materials treated to resist water

5.2.1.2 Container closure.- Closure shall be in accordance with the applicable container specification or appendix thereto.

5.2.1.3 Strapping and reinforcing.- Metal strapping conforming to QQ-S-781, Type I, Class B shall be applied to wood and wood-cleated shipping containers in accordance with the requirements of the applicable box specification or appendix thereto. Fiberboard shipping containers shall be reinforced by pressure sensitive filament tape or nonmetallic banding conforming to PPP-T-97, Type IV and PPP-S-760, grade B respectively; selection of the material shall be determined by the requirements of the appendix of the applicable box specification. Application shall be as prescribed in the applicable box specification.

5.2.2 Level C.- Radio Set AN/GRC-50 shall be packed in (a) shipping container(s) in a manner that will afford adequate protection against damage to the package and its contents during shipment from the supply source to the first receiving activity. Shipping containers shall comply with the rules and regulations of the common carrier as applicable to the mode of transportation.

5.3 Marking.-

Marking of unit packages and shipping containers shall be in accordance with MIL-STD-129.

5.4 Preproduction Pack.-

Preproduction inspection including rough handling test is required and shall be in accordance with USAECOM Exhibit PMD-38. Functional testing of items or operation of equipment after rough handling testing to determine freedom from operational malfunction shall be required. Approval shall be in accordance with Section 4 of USAECOM Exhibit PMD-38. In addition, the contractor shall submit a preproduction sample pack(s) of the specified commodity for approval of selected materials and methods and for the witness of all necessary inspection by a Packaging Specialist from the U. S. Army Electronics Packaging Activity or designated representative.

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

5.5 ECOM Drawing SC-A-362100.-

Bills of material and illustration(s) are required in accordance with ECOM Drawing SC-A-362100. The contractor shall submit completed illustration(s) and bills of material through the Activity performing the preproduction pack review to assure that all requirements have been satisfactorily met prior to forwarding to the Packaging Activity of the U. S. Army Electronics Command.

6. NOTES

6.1 Intended use.- This equipment is intended for use either as a terminal or modulation-frequency patch repeater in tactical radio relay systems. It may be used in connection with four, twelve, or twenty-four channel frequency division multiplexer equipments, AN/TCC-3( ), -7( ), -42( ), or -43( ); or twelve-or twenty-four channel pulse-code modulation equipment, AN/TCC-41 ( ). The equipment is intended to be reliable and suitable for operation by personnel with minimum training on a 24 hour-per-day basis. It is to function as a primary communication medium and alternate for wide-band spiral-four or video cable circuits, and serve as a two-way eight jump radio relay system providing adequate quality for via trunk use. Vehicular mounting is as Radio Terminal Set AN/MRC-54( ) or AN/MRC-73( ).

6.2 Ordering data. - Procurement documents should specify the following:

(a) Title, number, and date of this specification and any amendment thereto.

(b) Levels of preservation, packaging, and packing required. (Level A, level B, or level C.)

(c) If rough handling test or cyclic exposure test is required.

(d) Marking and shipping of samples.

(e) Place of final inspection.

(f) Technical literature required. (See 3.12.)

(g) Quantity of running spare parts required. (See 3.12.)

(h) Submission of the statement of treatment referenced in 3.5.16.4, as soon as possible after award of contract. This statement should be submitted to the contracting officer.

6.3 Color (see 3.5.2).- The color chip furnished by the procuring Agency will be color chip No. X-24087, and may be obtained upon request to Commanding General, U. S. Army Electronics Command, 225 South Eighteenth Street, Philadelphia, Pennsylvania 19103

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

6.4 Definitions.-

6.4.1 Equipment divisions.- The terms used herein for equipment divisions conform to Standard MIL-STD-280.

6.4.2 Definitions of inspection terms.- The inspection terms used herein conform to Standard MIL-STD-109. The following are abbreviated definitions selected from those given therein for immediate reference:

6.4.2.1 Inspection.- Inspection is the examination or testing, or both, of supplies to determine compliance with applicable requirements. Sampling is an element of inspection.

6.4.2.2 Examination.- Examination consists of simple, generally non-destructive determinations of compliance, without use of special testing equipment.

6.4.2.3 Testing.- Testing consists of determinations of compliance, using technical means.

6.5 Group C inspection.- Approval to ship may be withheld, at the discretion of the Government, pending the decision from the contracting officer on the adequacy of corrective action. (See 4.5.3.2)

6.6 Nomenclature (see 1.1).- The parentheses in the nomenclature will be deleted or replaced by a number identifying the particular design; for example: AN/GRC-50 (V) (99). The contractor should apply for nomenclature in accordance with the applicable clause in the contract.

6.7 Color of textiles (see 3.5.17).- Standard samples of Olive Drab No.7 may be obtained from the Sample Loan Unit, Military Clothing and Textile Supply Agency, Philadelphia, Pennsylvania. When requesting samples, indication should be given as to type of material concerned.

6.8. Verification inspection (see 4.5).- Verification by the Government will be limited to the amount deemed necessary to determine compliance with the contract and will be limited in severity to the definitive quality assurance provisions established in this specification and the contract. The amount of verification inspection by the Government will be adjusted to make maximum utilization of the contractor's quality control system and the quality history of the product.

6.9 Humidity.- Natural climatic extremes may include 100-percent relative humidity, with condensation, up to 4 hours at a time. This requirement (see 3.6.3) is specified by Standard MIL-STD-210. The moisture-resistance test (see 4.7), using Standard MIL-STD-170, is a procedure that gives a practical test approximating the natural conditions.

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the currentversion before use.

6.10 Radio Set AN/VRC-50(V) ( ) orderwire.- The orderwire for this set contains the following:

(a) Provisions for orderwire communications between a multiplex, spaced up to one mile from a terminal radio set, and the terminal radio set, and all other radio sets or multiplexers in the system by utilizing the portion of the broadband transmission facilities below 4 kc.

(b) Provisions for bridging a telephone handset to the transmission circuits of the radio set in both directions at a repeater station so that communications may be had in both directions simultaneously and between two halves in a repeater station when they are separated by up to one mile of spiral-four cable.

(c) Provision for through orderwire communications between terminal radio set and all other radio sets in the system using PCM multiplexing (reshaping and combining at each radio set). In view of the use of combiners between radio sets, and tandem connection f or orderwire communications are obtained by separate patching other than the PCM connections through the combiners.

(d) Provision for remote orderwire operation on a 2-wire basis for distances up to one mile; this facility is available during normal multi-channel operation.

(e) Provisions for signalling to or being signalled by the attendant at all other radio or multiplex stations in the system.

(f) A 1,600 cps tone generator is incorporated into the radio set to provide the facilities of (a) above. The tone generator is controlled by the operation of a non-locking switch. A buzzer and lamp are provided in the receiver; these are energized when a 1,600 cps tone is being received in the orderwire channel.

6.11 Noise figure testing.- Noise figure testing of receivers has been described in the RCA Review, March 1947, Volume VIII, No. 1, PPP-169-185, "A Coaxial: Line Diode Noise Source for UHF" by Howard Johnson.

6.12 Identification of Changes of Revised Specification.- Asterisks are not used in this revision to identify changes with respect to the previous issue, due to the extensiveness of the changes.

PREPARING ACTIVITY
Army-EL

CUSTODIAN
ARMY-EL

PROJECT NUMBER
5820-A322

Source: https://assist.dla.mil -- Downloaded: 202610-26T04:14Z
Check the source to verify that this is the current version before use.

FOLD

HEADQUARTERS
U. S. ARMY ELECTRONICS COMMAND
FORT MONMOUTH, NEW JERSEY 07703

POSTAGE AND FEES PAID
DEPARTMENT OF THE ARMY

OFFICIAL BUSINESS

Commanding General
U. S. Army Electronics Command
ATTN: AMSEL-PP-
Fort Monmouth, New Jersey 07703

FOLD

To detach this form, cut along this line

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.

To detach this form, cut along this line

SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION

ORGANIZATION | CITY AND STATE

CONTRACT NO. | QUANTITY OF ITEMS PROCURED | DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)* | DATE

DD FORM 1426
1 APR 63

Source: https://assist.dla.mil -- Downloaded: 2026-10-26T04:14Z
Check the source to verify that this is the current version before use.